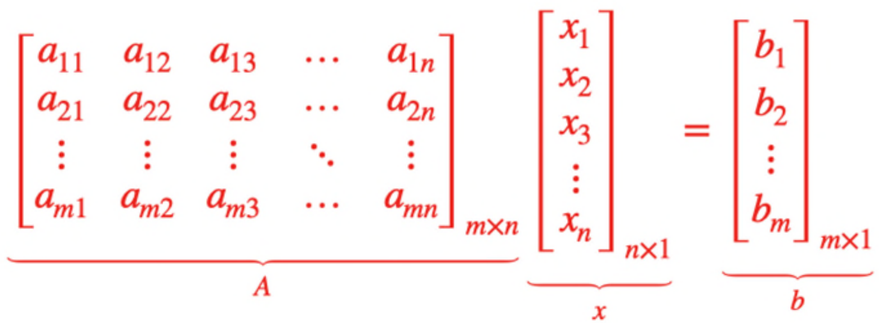


CX1104: **Linear Algebra for Computing**

Chap. No : **8A.0**

Lecture : **Complex Numbers**

Topic : **Overview of this chapter**

Concept :

Instructor: **A/P Chng Eng Siong**

TAs: **Shreyas Gopal**

Rev: 23rd July 2024

Content

8A.1.0: **Overview**

8A.1.x: **Introduction to complex numbers+ Arithmetic Operations + Eulers formula (1.5 hour)**

- definition and motivation
- geometric interpretation
- rectangular and Polar form
- add, subtract, multiply, and divide,

8A.2.x: **Complex exponential form (1.5 hour)**

DeMoivre's Theorem: power and roots of complex number, n^th root of unity

8A.3.x: **DFT transformation matrix $\mathbf{W}$ and $e^{j\omega n}$ (3 hours)**

- introducing $\boldsymbol{x} \in C^n = [e^{j\omega n}]$ a complex sequence vector with N elements
- scaling $\alpha \boldsymbol{x}$,where $\alpha \in C$
- introducing Fourier basis DFT W matrix and its inverse:
  - Analysis : $\mathrm{X} = Wx$: if A is square, this can be viewed as a change of basis (analysis)
  - Synthesis: $x = \frac{1}{N} W^H X$

References Complex numbers: some notes and YouTube

1) CUHK: https://www.math.cuhk.edu.hk/course_builder/1920/math1510d/C10_ComplexNumber.pdf
2) Paul Lamar online primer: https://tutorial.math.lamar.edu/Extras/ComplexPrimer/ComplexNumbers.aspx, http://tutorial-math.wip.lamar.edu/Classes/Alg/ComplexNumbers.aspx

3) Harvey Mudd: https://math.hmc.edu/calculus/hmc-mathematics-calculus-online-tutorials/precalculus/complex-numbers/#top
4) Some Chapters on complex numbers:
   A. https://ncert.nic.in/ncerts/l/kemh105.pdf
   B. https://www.cimt.org.uk/projects/mepres/alevel/fpure_ch3.pdf
   C. https://ms.mcmaster.ca/~vantuyl/courses/oldcourses/Chap10.pdf

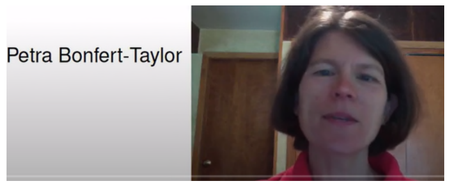
5) Online courses:
   A. Petra's Week1,Lect 1~4: https://www.youtube.com/watch?v=CVpMpZpd-5s&list=PLi7yHjesblV0sSfZzWdSUXGO683n_nJdQ

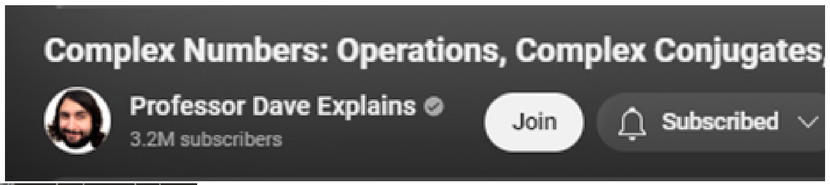
   B. Professor Dave Explains:
      https://www.youtube.com/watch?v=Yddjxj49C_M

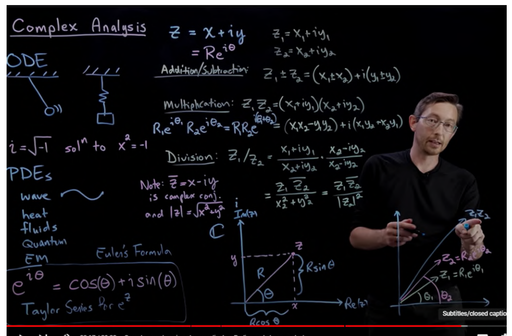
   C. MIT Gilbert Strang:
      https://www.youtube.com/watch?v=Jkv-55ndVYY

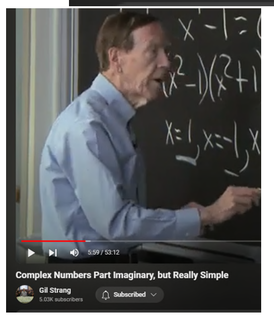
   D. Oxford Vicky Neale: An Intro to Complex Numbers
      https://youtu.be/BP7Ujbyu-NE

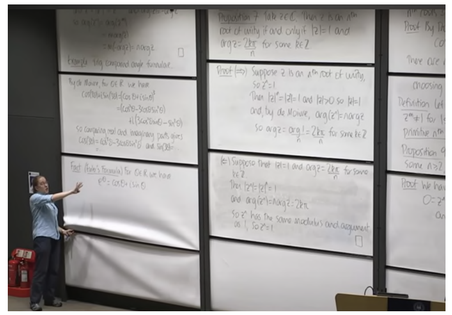
   E. Steve Brunton: Complex Analysis L01:
      https://www.youtube.com/watch?v=_mv0q7-WF4E











More References for Complex Numbers

Chng Eng Siong

10 Aug 2024

Last edited: 23 May 2024

DFT and Linear Algebra: some nice links

Concepts:

1) An application of compression using DFT

Can you guess the song? Fourier Music Decomposition – YouTube

2) Reducible = https://youtu.be/yYEMxqreA10

Good Overview:

1) Simon Xu: Discrete Fourier Transform - Simple Step by Step (youtube.com) (from:3:30)

3) Matlab: https://www.youtube.com/watch?v=QmgJmh2I3Fw

Some Maths:

1) Brunton introduces DFT: The Discrete Fourier Transform (DFT) (youtube.com)
2) Brunton: The DFT Matrix: https://www.youtube.com/watch?v=Xw4voABxU5c
3) SigFy: The Linear Algebra of Fourier Transforms (youtube.com)
4) Van Veen (signal representation using bases): https://www.youtube.com/watch?v=a4Atmssz8-A
5) Jack Gunther: https://www.youtube.com/watch?v=Er-FcErLXrQ

Work examples:

MIT Mysore = https://youtu.be/BQRmJYoFR3M
Exploring technologies (DFT) = https://youtu.be/50_VnwA3LEk
Easy Electronics = https://youtu.be/dE9g72LlPdM
RLC-EEE = https://youtu.be/7VHE3v57XHU
Thansi = https://youtu.be/M0Ez9Pa-6xQ

Inner Product: linear in slot 1

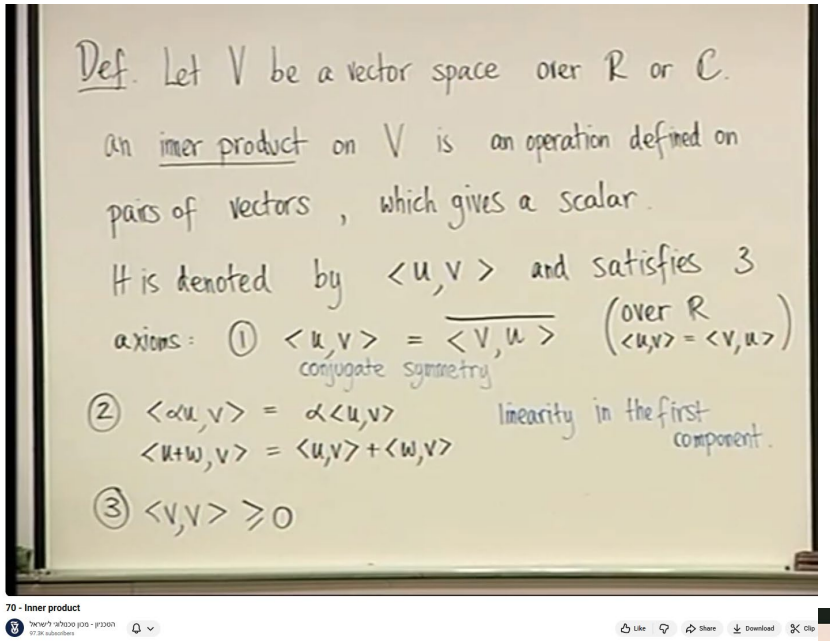
1) Technion: https://www.youtube.com/watch?v=raW8dUc1Ts0

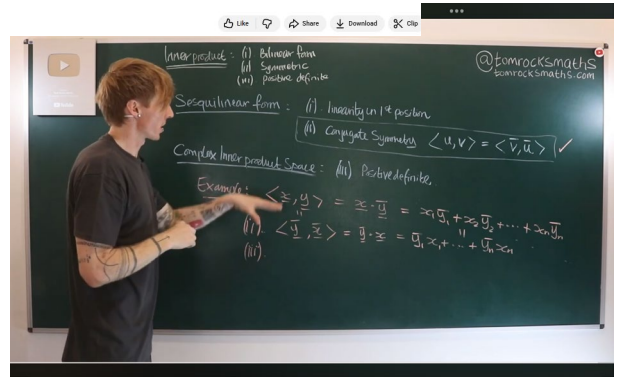
2) Oxford: https://www.youtube.com/watch?v=ywjA25xFuew

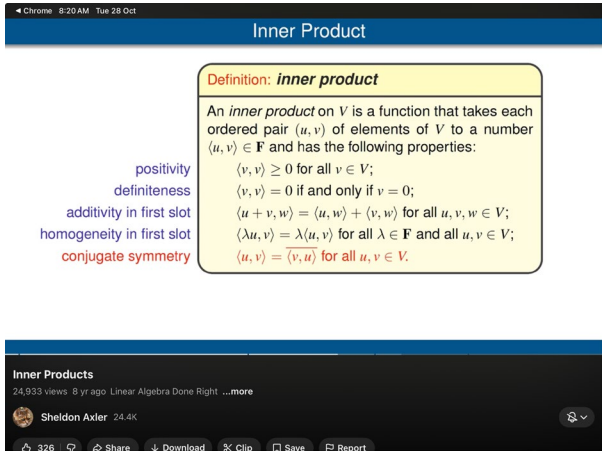
3) Sheldon Axler: https://www.youtube.com/watch?v=MkYzm4W9p6U

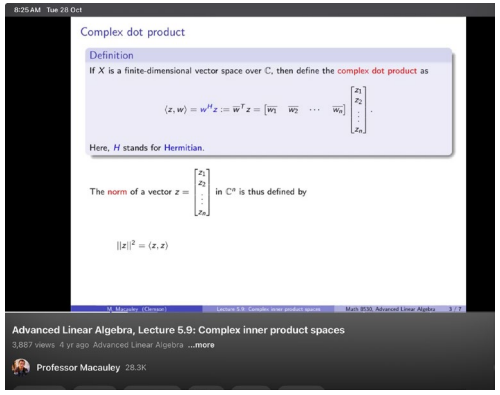
4) Professor Macauley: https://www.youtube.com/watch?v=uXAaRKVoxqI

5) Krishna: https://www.youtube.com/live/f7OJ6WKiOyc



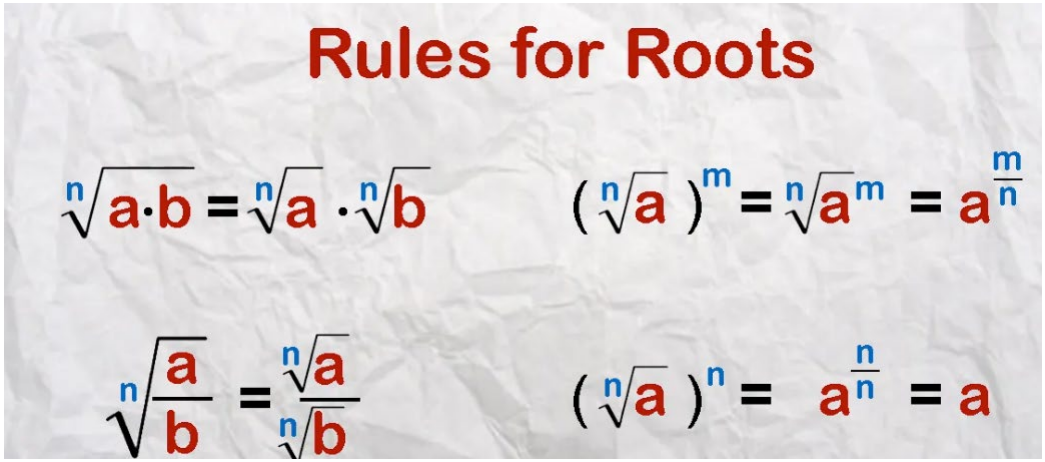




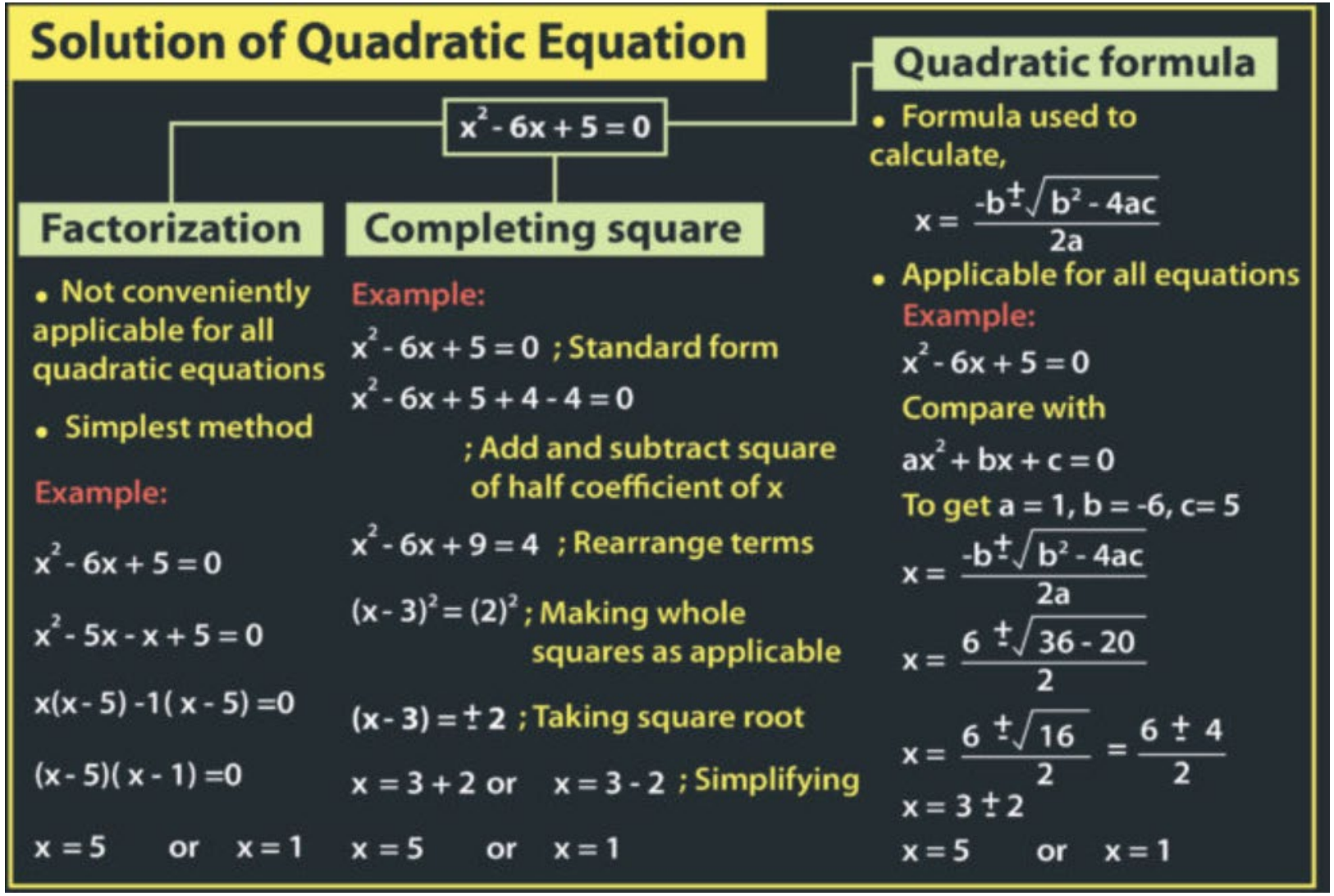


Ref: Properties of Exponents (for real)

Exponent Rules For $a \neq 0, b \neq 0$	
Product Rule	$a^x \times a^y = a^{x+y}$
Quotient Rule	$a^x \div a^y = a^{x-y}$
Power Rule	$\left(a^x\right)^y = a^{xy}$
Power of a Product Rule	$(ab)^x = a^x b^x$
Power of a Fraction Rule	$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
Zero Exponent	$a^0 = 1$
Negative Exponent	$a^{-x} = \frac{1}{a^x}$
Fractional Exponent	$a^{\frac{x}{y}} = \sqrt[y]{a^x}$



Ref: useful maths background: roots of quadratic functions



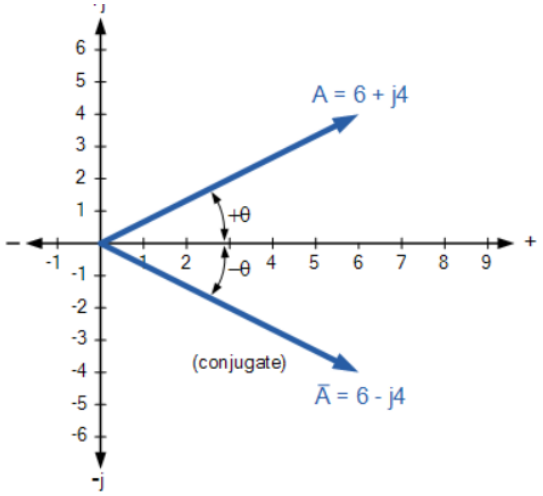
https://www.toppr.com/guides/maths/quadratic-equations/nature-of-roots/

Ref: useful maths background - exponent

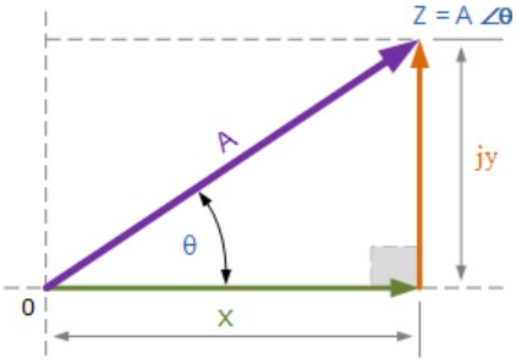
Tip: Easy-to-remember property name	Property expression
Same-base product	$x^a \cdot x^b = x^{a+b}$
Same-base division	$x^a \div x^b = x^{a-b}$
Same-exponent product	$x^a \cdot y^a = (xy)^a$
Same-exponent division	$x^a \div y^a = (x \div y)^a$
Double exponent	$(x^a)^b = x^{a \times b}$
Zero exponent	$x^0 = 1$
Negative exponent	$x^{-a} = \frac{1}{x^a}$
Fractional exponent	$x^{a/b} = \sqrt[b]{x^a}$

3 Different ways to represent complex numbers

Rectangular form



Polar Form Representation of a Complex Number



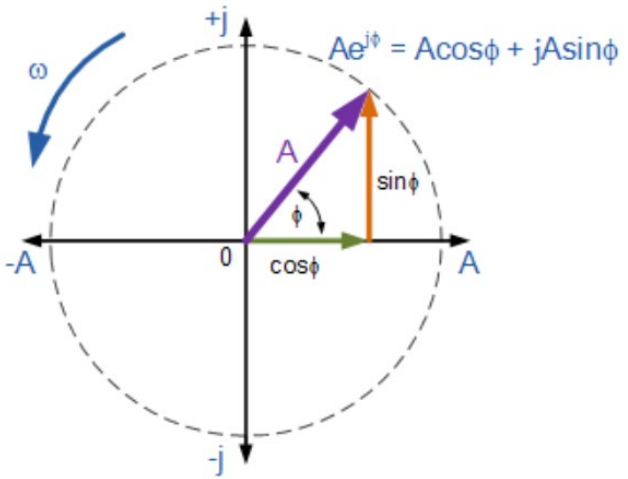
$$A = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\frac{y}{x}$$

$$x = A.\cos\theta$$

$$y = A.\sin\theta$$

Complex exponential form



Euler's Formula

$$e^{i\phi} = \cos\phi + i\sin\phi$$

$$Z = Ae^{j\phi}$$

$$Z = A(\cos\phi + j\sin\phi)$$

https://www.electronics-tutorials.ws/accircuits/complex-numbers.html

https://muthu.co/deriving-the-famous-eulers-formula-through-taylor-series/

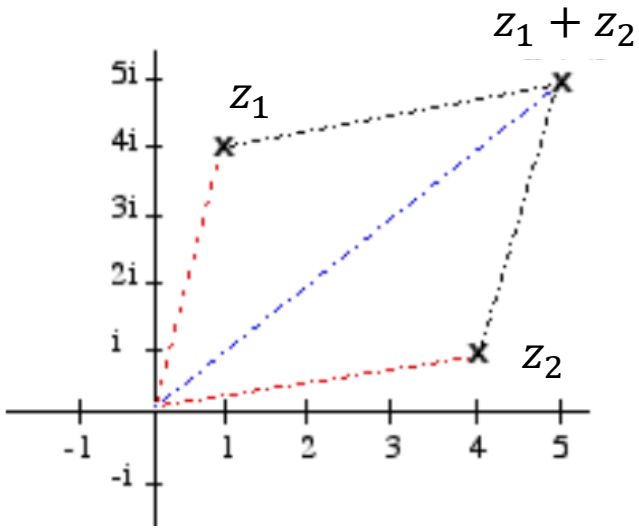
Visualisations: Addition and Subtraction

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

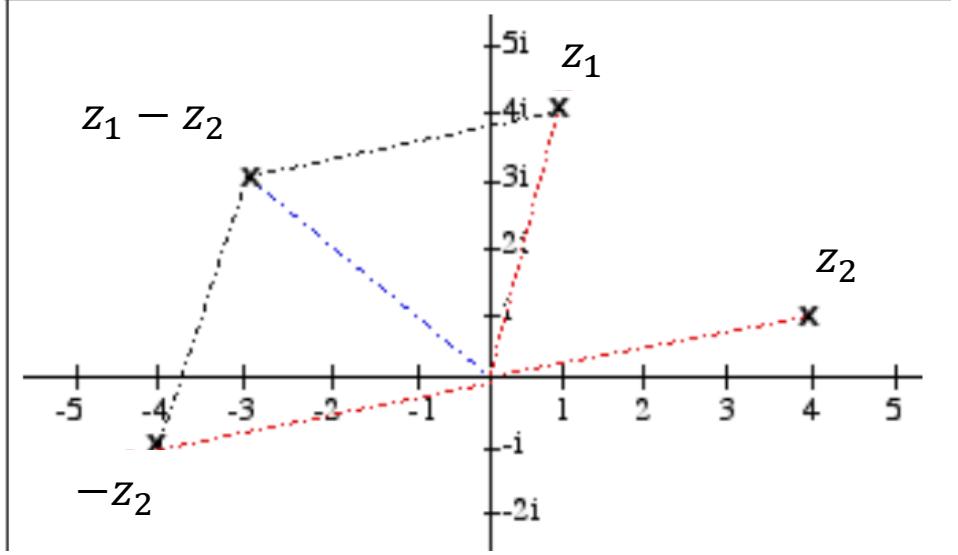
$$(a + ib) - (c + id) = (a - c) + i(b - d)$$

$z_1 = 1 + 4i, \qquad z_2 = 4 + i$

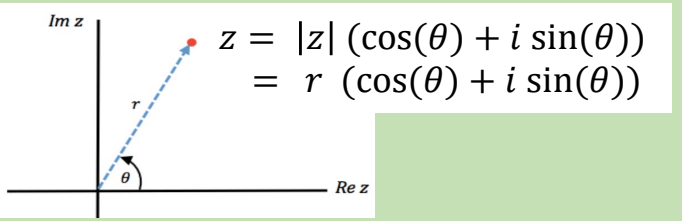
$z_1 + z_2 = (1 + 4i) + (4 + i) = 5 + 5i$



$z_1 - z_2 = (1 + 4i) - (4 + i) = -3 + 3i$



Visualisations: Multiplication of 2 complex number



$$z = |z|\,(\cos(\theta) + i\sin(\theta)) = r\,(\cos(\theta) + i\sin(\theta))$$

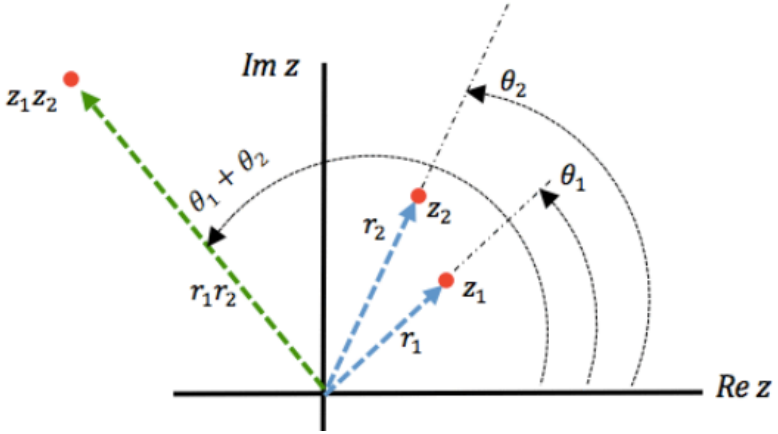
modulus $r = |z|$ (depicted as the length of the arrow)

argument θ the angle the arrow makes with the positive real axis

Multiplication in Cartesian Form

Let $z_1 = a + bi$ and $z_2 = c + di$ be complex numbers, then:

$$\begin{aligned} z_1 \cdot z_2 &= (a + bi)(c + di) \\ &= ac + adi + bci + bdi^2 \\ &= (ac - bd) + (ad + bc)i. \end{aligned}$$



$$\begin{aligned} z_1z_2 &= r_1\,(\cos(\theta_1) + i\sin(\theta_1))\,r_2\,(\cos(\theta_2) + i\sin(\theta_2)) \\ &= r_1r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)) \\ &= r_1r_2e^{j(\theta_1+\theta_2)} \end{aligned}$$

Note: its much easier to perform multiplication of complex numbers in complex exponential form.

https://mathspace.co/textbooks/syllabuses/Syllabus-501/topics/Topic-9613/subtopics/Subtopic-127684/?activeTab=theory
https://www.houseofmath.com/encyclopedia/numbers-and-quantities/numbers/complex-numbers/calculations/how-to-multiply-complex-numbers

Examples: multiplication and division of complex numbers in polar form

if $z_1 = r_1\angle\theta_1$ and $z_2 = r_2\angle\theta_2$ then

Note: r_j is the the modulus (absolute value)of $|z_j|$

$$z_1 z_2 = r_1 r_2 \angle(\theta_1 + \theta_2), \qquad \frac{z_1}{z_2} = \frac{r_1}{r_2}\angle(\theta_1 - \theta_2)$$

Example

If $z_1 = 5\angle(\pi/6)$, and $z_2 = 4\angle(-\pi/4)$ find a) $z_1 z_2$, b) $\frac{z_1}{z_2}$, c) $\frac{z_2}{z_1}$

Solution

a) To multiply the two complex numbers we multiply their moduli and add their arguments. Therefore

$$z_1 z_2 = 20\angle\left(\frac{\pi}{6} + \left(-\frac{\pi}{4}\right)\right) = 20\angle\left(-\frac{\pi}{12}\right)$$

b) To divide the two complex numbers we divide their moduli and subtract their arguments.

$$\frac{z_1}{z_2} = \frac{5}{4}\angle\left(\frac{\pi}{6} - \left(-\frac{\pi}{4}\right)\right) = \frac{5}{4}\angle\frac{5\pi}{12}$$

c)

$$\frac{z_2}{z_1} = \frac{4}{5}\angle\left(-\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{4}{5}\angle\left(-\frac{5\pi}{12}\right)$$

https://harshasnmp.wordpress.com/2017/11/15/multiplication-division-of-complex-numbers-in-polar-form/

DeMoivre Theorem: finding powers of z

$$z = |z|e^{j\theta} = |z|(\cos(n\theta) + i\sin(n\theta)$$

$$z^n = \left(|z|e^{j\theta}\right) \ldots \left(|z|e^{j\theta}\right) \quad ; \textit{(n of them)}$$
$$= |z|^n e^{jn\theta}$$
$$= |z|^n(\cos(n\theta) + i\sin(n\theta)$$

// Fractional power

$$z^{1/n} = \sqrt[n]{|z|e^{j\theta}}$$

$$= \sqrt[n]{|z|}\, e^{j\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right)} \quad , k = 0 \cdots (n-1)$$

Uses of De Moivre's Theorem

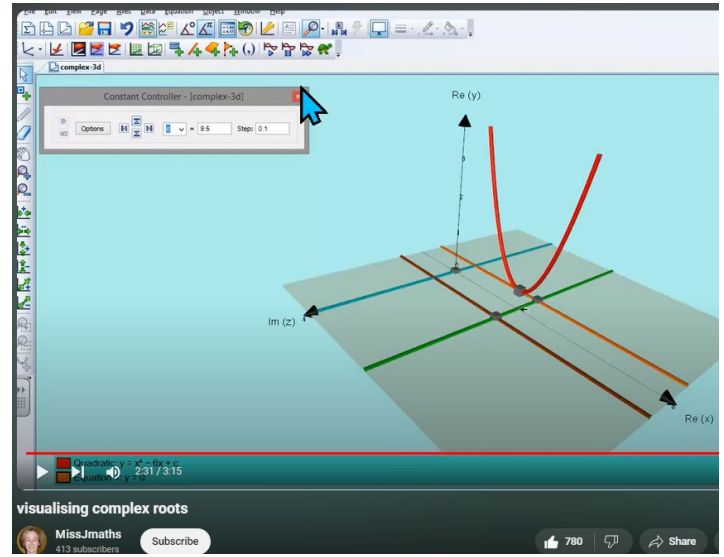
De Moivre's Theorem is used for various purposes. Some of its most important uses are,

- Finding the Roots of Complex Numbers.
- Finding the relationships between Powers of Trigonometric Functions and Trigonometric Angles.
- Solving the Power of Complex Numbers.

1) Mark Willis: Proof: https://www.youtube.com/watch?v=6UecqjGHR1w
2) https://ccrma.stanford.edu/~jos/st/Direct_Proof_De_Moivre_s.html

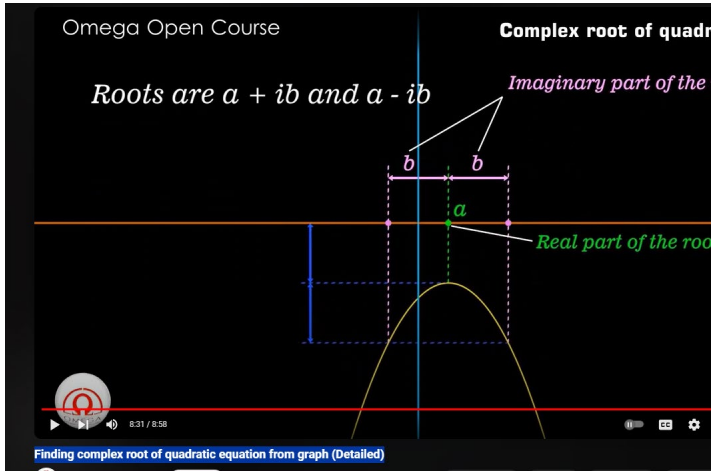
Visualizing complex roots

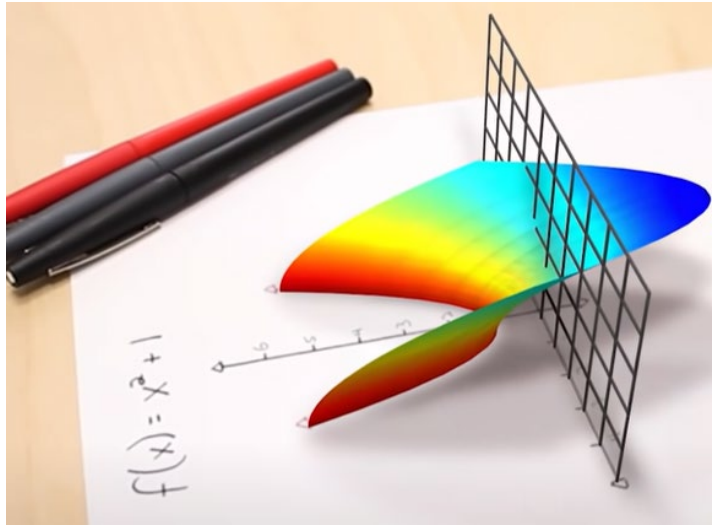
1) https://www.youtube.com/watch?v=5g80j17w0CE



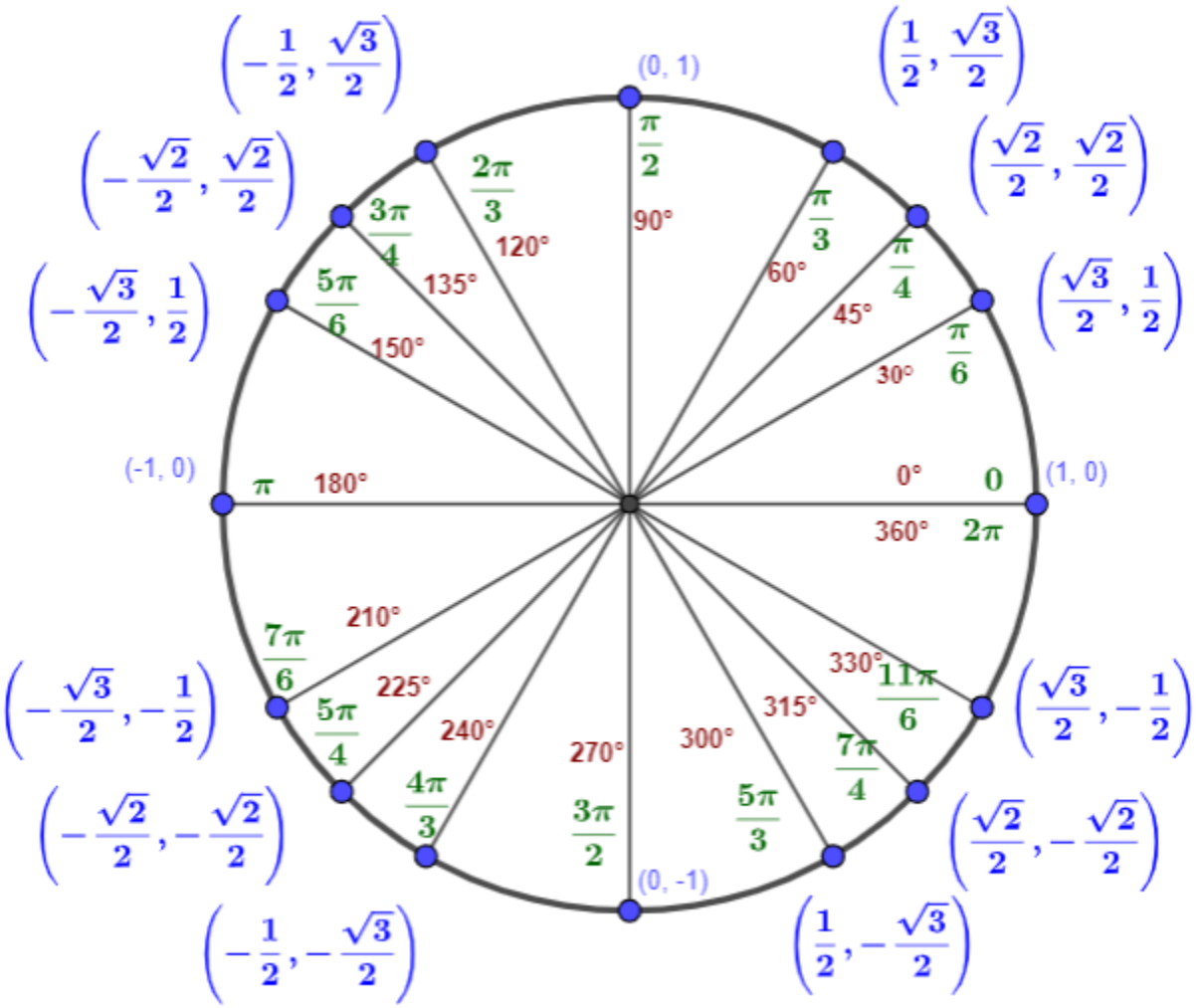
2) Welch Lab: **Imaginary Numbers Are Real [Part 1: Introduction]**
https://youtu.be/T647CGsuOVU?si=dMXaQ1rL7sOoPrcO

3) Omega Open Course:
https://www.youtube.com/watch?v=i0YkkVrew54&t=322s





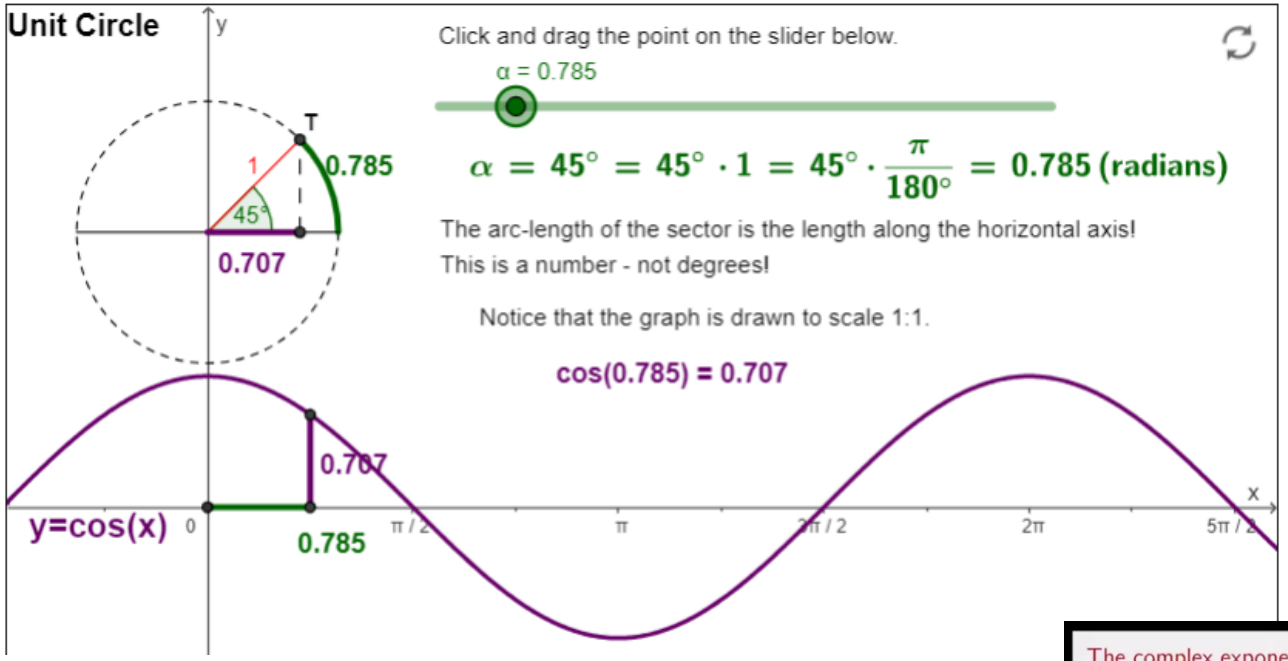
Ref: radian vs degree, and values on the unit circle (cartesian plane)



Manocha Academy: https://www.youtube.com/watch?v=LFY5JuQ49Kg

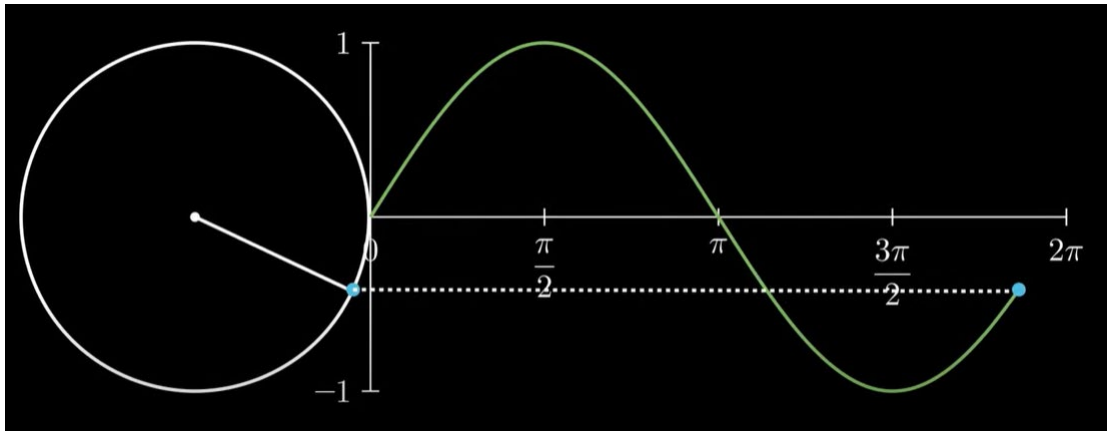
Organic chemistry: https://www.youtube.com/watch?app=desktop&v=V5A

Viewing the rotating phasor e^{it} and sine/cosine

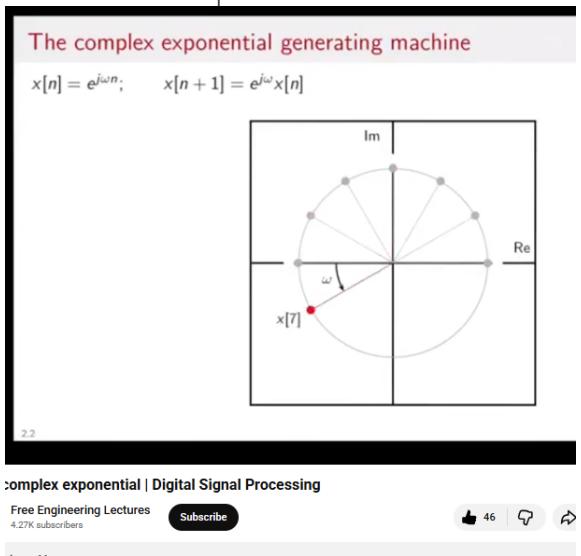


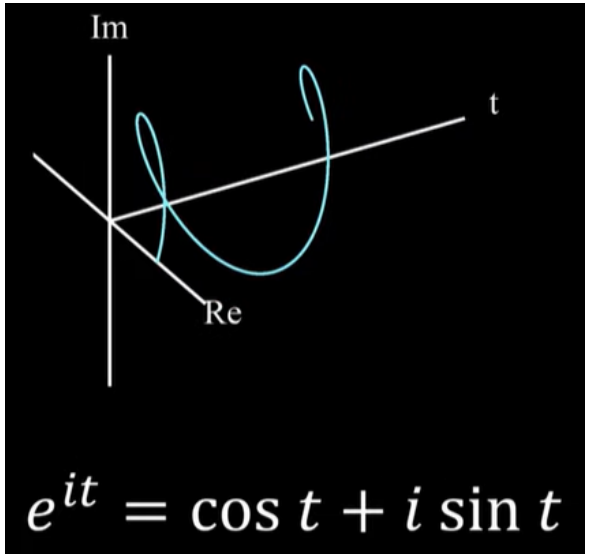
https://www.geogebra.org/m/wARRxDYf

Vitterli:
https://www.youtube.com/watch?v=ZzP63nHToa4



https://www.youtube.com/watch?v=9L57ygCwM4w
https://www.youtube.com/shorts/QfxBGWb96r8

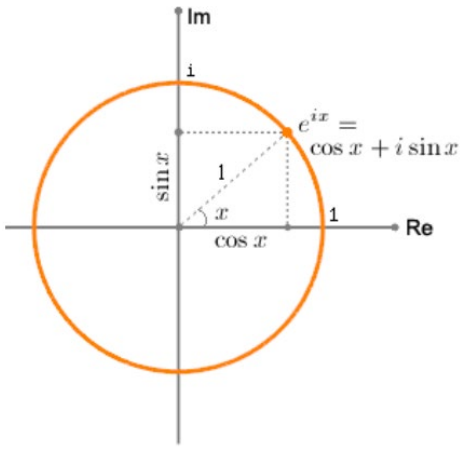




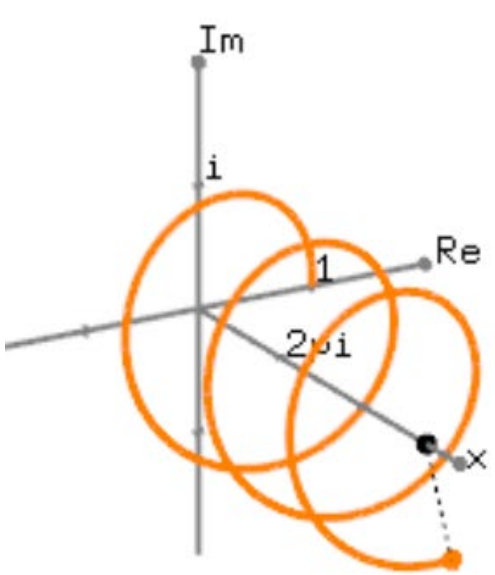
Euler Equation

$$e^{ix} = \cos(x) + i\,sin(x)$$

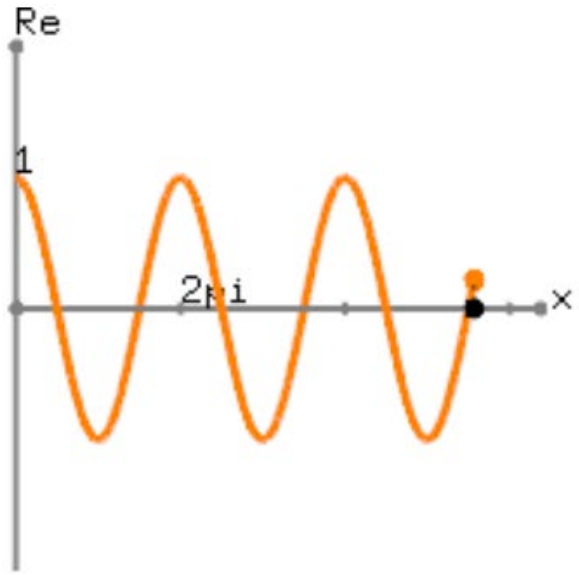
$$e^{-ix} = \cos(x) - i\,sin(x)$$



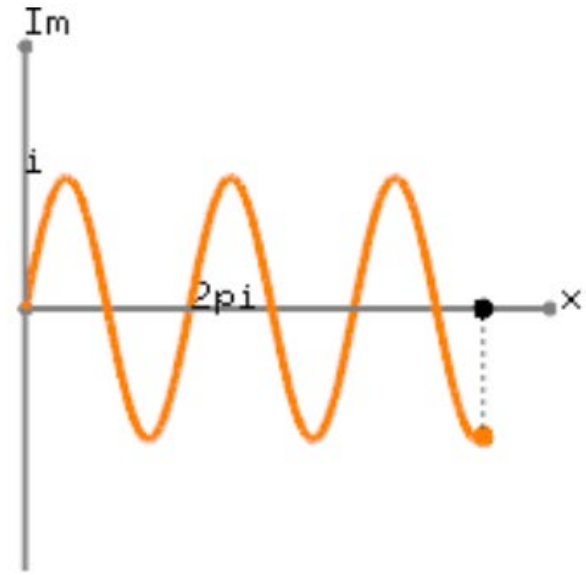
Graph of e^{ix} on the complex plane



Graph of e^{ix}



Real number part of e^{ix}



Imaginary number part of e^{ix}

Graph of e^{ix}, *as x increases, the phasor is rotating anti-clockwise.*

https://www.songho.ca/math/euler/euler.html

Time vs Frequency Representations: what is $e^{j(\frac{2\pi}{N}k)n}$,

k^{th} harmonic: $b_{k,n} = e^{j\frac{2\pi}{N}kn}$

The complex exponential at digital angular frequency $\Omega = \frac{2\pi}{N}k\ (\frac{rad}{sample})$

$e^{j(\frac{2\pi}{N}k)n}, \qquad n = 0..(N-1)$

To appreciate, represent $c_k = |c_k|e^{j\theta}$ (complex exponential form)

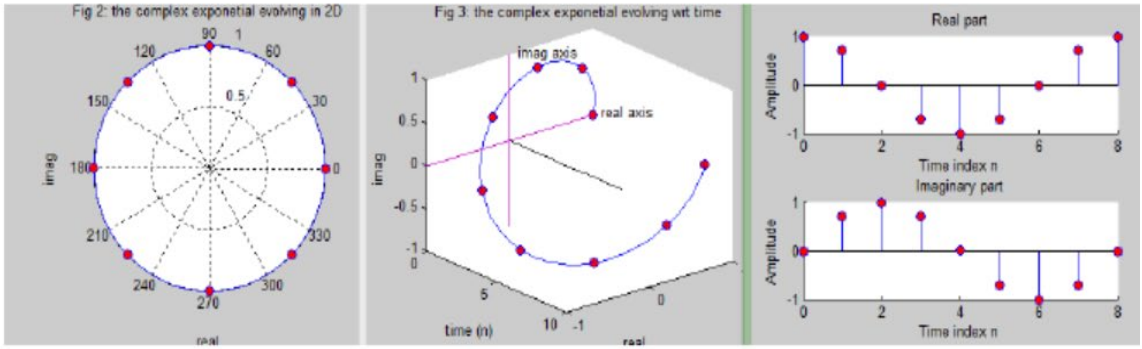
See that to reconstruct $x[n]$, the complex exponential at $\Omega = \frac{2\pi}{N}k$ is modified by c_k in its magnitude and phase

$c_k e^{j(\frac{2\pi}{N}k)n} = |c_k|e^{j\theta}e^{j(\frac{2\pi}{N}k)n}$

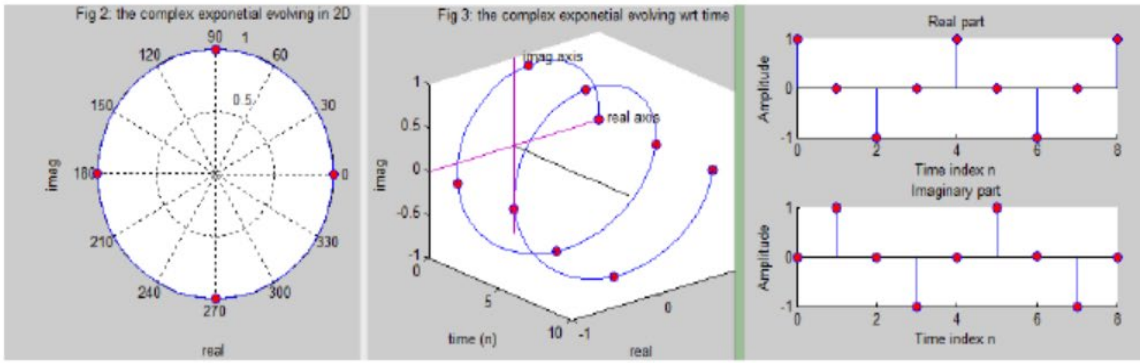
$= |c_k|e^{j(\left(\frac{2\pi}{N}k\right)n+\theta)}$

$N = 8,$
$n = 0..(N-1)$

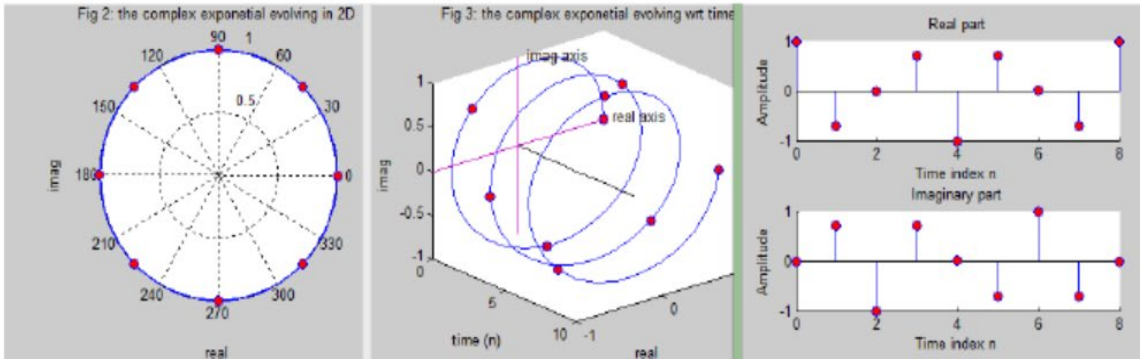
$b_{1,n} = e^{j\frac{2\pi}{N}(1)n}$



$b_{2,n} = e^{j\frac{2\pi}{N}(2)n}$

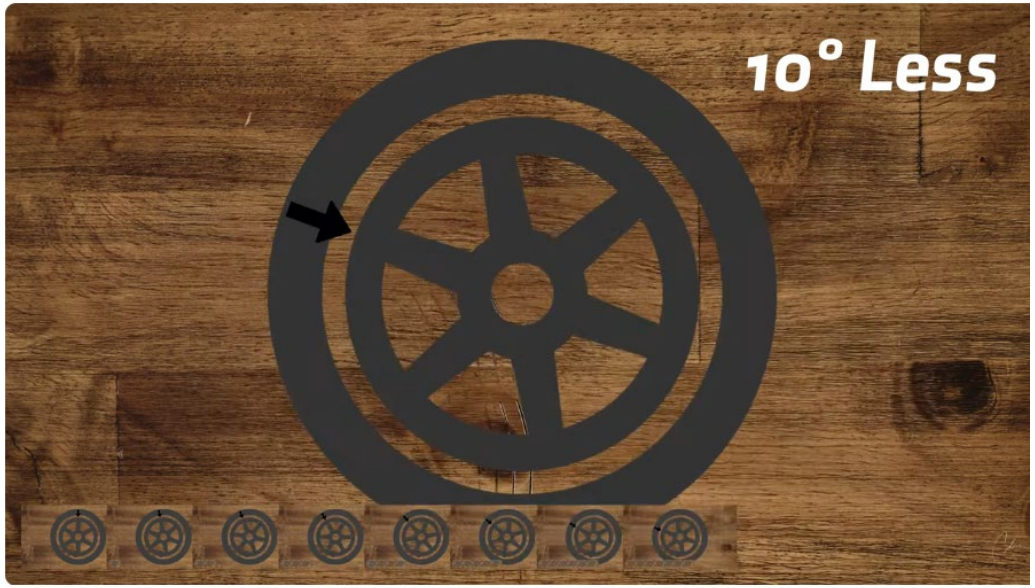


$b_{3,n} = e^{j\frac{2\pi}{N}(3)n}$



Aliasing – wagon wheel

youtube.com/watch?v=BFeUb1mnKyQ&feature=youtu.be



Prof Fowler: Fourier Basis

- Matrix and Vectors – good to understand basis for DFT, DTFS http://www.ws.binghamton.edu/fowler/fowler%20personal%20page/EE523_files/Ch_11_1%20Matrices%20and%20Vectors.pdf

DFT Coefficients as Inner Product Results:

Now lets see how these ideas relate to the DFT. We've already seen that we can interpret the N-pt. IDFT as an expansion of the signal vector in terms of the linear combination of N ON vectors $\mathbf{d}_k$ defined above. (Note that for the IDFT case this is just saying that we are building the signal out of a sum of complex sinusoids.) From the above theory, we now know how to compute the required coefficients for any ON expansion so we should see what this idea gives for the IDFT case. Our main general result above was that the expansion coefficients are found from the inner products of the

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vector to be expanded and the various ON basis vectors: so for the IDFT case the coefficients are $\langle \mathbf{x}, \mathbf{d}_k \rangle$. But, we know from standard DFT theory that the coefficients of the IDFT expansion are just the DFT values $X[k]$. Combining these two points of view gives

$$\boxed{X[k] = \langle \mathbf{x}, \mathbf{d}_k \rangle}.$$

Let's take a look at this an verify that the right-hand side of this is consistent with what we know from standard DFT theory. From vector inner product theory we know that the right side of this is

$$\langle \mathbf{x}, \mathbf{d}_k \rangle = \sum_{n=0}^{N-1} x[n] d_k^*[n]$$

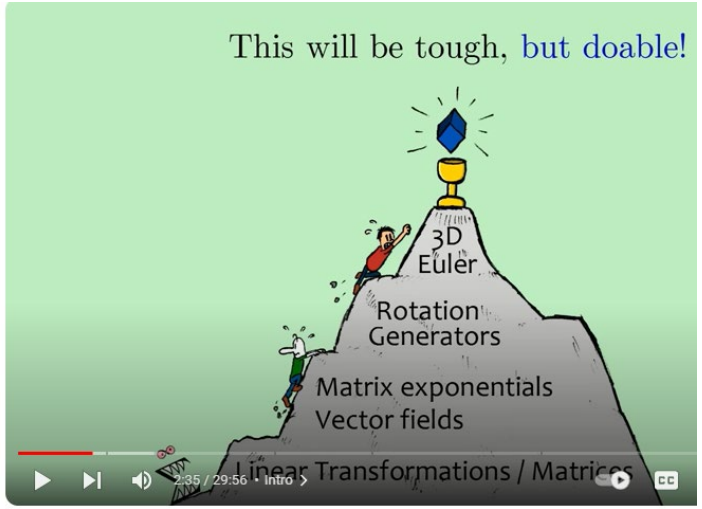
$$= \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

which is the DFT of the signal, so we see that things work as stated. Actually, we have been a little bit loose here – the $\mathbf{d}_k$ vectors as given above are orthogonal but not orthonormal so there is a 1/N term in the IDFT equation used back in the "DFT from Basis Viewpoint" section. To really do this using ON basis vectors we would have to put a $1/\sqrt{N}$ term in front of the $\mathbf{d}_k$ vectors, but that would lead to forms that aren't exactly the same as the conventional DFT/IDFT expressions; try as an exercise redoing this development with true ON basis vectors.

Ref Advance: Interesting videos

1) Euler's Formula Beyond Complex Numbers

https://www.youtube.com/watch?v=Y1gOYtQYRXo



Euler's Formula Beyond Complex Numbers

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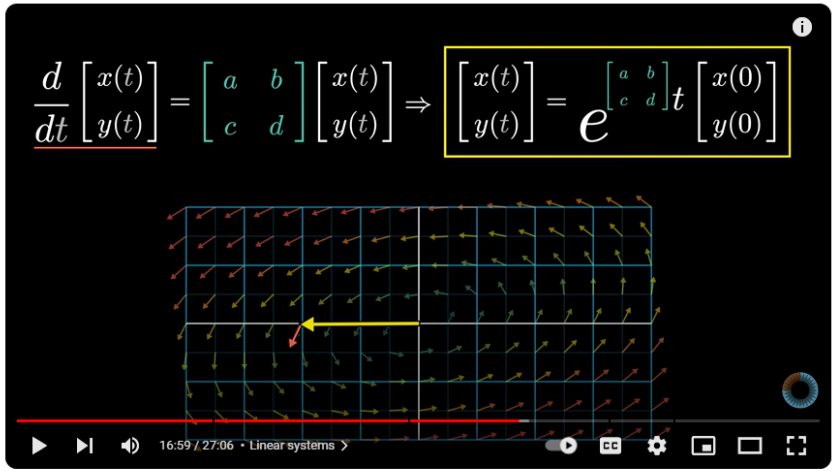
for matrix A

$$e^A := A^0 + \frac{A^1}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

2) How (and why) to raise e to the power of a matrix | DE6



How (and why) to raise e to the power of a matrix | DE6

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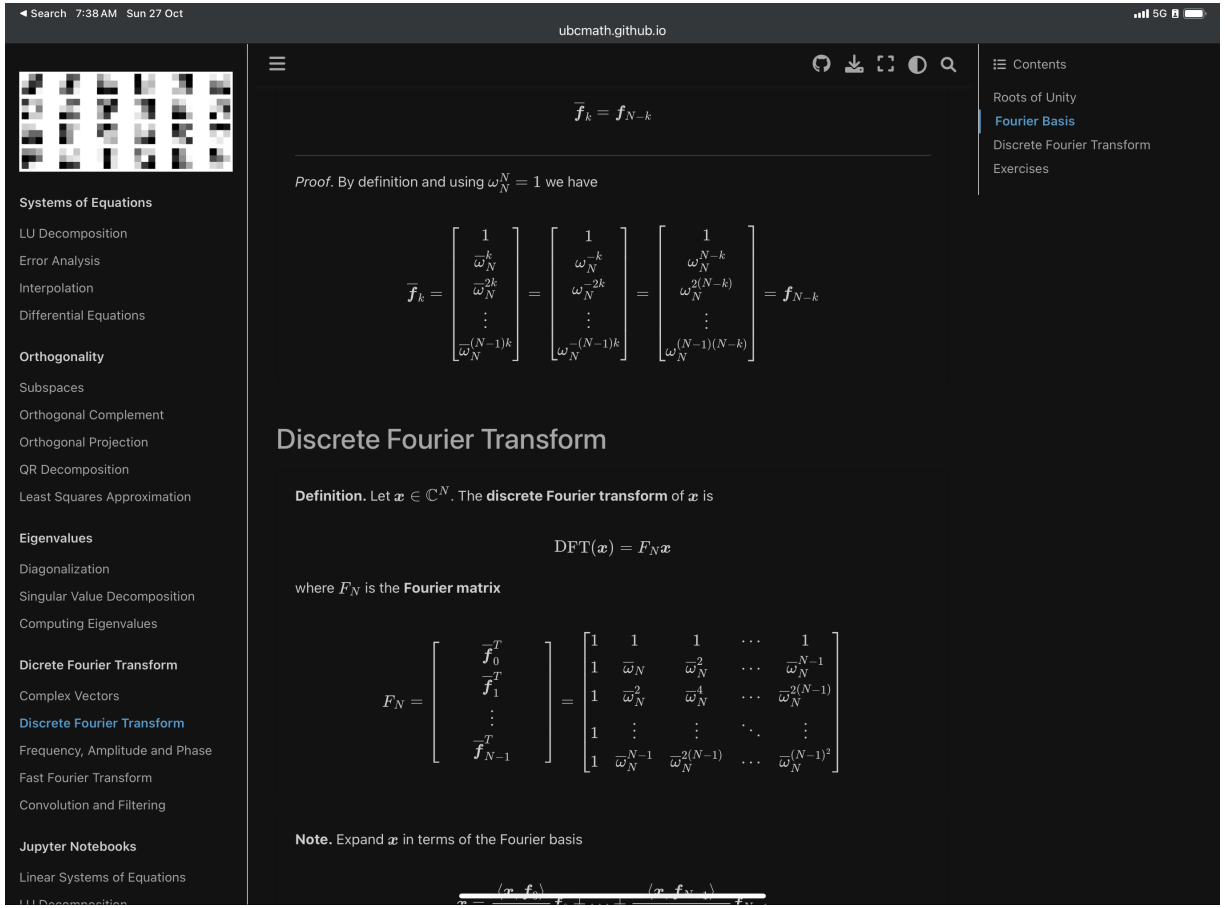
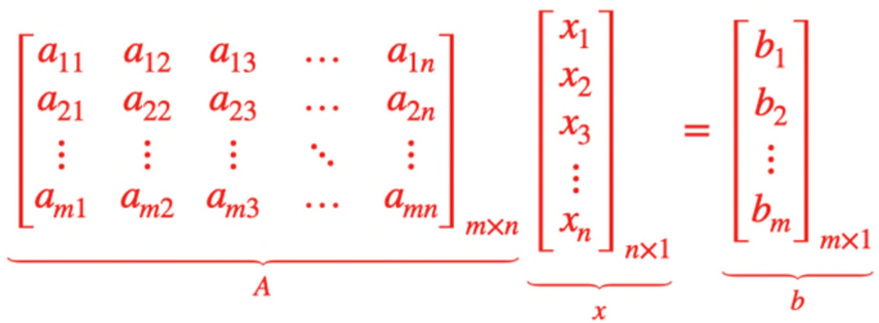
Some good references: DFT

1) https://ubcmath.github.io/MATH307/dft/dft.html
2) https://math.stackexchange.com/questions/2413218/proof-of-orthonormality-of-basis-of-dft
3) https://en.wikipedia.org/wiki/DFT_matrix
4) https://www2.seas.gwu.edu/~simhaweb/quantum/modules/appendix/dft.html#:~:text=The%20DFT%20is%20a%20linear,could%20be%20considered%20an%20algorithm.
5) https://www.youtube.com/watch?v=W5iXRA4ro1g
6) https://www.statlect.com/matrix-algebra/discrete-Fourier-transform#:~:text=they%20are%20orthogonal%3B,function%20taken%20at%20different%20frequencies.





CX1104: **Linear Algebra for Computing**

Chap. No : **8A.1**

Lecture : **Complex Numbers**

Topic : **Overview of Complex Numbers**

Concept : **Definitions, Motivations and Arithmetic**

Instructor: **A/P Chng Eng Siong**

TA: Shreyas Gopal

Rev: 23rd July 2024

Content: Complex Numbers

8A.1: Introduction to complex numbers+ Arithmetic Operations (1.5 hour)

8A.1.1 - Definitions and Motivations
- number system
- solving quadratic equation for complex roots

8A.1.2 - The argand diagram, and geometric interpretation
- The polar form, The unit circle

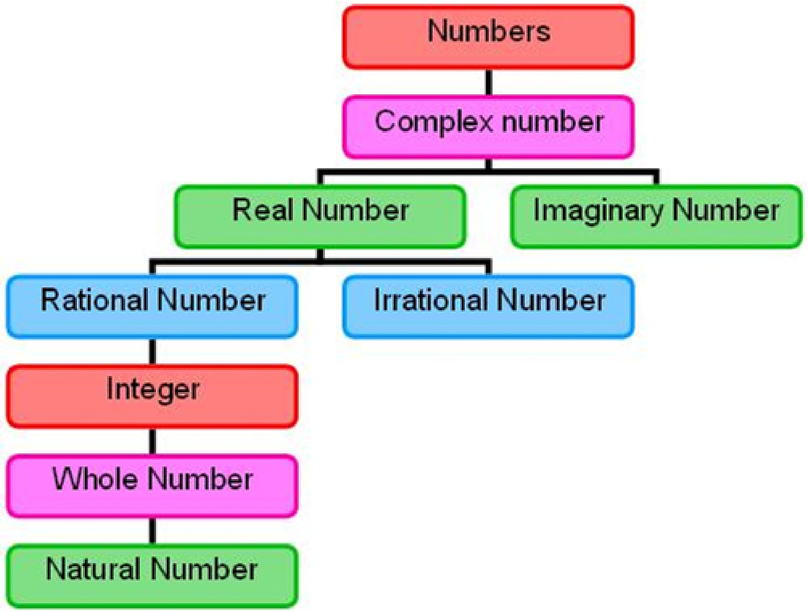
8A.1.3 - Arithmetic of Complex numbers
- Add, subtract, multiply, and divide,
- Properties
- Multiplication and Division in polar form

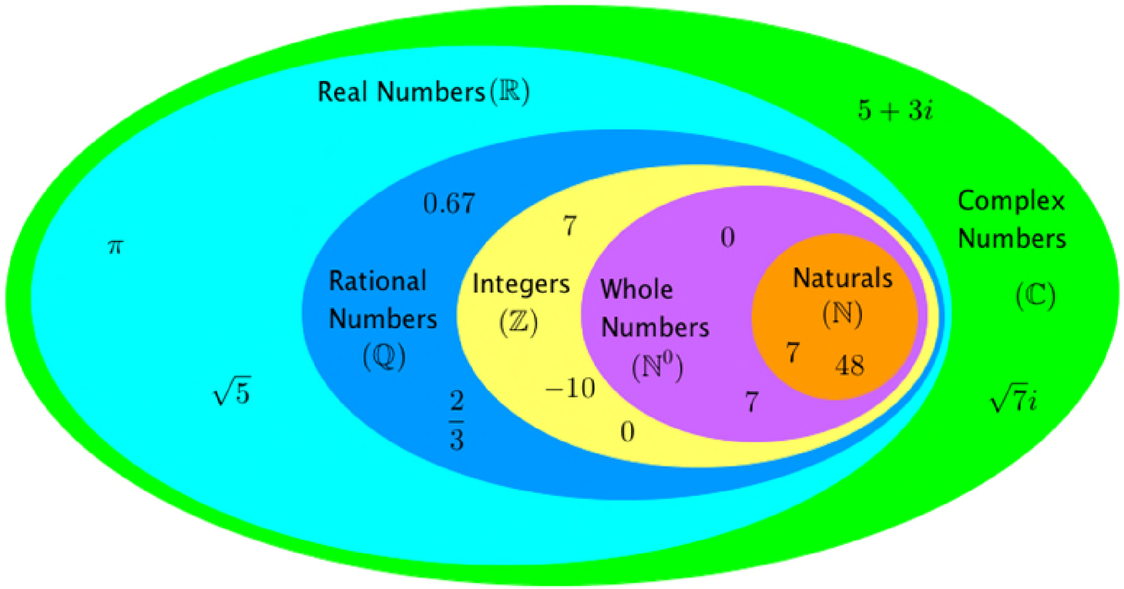
8A.1.4 - Complex conjugation

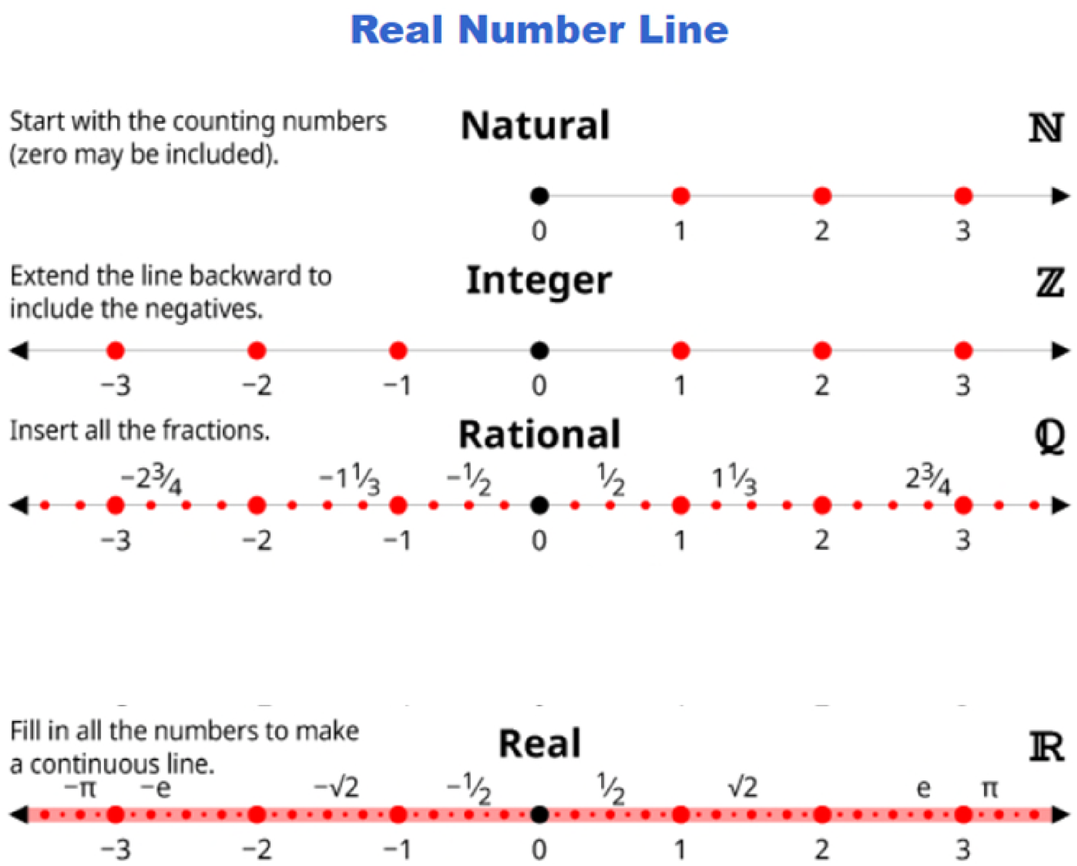
8A.1.1 Complex numbers

- Definitions and Motivations
- number system
- solving quadratic equation for complex roots

8A.1.1 - Definitions and Motivations: Number sets







Ref: https://www.ck12.org/book/ck-12-algebra-ii-with-trigonometry-concepts/r8/section/5.8/
https://www.mathsisfun.com/sets/number-types.html

8A.1.1 - Definitions and Motivations

Define a complex number z to be of the form

Define i or $j = \sqrt{-1}$

$$z = a + jb,$$

real part

imaginary part

where a is called the **real part** of z $(\mathrm{Re}(z))$and b its **imaginary part** $(\mathrm{Im}(z))$.

Example: $3 + j9$ or $3 + 9j$

If $a = 0$, the complex number is said to be **purely imaginary** and if $b = 0$, it is said to be **purely real**.

In this form, the complex number $z = a + jb$
is called the Rectangular or Cartesian form.

Note: there are 3 forms, the other 2 forms
are called the polar form, and complex exponential form.

8A.1.1 - Definition and Motivation
Example: Why need for complex numbers

There is no real number that solves $x^2 + 4 = 0$.

$$x^2 = -4 \Rightarrow x = \sqrt{(-4)} \Rightarrow x = \sqrt{(-1)(4)} \Rightarrow x = \sqrt{-1}\sqrt{4}$$

Define an imaginary unit j

$$\boxed{j \stackrel{\text{def}}{=} \sqrt{-1}} \quad \Leftrightarrow \quad j^2 = \sqrt{-1}\sqrt{-1} = -1.$$

More generally, define an imaginary number z to be of the form

$$z = jy, \qquad y \in \mathbb{R}, \text{the set of real numbers}$$

so that

$$z^2 = (jy)^2 = j^2y^2 = (-1)y^2 = -y^2.$$

In the above example, $x = \sqrt{-1}\sqrt{4} = j(\pm 2) = \pm 2j$ (an imaginary number).

8A.1.1 - Definitions and Motivations: Why need for complex numbers

- Solving quadratic equations: $ax^2+bx+c=0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

D = b^2 - 4ac

D>0, Two real solutions
D=0, One real solution
D<0, Complex Solutions

Introduce $i \ or \ j \ = \sqrt{-1}$
To have complex solutions!

- Solving cubic equations
  - square roots of negative numbers appear in intermediate steps even when the roots are real
- Fundamental Theorem of Algebra- every polynomial of degree *n* has *n* roots

Ref:https://cyberspacesystem.blogspot.com/2012/03/complex-numbers-quadratic-formula.html

8A.1.1 - Definitions and Motivations: Fundamental Theorem of Algebra

Fundamental Theorem of Algebra

The polynomial equation of degree n with $a_i \in \mathbb{C}$

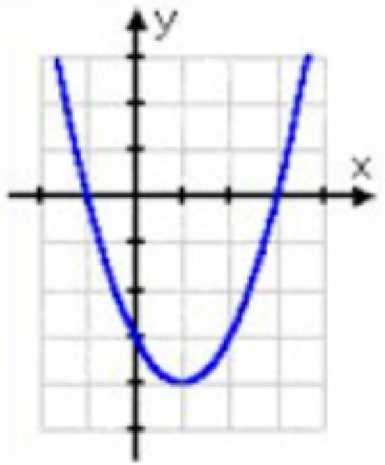
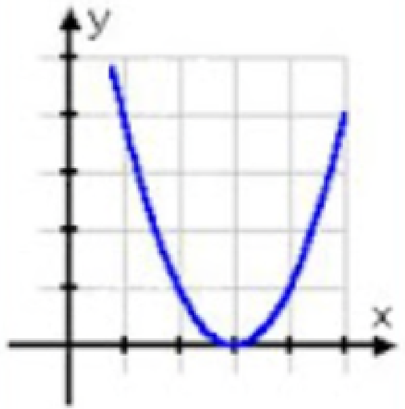
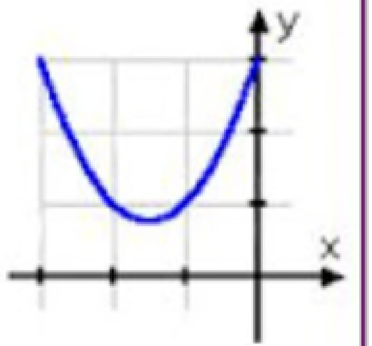
$$P(z) = z^n + a_{n-1}z^{n-1} + \ldots + a_1 z + a_0 = 0$$

has at least 1 solution in $\mathbb{C}$ and the polynomial can be factorised as

$$P(z) = (z - \alpha_1)(z - \alpha_2)\ldots(z - \alpha_n)$$

where $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ are the roots of the polynomial

Ref:
https://cyberspacesystem.blogspot.com/2012/03/complex-numbers-quadratic-formula.html

$x^2 - 2x - 3$	$x^2 - 6x + 9$	$x^2 + 3x + 3$
$x = \frac{2 \pm \sqrt{(-2)^2 - 4(-3)}}{2}$ $= \frac{2 \pm \sqrt{4+12}}{2}$ $= \frac{2 \pm \sqrt{16}}{2} = \frac{2 \pm 4}{2}$ $= \frac{-2}{2}, \frac{6}{2} = -1, 3$	$x = \frac{6 \pm \sqrt{(-6)^2 - 4(9)}}{2}$ $= \frac{6 \pm \sqrt{36-36}}{2}$ $= \frac{6 \pm \sqrt{0}}{2} = \frac{6 \pm 0}{2} = 3$	$x = \frac{-3 \pm \sqrt{(3)^2 - 4(3)}}{2}$ $= \frac{-3 \pm \sqrt{9-12}}{2}$ $= \frac{-3 \pm \sqrt{-3}}{2}$ $= -\frac{3}{2} \pm \frac{\sqrt{3}i}{2}$
a positive number inside the square root	zero inside the square root	a negative number inside the square root
two real solutions	one (repeated) real solution	two complex solutions
		
two distinct x-intercepts	one (repeated) x-intercept	no x-intercepts

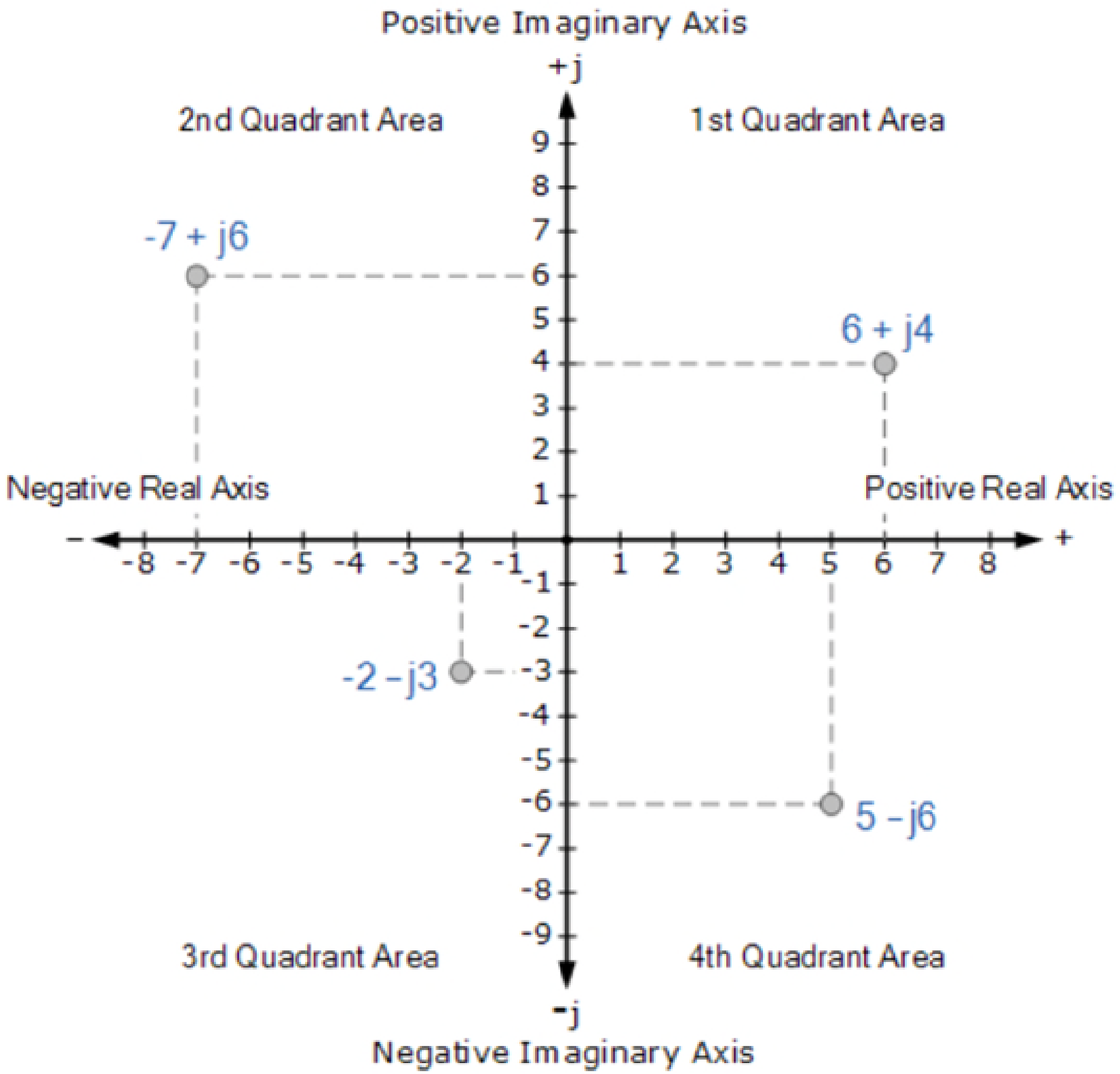
8A.1.2 - The argand diagram, and geometric interpretation
- The polar form, The unit circle

8A.1.2 - Geometric interpretation of the complex number
The Argand diagram (complex plane)

An Argand diagram, also known as the complex plane or Argand plane, is a graphical representation used to visualize complex numbers.

It is named after the Swiss mathematician Jean-Robert Argand.

Four Quadrant Argand Diagram



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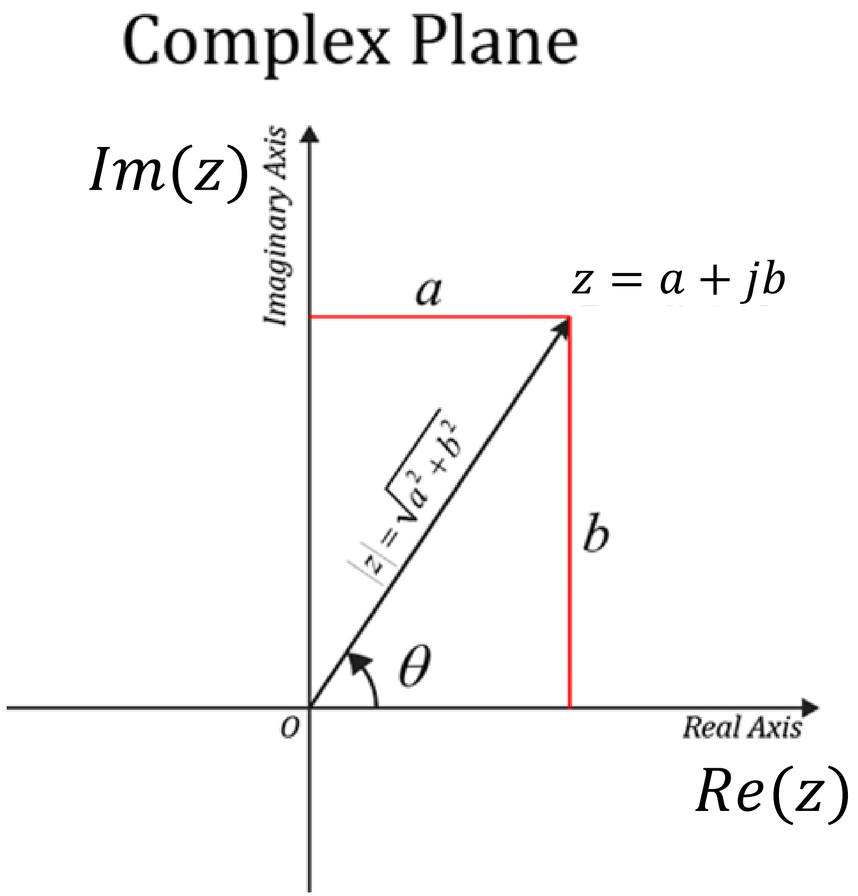
8A.1.2 - Geometric interpretation of the complex number The Argand diagram (complex plane)



The distance of the complex number $z = a + jb$ from the origin $0 + j0$ is termed the **modulus** or **magnitude** of the complex number, denoted as $|z|$, and is computed as $\sqrt{a^2 + b^2}$.

The angle θ, formed by the line connecting the complex number to the origin with the positive x-axis, is known as the argument or amplitude of the complex number. It is symbolized by $\arg(z)$

$$\arg z = \theta, \text{ where } \tan\theta = b/a$$

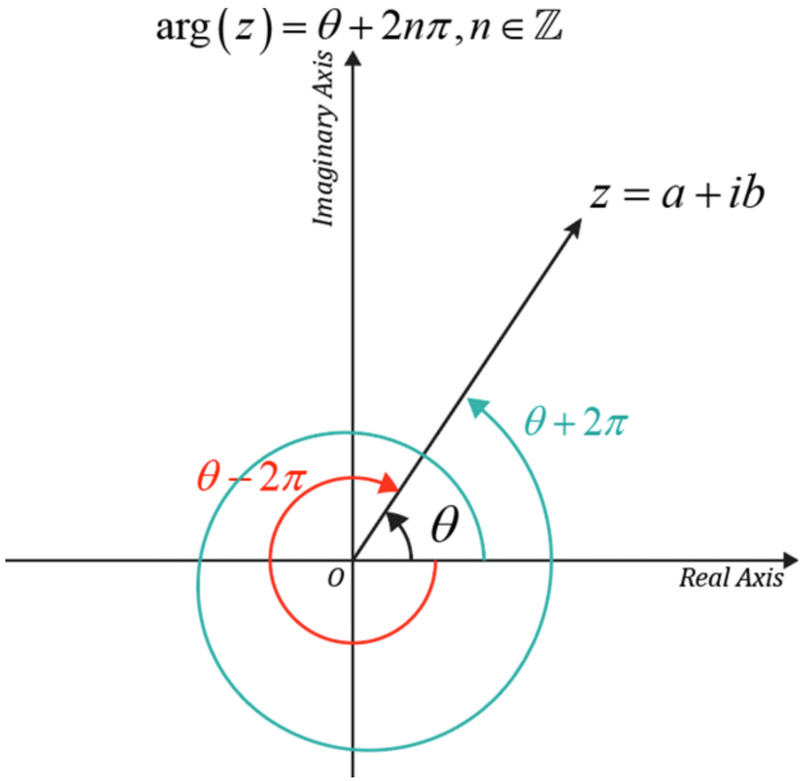
θ is measured from the positive real axis.
$-\pi < \theta \leq \pi$
Important: The value of θ depends on the quadrant of the complex plane that z is in.

Note: angle θ 's unit is in radian and measured against the positive real axis, where rotation of
anti-clockwise for +ve angle
and clockwise for –ve angle

8A.1.2 - The non-uniqueness of θ and Principal Argument

Non-Uniqueness of Argument

Argument of a complex number is not unique. If θ is the argument of the complex number so can be $\theta + 2\pi$ or $\theta - 2\pi$ or $\theta + 4\pi$ and so on. In general the argument of a complex number is $\theta + 2n\pi$ where θ is any one of the argument.

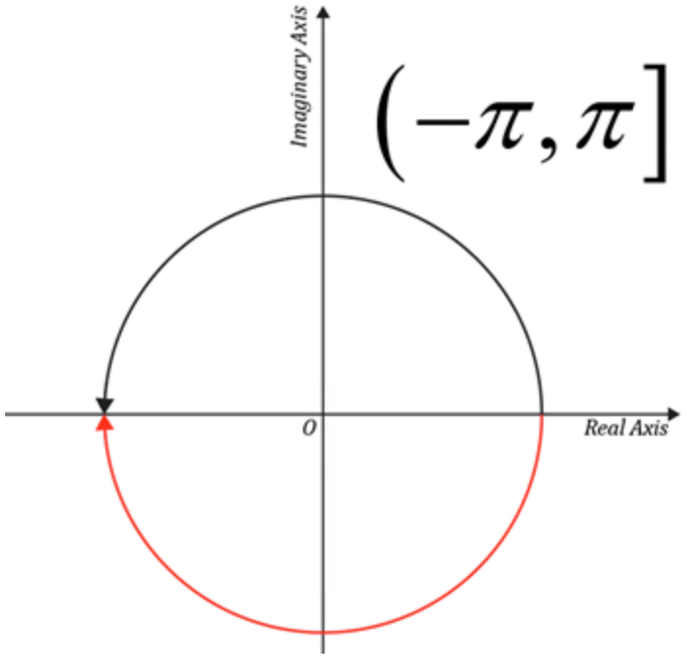


Principal Argument

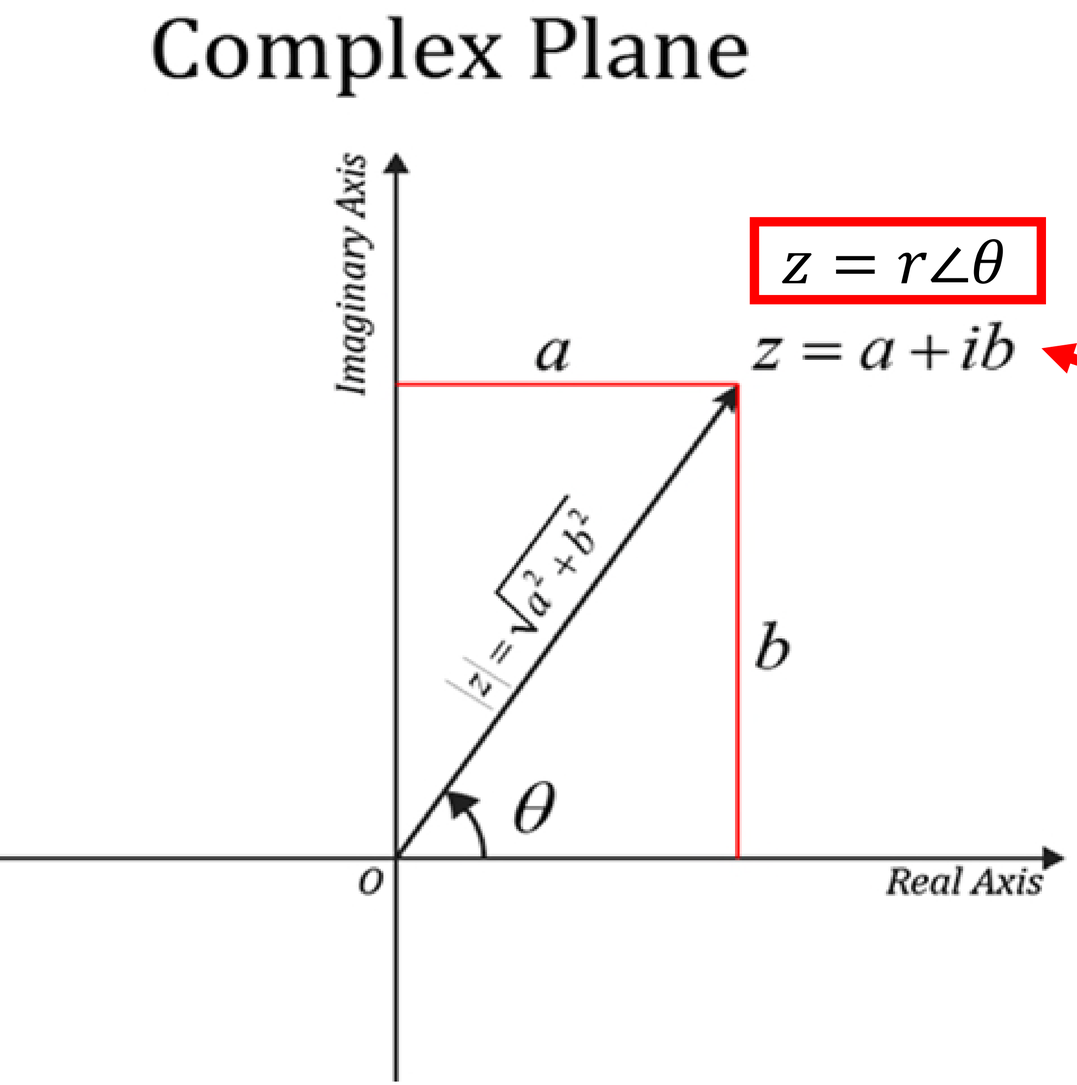
Arg(z) vs arg(z)

To make the argument unique we define principal argument.

Principal Argument of a complex number is that argument of the complex number which lies in the interval $(-\pi, \pi]$. Exactly one such argument out of infinite arguments lie in this interval. $(-\pi, \pi]$ denotes one complete revolution around the circle with centre at origin. An argument lying in $(-\pi, \pi]$ also has the smallest magnitude.



8A.1.2 - Definition and motivation
The polar form of complex number



In this form, the complex number $z = a + jb$ can be represented in polar forms:

i) $z = |z|\angle\theta = r\angle\theta$ is in (abbreviated) polar form

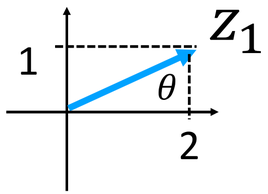
ii) $z = r(\cos(\theta) + jsin(\theta) = r\ cis\ \theta$
is in trigonometric polar form

$r = |z|$ (modulus of z)

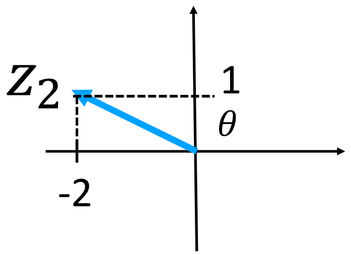
$\theta = arctan(\frac{b}{a})$, $a = |z|\cos(\theta)$, $b = |z|\sin(\theta)$,

Ref: https://math.stackexchange.com/questions/4293260/argument-of-a-complex-number-as-arctan

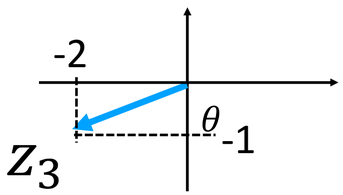
8A.1.2: Example of z in polar form for 4 quadrants



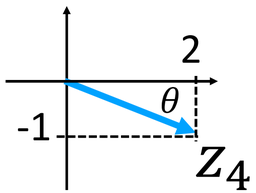
$$z_1 = 2 + 1j$$
$$= \sqrt{5}\angle(0.4636)(rad)$$



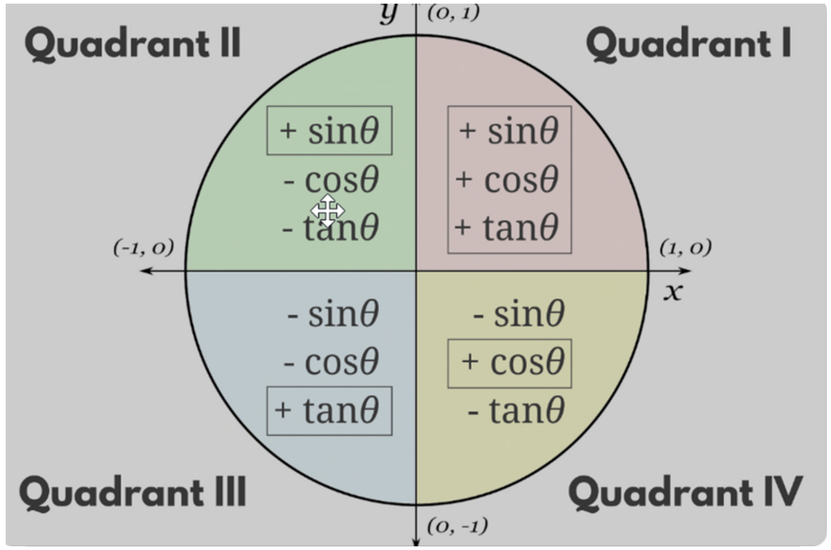
$$z_2 = -2 + 1j$$
$$= \sqrt{5}\angle(\pi - 0.4636)\ (rad)$$
$$= \sqrt{5}\angle(2.678)\ (rad)$$



$$z_3 = -2 - 1j$$
$$= \sqrt{5}\angle(-\pi + 0.4636)\ (rad)$$
$$= \sqrt{5}\angle(-2.678)\ (rad)$$

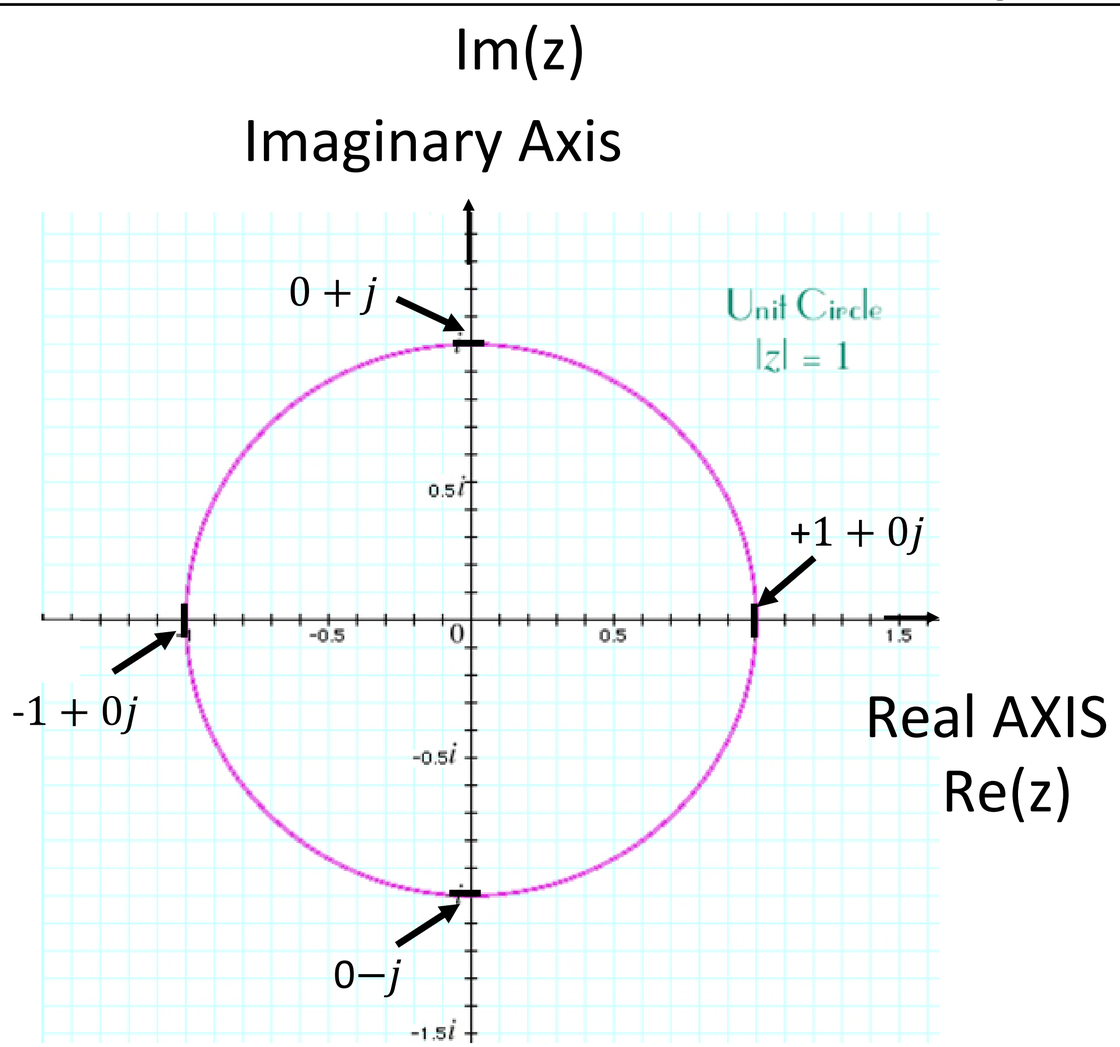


$$z_4 = 2 - 1j$$
$$= \sqrt{5}\angle(-0.4636)(rad)$$



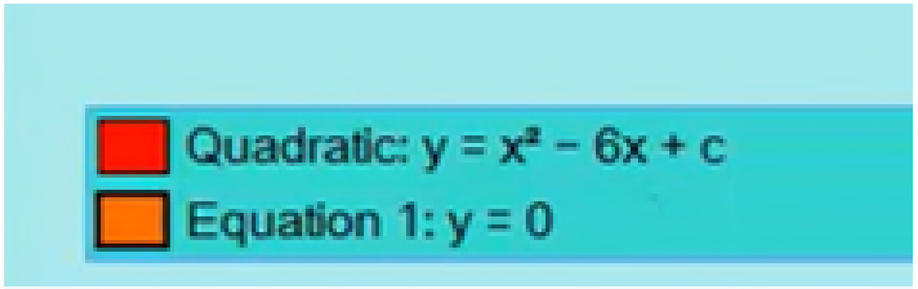
Ref: Reference angles and 4 quadrants
https://www.youtube.com/watch?v=ONPcD6FdoBI

8A.1.2 - definition and motivation
The unit circle in the complex plane.



The *unit circle* (in pink) is the circle of radius 1 centered at 0. It include all complex numbers of absolute value 1, so it has the equation $|z| = 1$

8A.1.2 Example: Visualizing the complex roots



visualising complex roots

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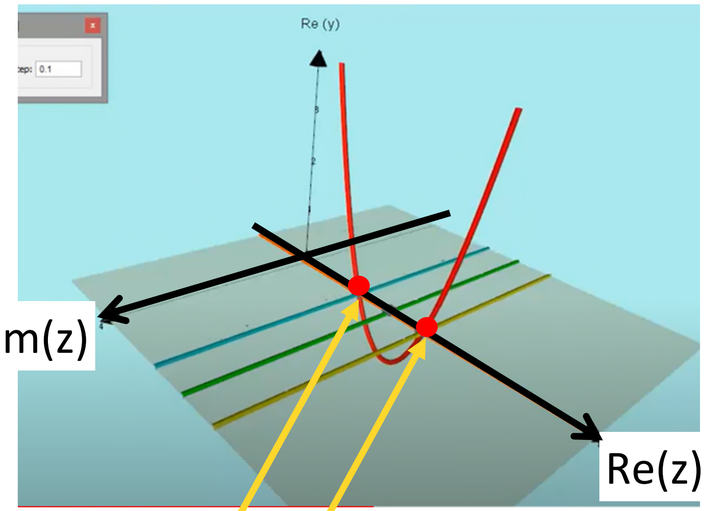
https://youtu.be/5g80j17w0CE?si=HMWc-l8HekEjptxg

c=8



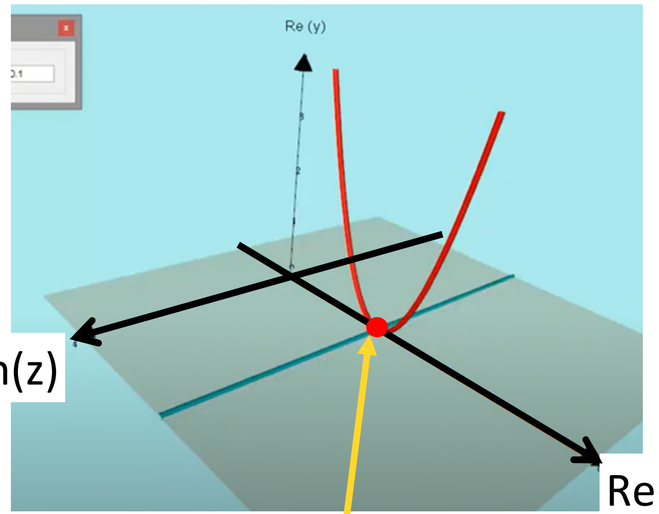
Roots at real values = 2, 4

```
>> roots([1 -6 8])

ans =
```

2, 4

c=9



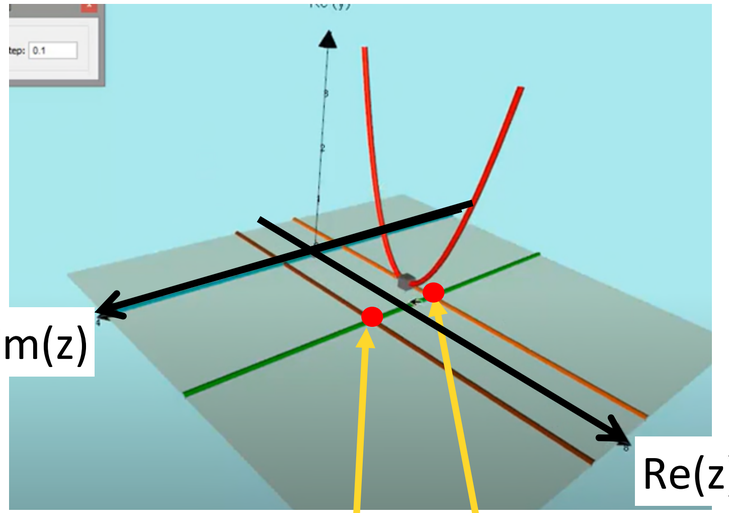
Roots at real value = 3

```
>> roots([1 -6 9])

ans =

   3.0000 + 0.0000i
   3.0000 - 0.0000i
```

c=9.5



Roots are complex numbers

```
>> roots([1 -6 9.5])

ans =
```

3+j0.707, 3-j0.707

8A.1.3 - Arithmetic of Complex numbers

- Add, subtract, multiply, and divide,
- Properties
- Multiplication and Division in polar form

8A.1.3 - Arithmetic of Complex numbers (in rectangular form)

Equality: $z_1 = a + ib$, and $z_2 = c + id$ are equal if and only if $a = c$ and $b = d$.

Arithmetic Operations Complex numbers can be added, subtracted, multiplied, and divided. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, these operations are defined as follows.

Addition: $$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

Subtraction: $$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$$

Multiplication: $$z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2)$$

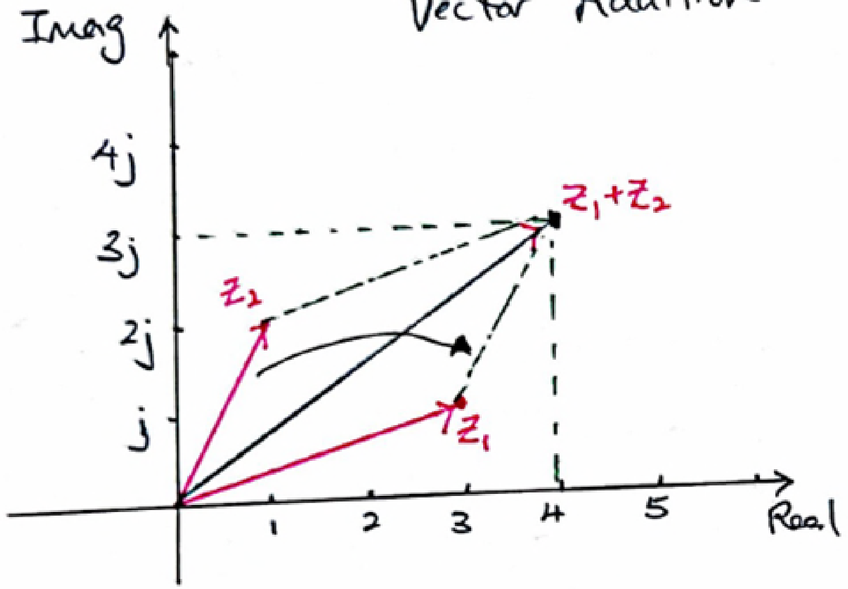
$$= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)$$

Division: $$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2}$$

$$= \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}$$

8A.1.3 - Arithmetic of Complex numbers (in Argand plane)

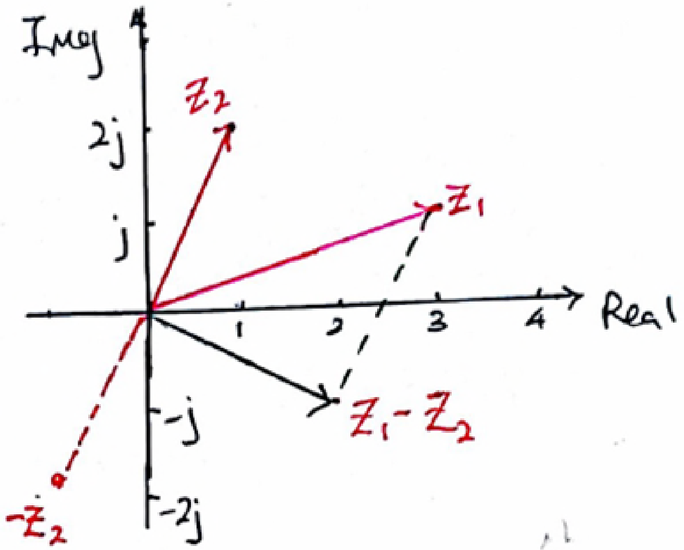
Vector Addition



$z_1 = 3+1j$
$z_2 = 1+2j$
$\therefore z_1+z_2 = 4+3j$

Note: adding 2 vectors visually is taking z_1's tail and put it at z_1's tip.

Vector Subtraction



$z_1 = 3+1j$
$z_2 = 1+2j$
$\therefore -z_2 = -1-2j$
$\therefore z_1-z_2 = 2-1j$

Note: vector subtraction visually, flip z_2 to $-z_2$, then take $-z_2$ vector and put it at z_1's tip!

8A.1.3 - Other properties of Complex numbers

The familiar commutative, associative, and distributive laws hold for complex numbers.

Commutative laws:

$$\begin{cases} z_1 + z_2 = z_2 + z_1 \\ z_1 z_2 = z_2 z_1 \end{cases}$$

Associative laws:

$$\begin{cases} z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3 \\ z_1(z_2 z_3) = (z_1 z_2) z_3 \end{cases}$$

Distributive law:

$$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$$

8A.1.3 - Examples when dealing with $\mathbf{j = \sqrt{-1}}$

- As in real number algebra, $-j \overset{\text{def}}{=} (-1)j$
- $j^2 = -1$; $j^3 = j.j^2 = -j$; $j^4 = j^2.j^2 = 1$; $j^5 = j^4.j = j$;
  $j^6 = j^4.j^2 = -1$; $j^7 = j^4.j^3 = -j$; and so on.
- $\frac{1}{j} = j^{-1} = -j \qquad (j.j = -1)$;
  $\frac{1}{j^2} = j^{-2} = -1$; $\frac{1}{j^3} = j$; $\frac{1}{j^4} = 1$; $\frac{1}{j^5} = -j$; and so on.
- $x(-j) = (-x)j = -jx$, for all $x \in \mathbb{R}$
- $\left(\sqrt{-x}\right)^2 = \sqrt{-x}.\sqrt{-x} = \sqrt{(-x).(-x)} = -x.$

  CAUTION: $\underset{-x}{\sqrt{(-x).(-x)}} \neq \underset{x}{\sqrt{x.x}}$

  and $\sqrt{x^2} = \sqrt{(-x).(-x)}$ OR $\sqrt{(x).(x)} = \pm x$

  Important: Always write $\sqrt{-x}$ as $\sqrt{x}.j$ prior to manipulation

8A.1.3 - Example

Now we also saw that if a and b were both positive then $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$. For a second let's forget that restriction and do the following.

$$\sqrt{-9} = \sqrt{(9)\,(-1)} = \sqrt{9}\sqrt{-1} = 3\sqrt{-1}$$

by using the following definition.

$$\boxed{i = \sqrt{-1}}$$

Note that if we square both sides of this we get,

$$\boxed{i^2 = -1}$$

Now, $\sqrt{-1}$ is not a real number, but if you think about it we can do this for any square root of a negative number. For instance,

$$\sqrt{-100} = \sqrt{100}\sqrt{-1} = 10\sqrt{-1}$$
$$\sqrt{-5} = \sqrt{5}\,\sqrt{-1}$$
$$\sqrt{-290} = \sqrt{290}\,\sqrt{-1} \quad etc.$$

Using this definition all the square roots above become,

$$\sqrt{-9} = 3\,i \qquad \sqrt{-100} = 10\,i$$
$$\sqrt{-5} = \sqrt{5}\,i \qquad \sqrt{-290} = \sqrt{290}\,i$$

In Ch 8A.2_DeMoivre, you will see that there are 2 solutions for $\sqrt{-9}$.

8A.1.3 - Multiplication in polar form

Angle sum and difference

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$
$$\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$$
$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$$
$$\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$$

Let $z_1 = r_1(\cos\theta_1 + j\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + j\sin\theta_2)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2(\cos\theta_1 + j\sin\theta_1)(\cos\theta_2 + j\sin\theta_2) \\ &= r_1 r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + j(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)] \\ &= r_1 r_2[\cos(\theta_1+\theta_2] + j[\sin(\theta_1+\theta_2)] \end{aligned}$$

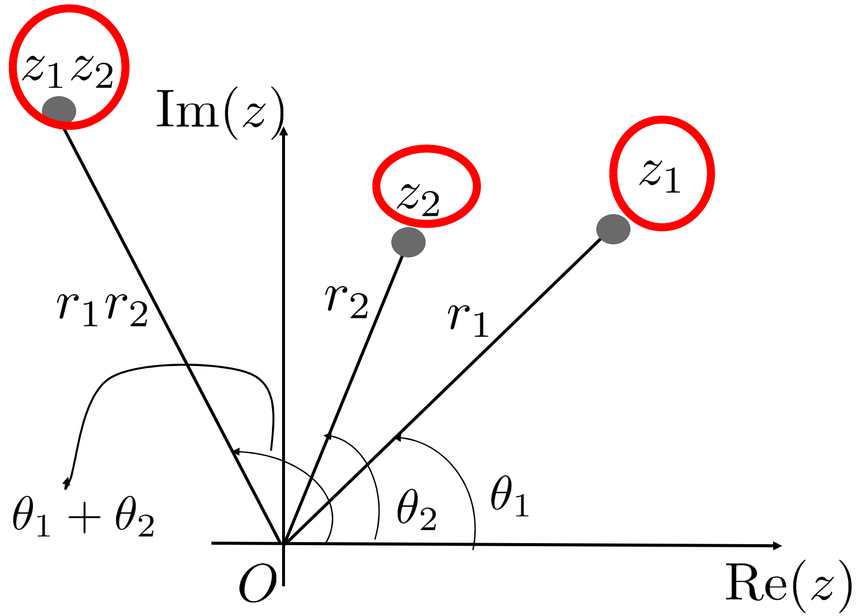
Hence,

$$|z_1 z_2| = r_1 r_2 = |z_1||z_2|$$

and

$$\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg z_1 + \arg z_2$$

$$-\pi < \arg(z_1 z_2) \le \pi$$



In summary: multiplication of complex number can be easily carried out in polar form by:

$$\begin{aligned} z_1 z_2 &= |z_1|\angle(\theta_1)\,|z_2|\angle(\theta_2) \\ &= |z_1||z_2|\ \angle(\theta_1+\theta_2) \end{aligned}$$

8A.1.3 - Division in polar form

Angle sum and difference

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

Let $z_1 = r_1(\cos\theta_1 + j\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + j\sin\theta_2)$. Then

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{r_1(\cos\theta_1 + j\sin\theta_1)}{r_2(\cos\theta_2 + j\sin\theta_2)} \\ &= \frac{r_1}{r_2}(\cos\theta_1 + j\sin\theta_1)(\cos\theta_2 - j\sin\theta_2) \\ &= \frac{r_1}{r_2}[\,(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + j(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)\,]\end{aligned}$$

Therefore,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1-\theta_2) + j(\sin(\theta_1-\theta_2)]$$

Hence,

$$\left|\frac{z_1}{z_2}\right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$$

and

$$\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg z_1 - \arg z_2,$$

$$-\pi < \arg(z_1/z_2) \le \pi$$

Summary: division of complex number can be easily carried out in polar form by:

$$\frac{z_1}{z_2} = \frac{|z_1|\angle(\theta_1)}{|z_2|\angle(\theta_2)}$$

$$= \frac{|z_1|}{|z_2|}\angle(\theta_1 - \theta_2)$$

8A.1.4 - Complex Conjugation

8A.1.4 - Complex conjugate

Define Complex Conjugate of $z = a + jb$ as $\bar{z} = a - jb$.
The complex conjugate of $z = a - jb$ is $\bar{z} = a + jb$. Thus $\bar{\bar{z}} = z$.

Some results from complex conjugate:

$$z + \bar{z} = 2a = 2\mathrm{Re}(z)$$
$$z - \bar{z} = 2jb = 2j\mathrm{Im}(z)$$

$$z\bar{z} = (a + jb)(a - jb) = a^2 + b^2$$
$$= |r|\angle\theta \; |r|\angle(-\theta)$$
$$= |r|^2\angle 0 = |r|^2$$

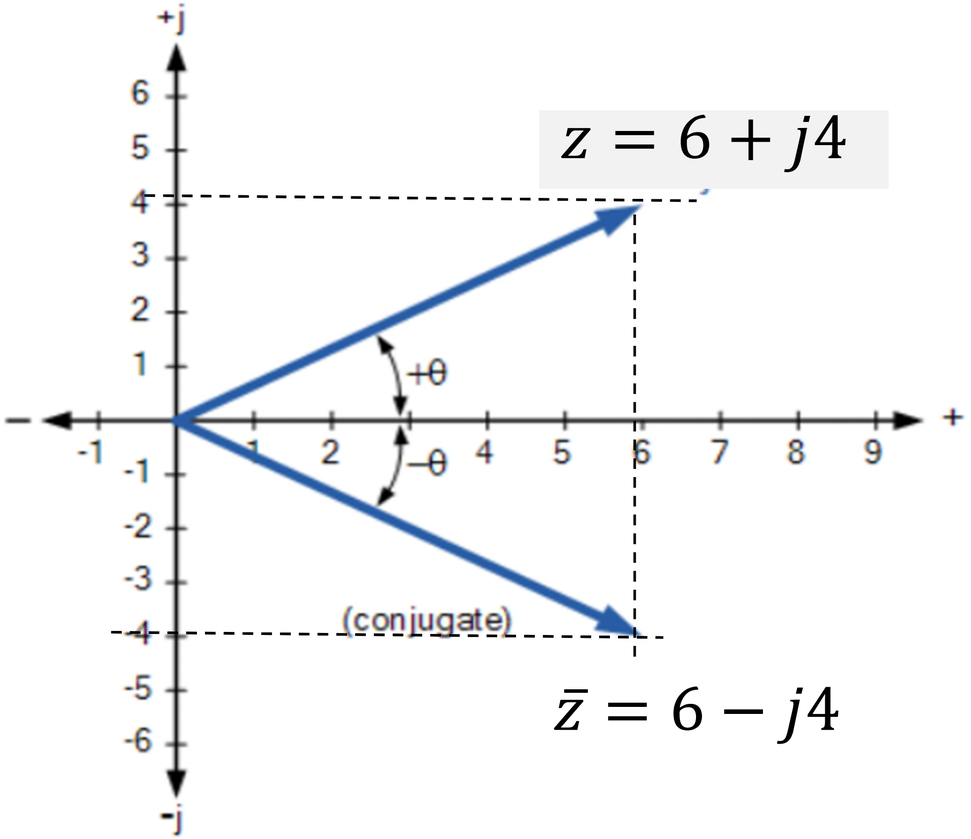
Complex conjugation is important as it allows us to convert a complex value to real value
when we multiply by its conjugate

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$$

$$= \overline{|r_1|\angle\theta_1 \; |r_2|\angle(\theta_2)}$$
$$= |r_1||r_2|\angle(\theta_1+\theta_2)$$
$$= |r_1||r_2|\angle -(\theta_1+\theta_2)$$

$$= \overline{|r_1|\angle\theta_1} \; \overline{|r_2|\angle\theta_2}$$
$$= |r_1|\angle(-\theta_1) \; |r_2|\angle(-\theta_2)$$
$$= |r_1||r_2|\angle(-\theta_1 - \theta_2)$$
$$= |r_1||r_2|\angle -(\theta_1 + \theta_2)$$

8A.1.4 - Geometric Interpretation : Example: Complex conjugate in the Complex plane (Argand Plane)



$z = 6 + j4$

$z = r\angle\theta = 7.211\angle 0.588$

$\bar{z} = 6 - j4$

$\bar{z} = r\angle(-\theta) = 7.211\angle(-0.588)$

Geometric Interpretation: Geometrically, taking the complex conjugate of a complex number corresponds to reflecting the number across the real axis (horizontal axis) in the Argand diagram (complex plane)

8A.1.4 - *More Examples and some identities*

Some examples:

Let $z_1 = 4 + 3i$ and $z_2 = 2 + 5i$.

$$\overline{(z_1 z_2)} = \overline{(4 + 3i)(2 + 5i)} = \overline{(-7 + 26i)} = -7 - 26i,$$

$$\bar{z}_1 \bar{z}_2 = (4 - 3i)(2 - 5i) = -7 - 26i.$$

Some identities:

$$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2, \qquad \overline{(z_1 - z_2)} = \bar{z}_1 - \bar{z}_2,$$

$$\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2, \qquad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}.$$

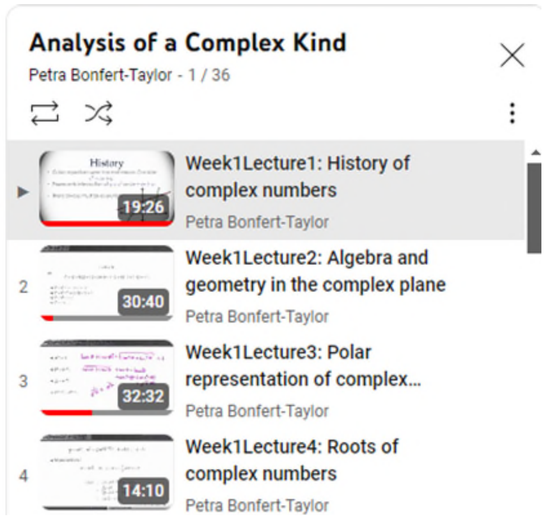
Relevant References for this section

1) U. of surrey course notes on complex numbers
https://www.surrey.ac.uk/sites/default/files/2021-07/10.1-introduction-to-complex-numbers.pdf

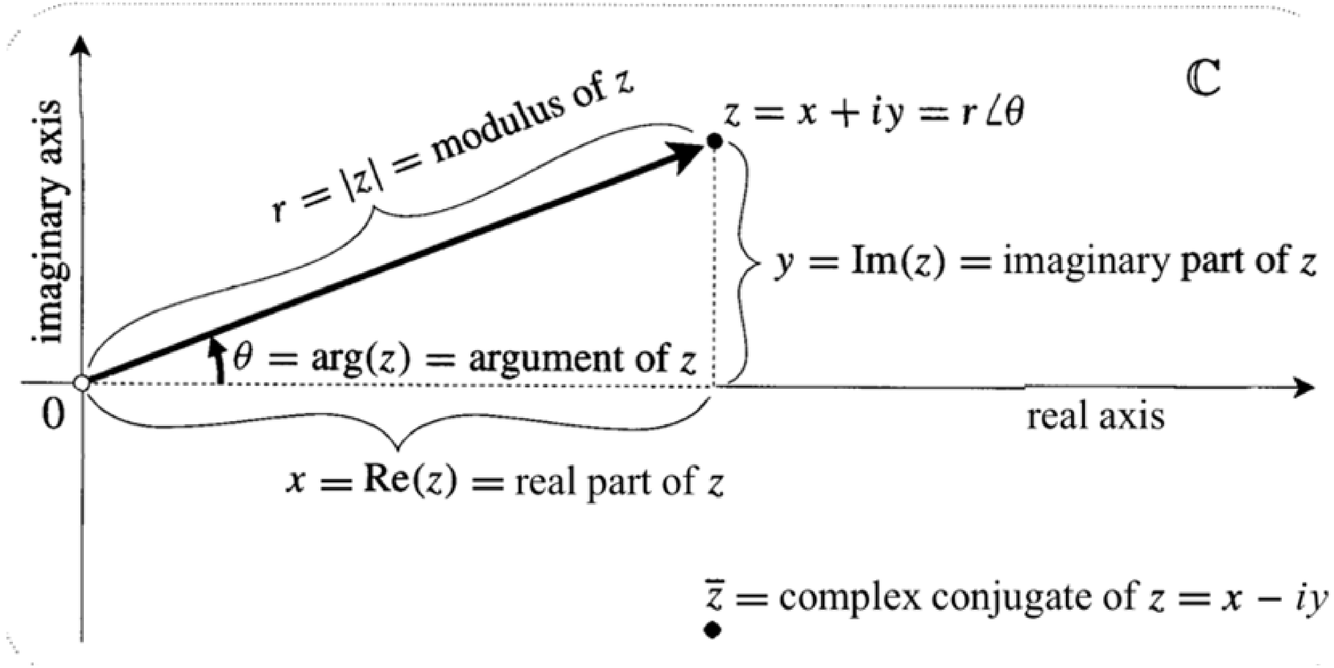
2) Beginner's guide:
https://www.matrix.edu.au/beginners-guide-maths-extension-2/introduction-to-complex-numbers/

3) Online book: UC Davis
https://math.libretexts.org/Bookshelves/Linear_Algebra/Book:_Linear_Algebra_(Schilling_Nachtergaele_and_Lankham)/02:_Introduction_to_Complex_Numbers

4) Petra's Bonfert Taylor complex analysis videos: Lect1~4
https://www.youtube.com/watch?v=CVpMpZpd-5s&list=PLi7yHjesblV0sSfZzWdSUXGO683n_nJdQ&index=1



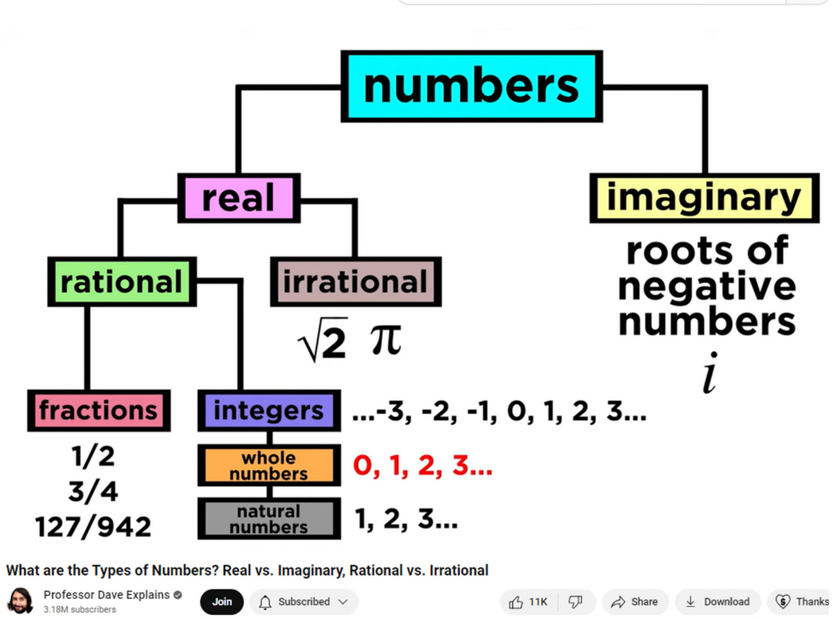
Ref: Terminologies



Name	Meaning	Notation
modulus of z	length r of z	$\|z\|$
argument of z	angle θ of z	$\arg(z)$
real part of z	x coordinate of z	$\mathrm{Re}(z)$
imaginary part of z	y coordinate of z	$\mathrm{Im}(z)$
imaginary number	real multiple of i	
real axis	set of real numbers	
imaginary axis	set of imaginary numbers	
complex conjugate of z	reflection of z in the real axis	$\bar{z}$

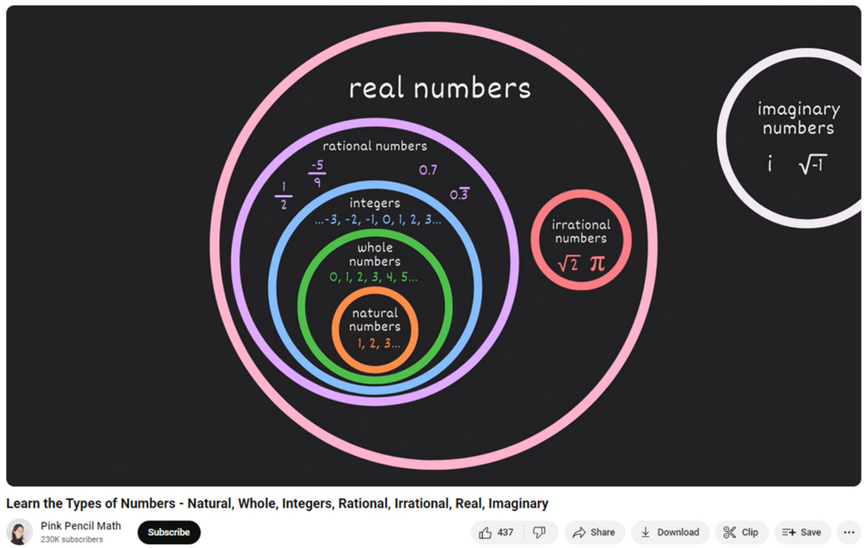
Other References

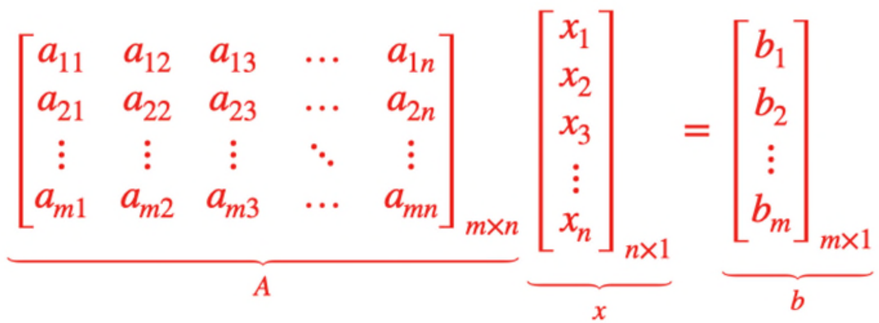
1) Professor Dave: Types of numbers
https://www.youtube.com/watch?v=QUGmwPwtbpg



Hoser: The story of (almost) All Numbers
https://www.youtube.com/watch?v=ggOPJ8gafPo

2) Pink Pencil Math: Types of numbers
https://www.youtube.com/watch?v=WxXZaP8Y8pI





CX1104: **Linear Algebra for Computing**

Chap. No : **8A.2**

Lecture : **Complex Numbers**

Topic : **Complex Exponential Form and DeMoivre theorem**

Concept : $\boldsymbol{Z^n}$, $\mathbf{z}^{\frac{1}{n}}$, $\boldsymbol{Z}^{\frac{-1}{n}}$, **where** $\boldsymbol{n} \in$ **integers**

Instructor: **A/P Chng Eng Siong**
TA: Shreyas Gopal

Rev: 23rd Jul 2024

8A.2.1

- Euler's formula: $e^{j\theta} = \cos(\theta) + jsin(\theta)$
- complex exponential form: $z = a + jb = |z|e^{j\theta}$
- some properties and identities
- redoing multiplication and division

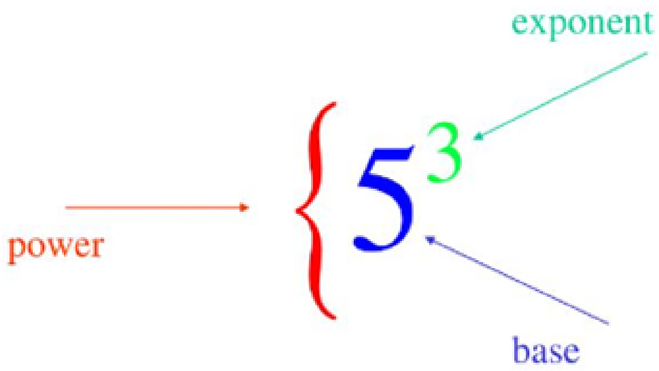
8A.2.2

- DeMoivre Theorem:
- power of z: $z^n = |z|^n e^{jn\theta}$

8A.2.3

- roots of z: $z^{1/n} = |z|^{\frac{1}{n}} e^{j\left(\frac{\theta+2\pi k}{n}\right)}$

8A.2.1 Revision: base and exponent



Example: $125 = 5^3$ means that 5^3 is the exponential form of the number 125.

Ex: the above can be stated as 125 is:

a) 5 to the exponent of 3.

b) The 3rd power of 5

c) (not so accurate) 5 to the power of 3 (see right hand side discussion)

Asking perplexity

how to say this in english a^n where a and n are given

Answer

When expressing "a^n" in English, where 'a' and 'n' are given values, we typically use the following phrasing:

"a to the power of n" or "a to the nth power"

Here are some specific examples to help illustrate:

1. For 2^3:
   "Two to the power of three" or "Two to the third power"

Confusion: 2 to the power of 3?

You are correct that, technically speaking, "a to the exponent of n" is a more precise way to describe a^n. Here's a breakdown to help clear things up:

1. Correct terminology:
   - a is called the "base"
   - n is called the "exponent" or "index"
   - a^n is called a "power"
2. Technically accurate way to read a^n:
   "a to the exponent of n" or "a to the nth power"
3. Common usage:
   While "a to the power of n" is widely used and understood, it's not the most technically precise way to read the expression.

The reason for this common usage is likely due to the fact that the entire expression a^n is referred to as "a power," so people often say "to the power of" when reading it aloud. However, this can indeed lead to confusion, as you've pointed out.

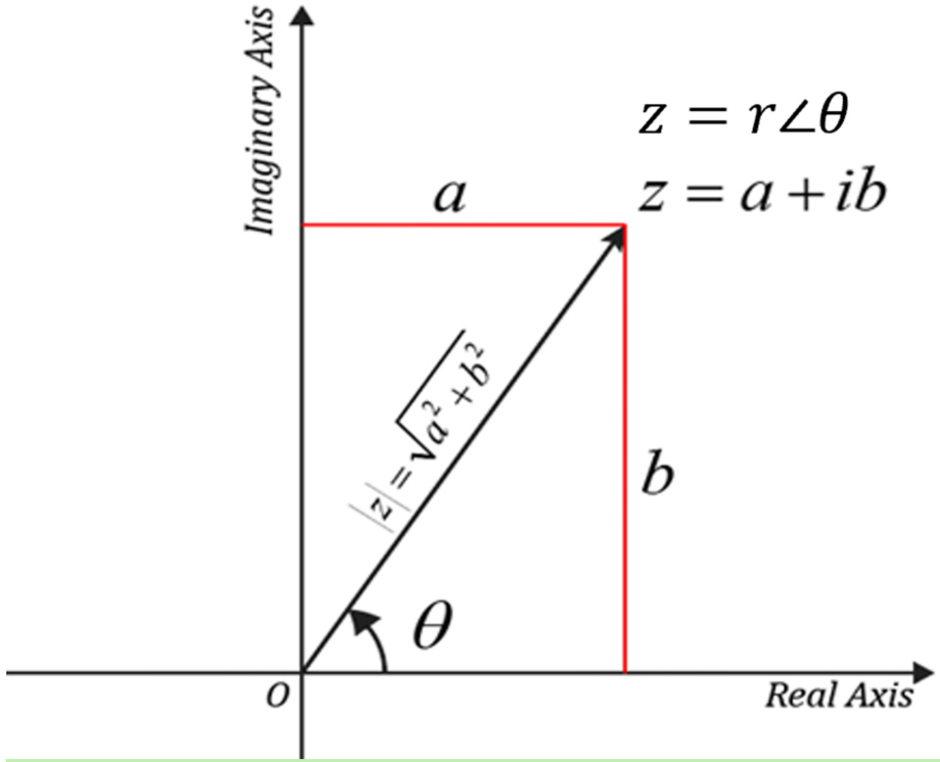
To be most precise:

- Say "a to the exponent of n" when referring specifically to the relationship between a and n.
- Use "the nth power of a" when referring to the entire expression a^n.

Thank you for bringing this up. It's an important distinction that often gets overlooked, and your attention to detail is commendable. In mathematics, precision in language is crucial, and your question helps promote that precision.

8A.2.1 - Euler's Formula to represent complex numbers

Complex Plane



Euler's formula:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

Using above, we can re-write z (the complex number) in the following forms:

$z = a + jb$ // rectangular or cartesian form of complex number

$= r(\cos\theta + j\sin\theta)$ // trigonometric polar form

$= \mathrm{r}\angle\theta$ // polar form

$= \mathrm{r}\, \mathrm{e}^{\mathrm{j}\theta}$ // complex exponential form

8A.2.1 - Proof : Deriving Euler formula $e^{j\theta} = cos(\theta) + jsin(\theta)$ through Taylor Series

Euler's formula:

$$e^{j\theta} = \cos(\theta) + jsin(\theta)$$

Firstly, recall the Taylor (or Maclaurin) series for $\sin x$, $\cos x$ and e^x:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \ldots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \ldots$$

Lets plug $x = ix$ into above

Remember that $i^2 = -1$. So, we see $i^3 = -i$, $i^4 = 1$, $i^5 = i$,

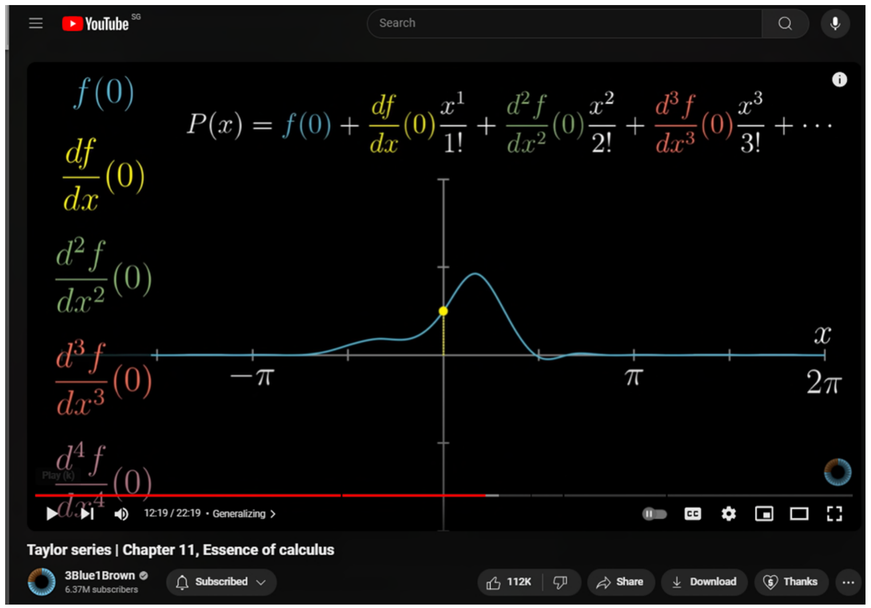
$$e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \ldots$$

$$= 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} + \ldots$$

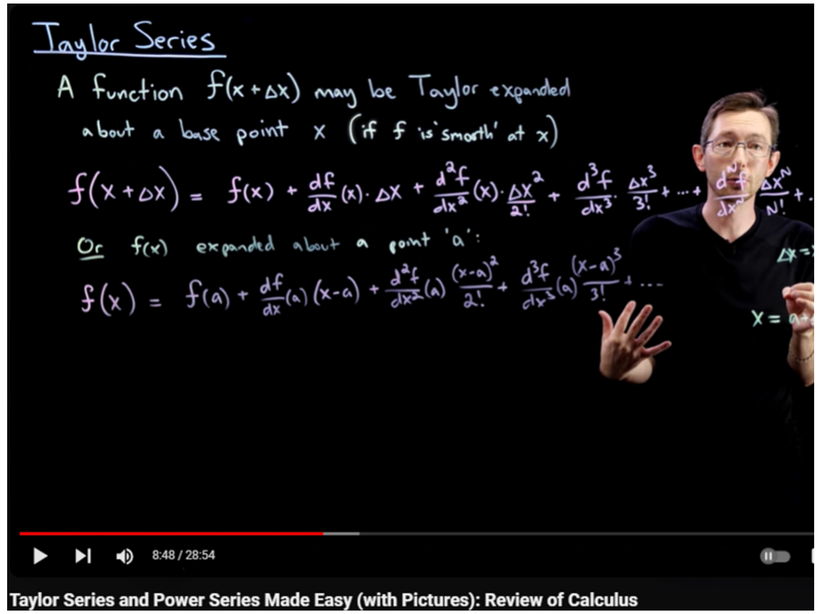
$$= \underbrace{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\right)}_{\cos(x)} + i\underbrace{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right)}_{i\sin(x)}$$

1) Brunton: https://www.youtube.com/watch?v=H0Zbg_CqMCs
2) https://mathsathawthorn.pbworks.com/f/De+Moivre%27s+Theorem+and+my+favourite+piece+of+maths.pdf

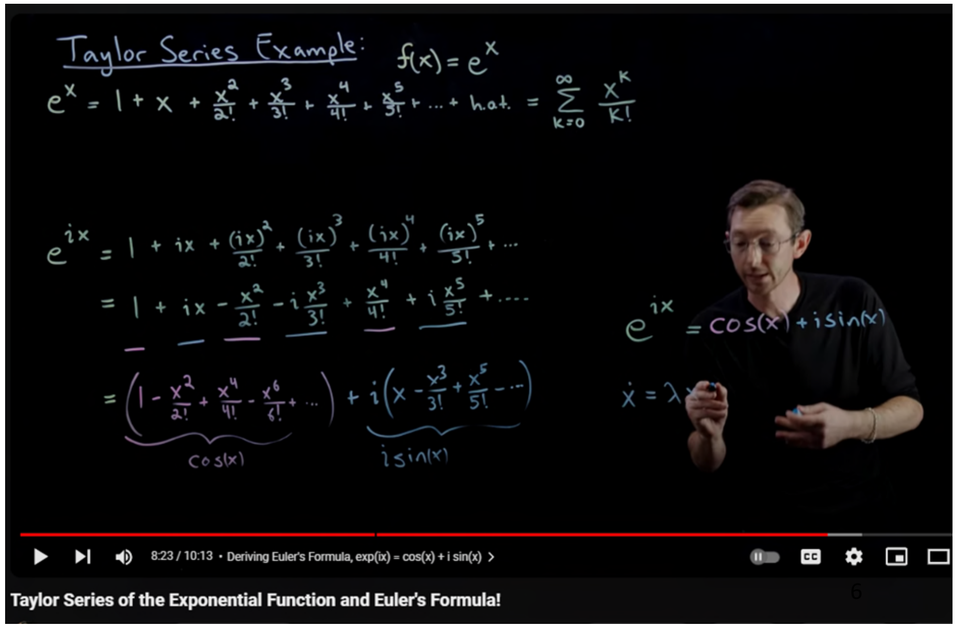
8A.2.1) References: Deriving the Euler's formula from Taylor Series: from Taylor Series to Euler's formula



1A) Blue 1 brown: Taylor Series:
https://www.youtube.com/watch?v=3d6DsjIBzJ4



1B) Brunton's Taylor Series
https://youtu.be/ebfOSDj4j3I?si=dZwrBXAssVJDBKT9



1C) Brunton's Taylor Series of Euler's formula
https://www.youtube.com/watch?v=H0Zbg_CqMCs

8A.2.1 - Euler's identity and examples

Euler's identity $e^{i\pi} + 1 = 0$

Some identities:

- $e^{-j\theta} = \cos(-\theta) + j\sin(-\theta) = \cos\theta - j\sin\theta$
- $e^{j\theta} + e^{-j\theta} = 2\cos\theta \quad \text{and} \quad e^{j\theta} - e^{-j\theta} = 2j\sin\theta$
- $re^{-j\theta} = r(\cos\theta - j\sin\theta) = \bar{z}$

Example: $\mathrm{Re}(e^{j\frac{\pi}{4}}) = \mathrm{Re}(\cos\frac{\pi}{4} + j\sin\frac{\pi}{4}) = \mathrm{Re}(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$

8A.2.1 - Some Rules and identities of exponents for complex numbers

Multiplication:

$$e^{j\theta} \cdot e^{j\psi} = e^{j\theta+j\psi} = e^{j(\theta+\psi)}$$

$$e^{j\theta} \cdot e^{-j\theta} = e^{-j0} = 1$$

Division: $z = x + iy$

$$e^{-z} = \frac{1}{e^z} \qquad \frac{e^{j\theta}}{e^{j\psi}} = e^{j\theta-j\psi} = e^{j(\theta-\psi)}$$

Some identities:

$$e^{z+2\pi i} = e^z \quad \text{for all complex numbers } z.$$

$$e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y) = (e^x \cos y) + i(e^x \sin y).$$

$$|e^z| = e^x \quad \text{and} \quad \arg e^z = y.$$

8A.2.1

- Euler's formula: $e^{j\theta} = \cos(\theta) + jsin(\theta)$
- complex exponential form: $z = a + jb = |z|e^{j\theta}$
- some properties and identities
- redoing multiplication and division

8A.2.2

- DeMoivre Theorem:
- power of z: $z^n = |z|^n e^{jn\theta}$

8A.2.3

- roots of z: $z^{1/n} = |z|^{\frac{1}{n}} e^{j\left(\frac{\theta+2\pi k}{n}\right)}$
- roots of z: $z^{-1/n}$

8A.2.2- Proof: DeMoivre Theorem

In mathematics, **de Moivre's formula** (also known as **de Moivre's theorem** and **de Moivre's identity**) states that for any real number x and integer n it holds that

$$\left(\cos x + i \sin x\right)^n = \cos nx + i \sin nx$$

...

Given:

$$z = a + jb = re^{j\theta}$$

DeMoivre Theorem: for $n \in Z$ $(integers)$

$$z^n = \left(re^{j\theta}\right)^n = \mathrm{r^n e^{j(n\theta)}} = \mathrm{r^n}(\cos(\mathrm{n}\theta) + jsin(n\theta))$$

...

Ref: https://en.wikipedia.org/wiki/De_Moivre%27s_formula#

8A.2.2 Proof of DeMoivre theorem by Induction

😎 The classic proof

RTP: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ for $n = 0, 1, 2 \ldots$

TEST $n = 0$:

$\text{LHS} = (\cos\theta + i\sin\theta)^0$

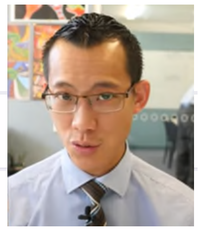
$= 1$

$\text{RHS} = \cos(0\theta) + i\sin(0\theta)$

$= \cos 0 + i\sin 0$

$= 1 + 0i$

$= 1$ as required.



ASSUME $(\cos\theta + i\sin\theta)^k = \cos(k\theta) + i\sin(k\theta)$

PROVE $(\cos\theta + i\sin\theta)^{k+1} = \cos[(k+1)\theta] + i\sin[(k+1)\theta]$

Proof: $\text{LHS} = (\cos\theta + i\sin\theta)^{k+1}$

$= (\cos\theta + i\sin\theta)^1 (\cos\theta + i\sin\theta)^k$

$= (\cos\theta + i\sin\theta)[\cos(k\theta) + i\sin(k\theta)]$ by assumption

$= \cos\theta\cos(k\theta) + i\cos\theta\sin(k\theta) + i\sin\theta\cos(k\theta) + i^2\sin\theta\sin(k\theta)$

$= \cos\theta\cos(k\theta) - \sin\theta\sin(k\theta) + i[\sin\theta\cos(k\theta) + \cos\theta\sin(k\theta)]$

$= \cos(\theta + k\theta) + i\sin(\theta + k\theta)$

$= \cos[(k+1)\theta] + i\sin[(k+1)\theta]$ as required.

$\therefore (\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ for $n = 0, 1, 2 \ldots$

by the principle of mathematical induction.

This slide proofs for n = 0,1,2,…
Next slide shows for n=-1,-2,…

Links of Proofs

1) Eddie Woo: https://youtu.be/Ih9xfnkUkGA?si=SdpGnFXgG0xVUY5O
2) Simple Science & Maths: https://youtu.be/gXhGwSIU0IU

8A.2.2 Proof of DeMoivre theorem by Induction (-ve index)

Previous page shows for n= 1,2,3..
Now lets extend to n= -1,-2,-3,…

If $z = \cos\theta + i\sin\theta$

$$z^{-1} = \frac{1}{\cos\theta + i\sin\theta} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta}$$

$$= \frac{\cos\theta - i\sin\theta}{\cos^2\theta - i^2\sin^2\theta}$$

$$= \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta}$$

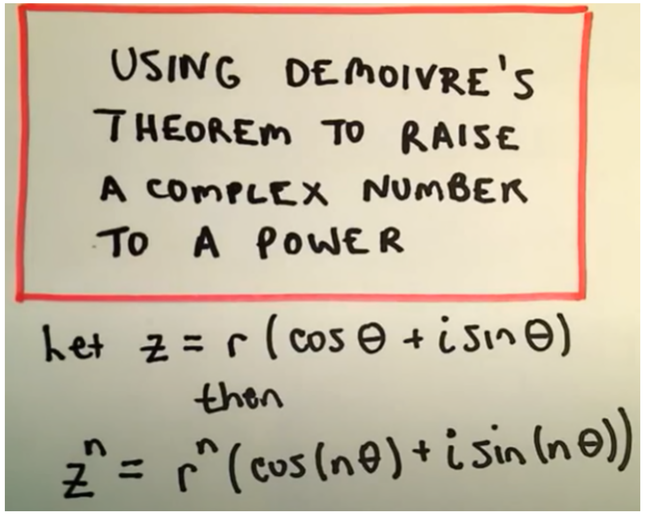
$$= \frac{\cos\theta - i\sin\theta}{1}$$

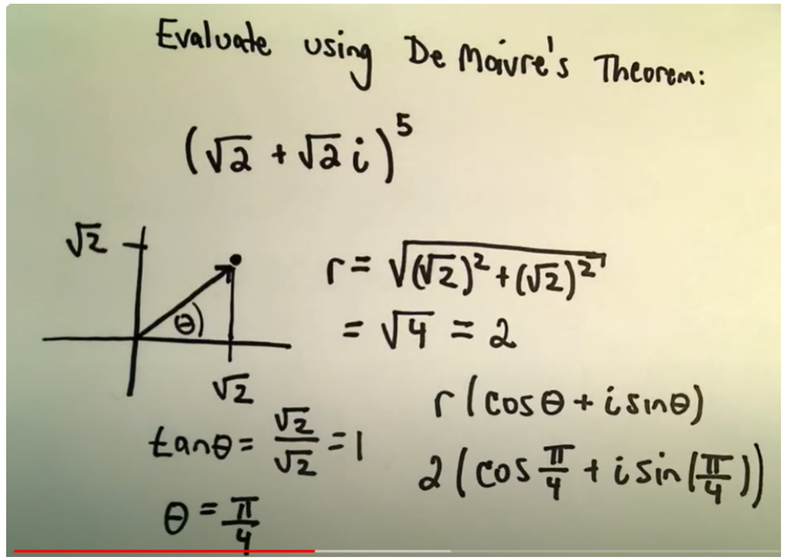
$$z^{-1} = \cos\theta - i\sin\theta \qquad = \cos(-\theta) + isin(-\theta)$$

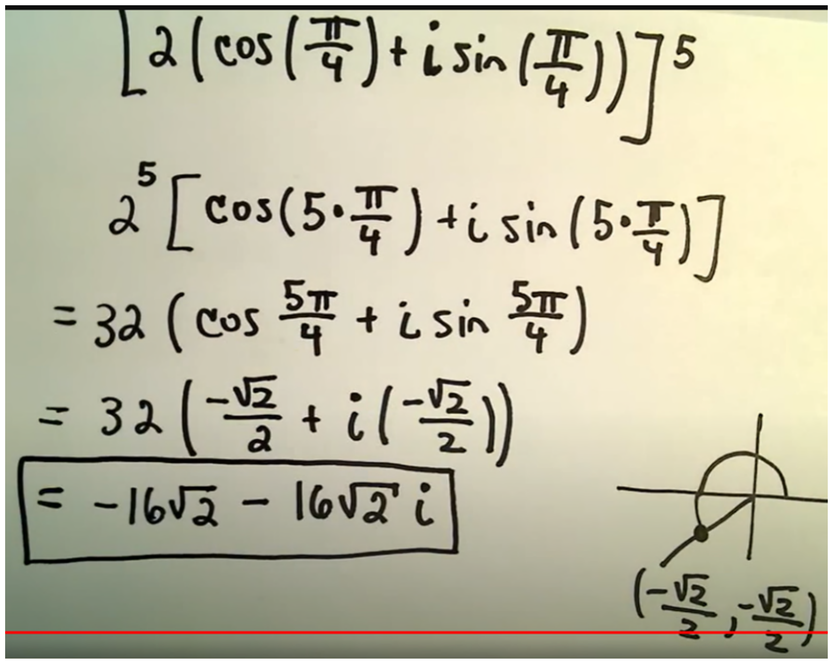
$$(z^{-1})^{\wedge} = \cos(-n\theta) + isin(-n\theta), \text{ for n=0,1,2…}$$

$$z^{-n} = 1e^{j(-n\theta)}$$

8A.2.2 DeMoivre Theorem Example 1: z^n , where $n \in$ integers







DeMoivre's Theorem: Raising a Complex Number to a Power, Ex 3

patrickJMT 1.38M subscribers Join Subscribed 479

Example: https://www.youtube.com/watch?v=Srdh05cICiE

Insight: you can think of $\left(2\angle\left(\frac{\pi}{4}\right)\right)^5$ as a vector starting at $1\angle 0$ and growing at a rate of length*2, and angle rotating anti-clockwise at 45 (degrees) $=\frac{\pi}{4}$ rad with each step, where step =1..5.

8A.2.1

- Euler's formula: $e^{j\theta} = \cos(\theta) + jsin(\theta)$
- complex exponential form: $z = a + jb = |z|e^{j\theta}$
- some properties and identities
- redoing multiplication and division

8A.2.2

- DeMoivre Theorem:
- power of z: $z^n = |z|^n e^{jn\theta}$

8A.2.3

- roots of z: $z^{1/n} = |z|^{\frac{1}{n}} e^{j\left(\frac{\theta+2\pi k}{n}\right)}$
- roots of z: $z^{-1/n}$

8A.2.3- Proof: DeMoivre Theorem extension to root

A modest extension of the version of de Moivre's formula can be used to find the *n*-th roots of a complex number for a non-zero integer *n*. (This is equivalent to raising to a power of 1 / *n*)

The roots of a number is multivalued.
E.g, given $z = 4$, what is $w = \sqrt{z}$?
$w = \{+2, -2\}$ are the two solutions.

Then by DeMoivre theorem, we can use it to find the n values of $\sqrt[n]{z} = z^{1/n}$

DeMoivre Theorem extended (for the n roots)

$$z = a + jb = re^{j\theta}$$

$$\mathrm{w_k} = z^{1/n} = \left(re^{j\theta}\right)^{\frac{1}{n}}$$

$$= \mathrm{r}^{\frac{1}{n}}\, \mathrm{e}^{\mathrm{j}\left(\frac{\theta+2\pi k}{\mathrm{n}}\right)}$$

$$= \mathrm{r}^{\frac{1}{n}}\left(\cos\left(\frac{\theta + 2\pi\mathrm{k}}{n}\right) + j\sin\left(\frac{\theta + 2\pi\mathrm{k}}{n}\right)\right)$$

for $k = 0..(n-1)$

https://www.youtube.com/watch?v=27BnukijSdQ

8A.2.3: Proof - DeMoivre Theorem extension to root

Extension of DeMoivre's Theorem (to fractional powers)

Let $z = r(\cos\phi + i\sin\phi)$ & $n \in \mathbb{N}$.

Then $z^{1/n} = r^{1/n}\left(\cos\left(\frac{\phi}{n} + \frac{2k\pi}{n}\right) + i\sin\left(\frac{\phi}{n} + \frac{2k\pi}{n}\right)\right)$

where $0 \le k \le n-1$

Let $z = r(\cos\phi + i\sin\phi)$

What is $z^{1/n}$?

Want $z^{1/n}$ to satisfy $(z^{1/n})^n = z$.

Proof:

Let $z^{1/n} = \rho(\cos\theta + i\sin\theta)$.

Then $(z^{1/n})^n = \rho^n(\cos\theta + i\sin\theta)^n$

$z = \rho^n(\cos n\theta + i\sin n\theta)$ ← DeMoivre Theorem

$z = r(\cos\phi + i\sin\phi) = \rho^n(\cos n\theta + i\sin n\theta)$

By inspection, to identify ρ and θ of $z^{1/n}$

Take absolute value of both sides to see $r = \rho^n$ so $r^{1/n} = \rho$

How do ϕ & $n\theta$ relate?

See $n\theta = \phi + 2k\pi \quad (k \in \mathbb{Z})$

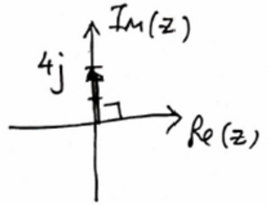
So $\theta = \frac{\phi}{n} + \frac{2k\pi}{n} \quad (k \in \mathbb{Z})$

1) Andrew McCrady: https://www.youtube.com/watch?v=27BnukijSdQ
time: 5:41

8A.2.3 Example: DeMoivre Theorem for roots , $z^{1/n}$

Example: Determine all the square roots of $4j$, i.e., $\sqrt[2]{4j}$.

8A.2.3) Example $z^{1/n}$



Given $z = 4j$

$= 4e^{j\pi/2}$

Find $z^{1/2}$.

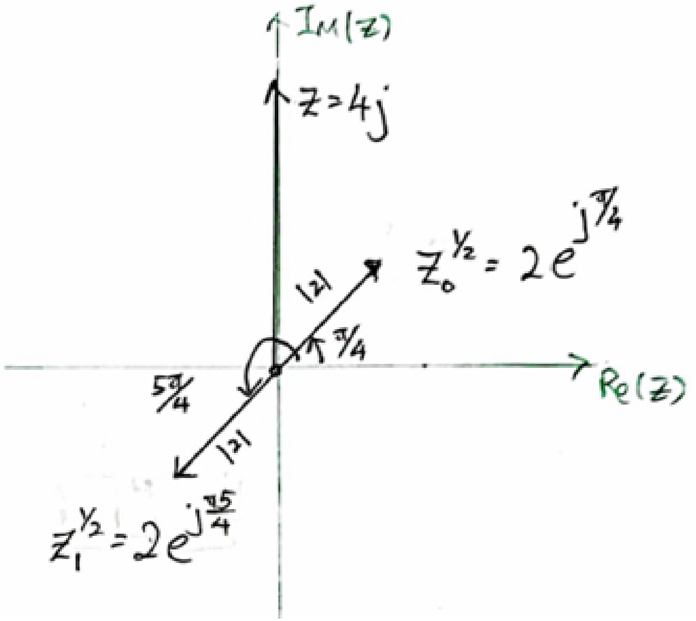
Using De Moivre Theorem:

$$z^{1/n} = r^{1/n} e^{j\left(\frac{\theta + 2\pi k}{n}\right)} \quad ; \quad k = 0..(n-1)$$

$$\therefore z^{1/2} = 4^{1/2} e^{j\left(\frac{\pi/2 + 2\pi k}{2}\right)}$$

$k=0$, $z_0^{1/2} = 2e^{j\pi/4}$.

$k=1$, $z_1^{1/2} = 2e^{j\left(\frac{5\pi}{4}\right)}$.



8A.2.3 Example: Find $(1+j)^{-1/3}$

Q: Given $z = 1+j$,

Find $z^{-1/3}$.

We take 2 steps:

a) Find $z^{1/3}$.

b) evaluate $\frac{1}{z^{1/3}}$.

a) Using DeMoivre Theorem.

if $z = re^{j\phi}$

$z^{\frac{1}{n}} = r^{1/n} e^{j\left(\frac{\phi+2\pi k}{n}\right)}$; $k = 0..(n-1)$

r = modulus of $z = \sqrt{1^2+1^2}$

$= \sqrt{2}$

$\theta = \arctan\left(\frac{1}{1}\right) = \pi/4$ rad.

$\therefore z^{1/n}$ has 3 soln in $n=3$.

$\omega_0 = r^{1/n} e^{j\left(\frac{\phi}{n}\right)} = (\sqrt{2})^{1/3} e^{j\left(\frac{\pi/4}{3}\right)}$

$= 1.1225 e^{j\pi/12}$

$\omega_1 = r^{1/n} e^{j\left(\frac{\phi+2\pi}{3}\right)} = 1.1225 e^{j\frac{9\pi}{12}}$

$\omega_2 = r^{1/n} e^{j\left(\frac{\phi+4\pi}{3}\right)} = 1.1225 e^{j\frac{17\pi}{12}}$

---

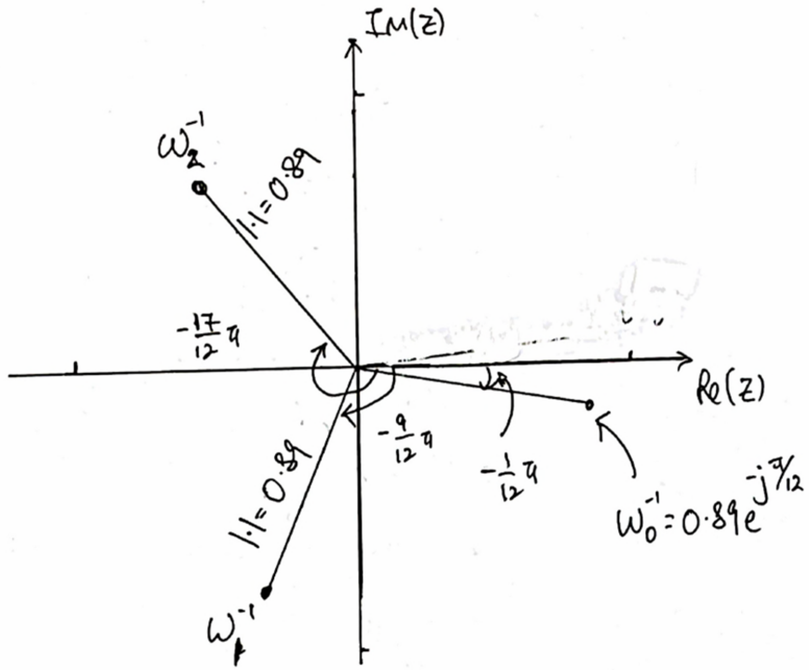
step b)

$\frac{1}{\omega_0} = \omega_0^{-1} = \left(\frac{1}{1.1225}\right) e^{-j\pi/12} = 0.891 e^{-j\pi/12}$

$\frac{1}{\rho e^{j\phi}} \cdot \frac{e^{-j\phi}}{e^{-j\phi}} = \frac{1}{\rho} e^{-j\phi}$

$\therefore \omega_1^{-1} = \left(\frac{1}{1.1225}\right) e^{-j\frac{9\pi}{12}}$

$\omega_2^{-1} = \left(\frac{1}{1.1225}\right) e^{-j\frac{17\pi}{12}}$



References

IIT Guwahati: DeMoivre's Proof

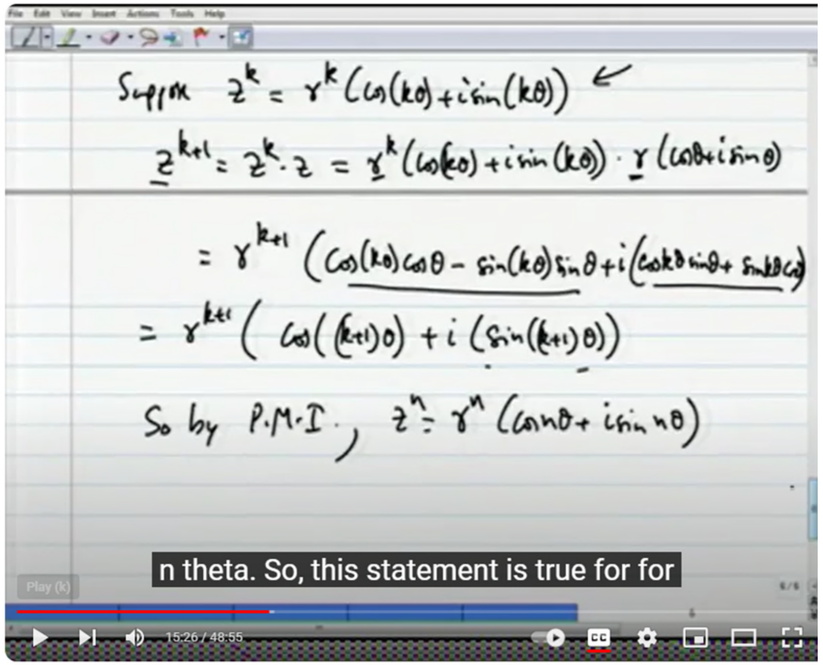
IIT Guwati https://youtu.be/08gUkfymPec

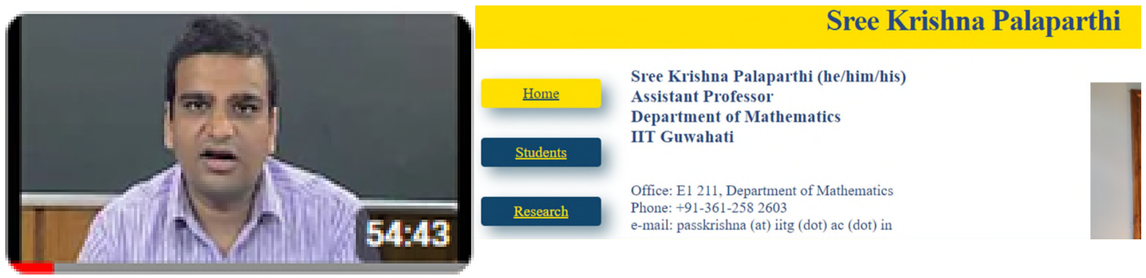
Time: 7:35 (geometric interpretation of z1*z2)

Time: 11:20 (deriving z^n by mathematical induction) for positive integer n

Time: 15:50 (now consider for n=0, and n = -ve values)
-> in deriving for 1/z , suggest we use complex exponential form instead of e(j\theta) = cos(theta)+jsin(theta)

Time: 20:10~27:50) DeMoivre formula is valid for any integer n
-> now some examples including 1/z, and z^(1/4)





Ref: When are the laws of exponent correct?

When are the Laws of Exponents correct?

Asked 6 years, 5 months ago Modified 4 years, 9 months ago Viewed 2k times

13

The rules of powers are in highschool books often briefly stated in the following way:

1. $a^n \cdot a^m = a^{n+m}$
2. $\frac{a^n}{a^m} = a^{n-m}$
3. $(a \cdot b)^n = a^n \cdot b^n$
4. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
5. $(a^n)^m = a^{n \cdot m}$

Steward's Review of Algebra states that those rules are true if a and b are positive (real) numbers, and n and m are rational numbers. This is of course very conservative. Those rules are also true if $a \neq, b \neq 0$ and n, m integers. Besides that I think many of those rules are also true if n, m are real numbers.

2 Answers Sorted by: Highest score (default)

13

Provided $a, b > 0$, all the rules are true for real a, b, m, n.

If $a = 0$ or $b = 0$, no negative power may appear.

For $a < 0$ or $b < 0$, irrational exponents are excluded. Rational ones are possible provided the denominator of the simplified fraction is odd. This can cause rule 5 to fail $((-1)^1 \neq ((-1)^{1/2})^2)$.

Share Cite Edit Follow edited Mar 16, 2018 at 9:25 answered Mar 16, 2018 at 9:19

user65203

- https://math.stackexchange.com/questions/2693503/when-are-the-laws-of-exponents-correct

8A.1.3: Ref: Exponent law for real vs exponent law for complex

The familiar exponent laws
is valid for positive real
values of x,y and real of m,n.
care must be used when base is -ve

But some of these laws are NOT applicable
when the exponent is complex.
See: Power of a power for complex exponent

Ref:
https://www.quora.com/Why-dont-the-algebraic-properties-of-exponents-work-on-complex-numbers
https://math.stackexchange.com/questions/4325029/exponent-law-for-complex-numbers
https://math.stackexchange.com/questions/2503005/how-can-i-intuitively-understand-complex-exponents
https://web.spms.ntu.edu.sg/~ydchong/teaching/08_branch_cuts.pdf

So we can say, for example, that

$$e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y) = (e^x \cos y) + i(e^x \sin y).$$

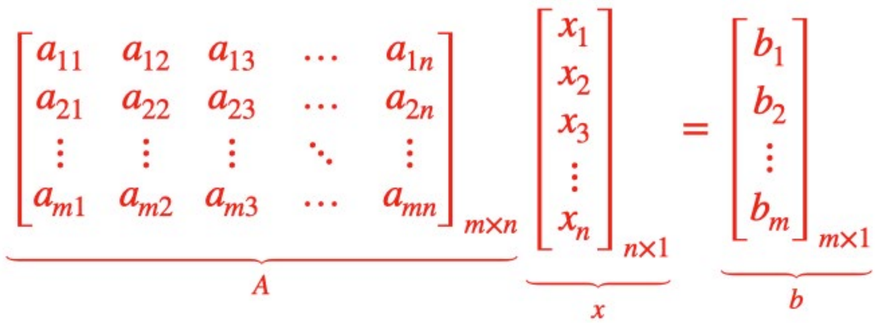
However, the exponential function does *not*, in general, satisfy

$$(e^{z_1})^{z_2} = e^{z_1 z_2}$$ Unless z_2 is an integer

if z_1 and/or z_2 are complex numbers. It is fine if z_2 is an integer, because then you can write $e^{z_1 z_2}$ as $e^{z_1} e^{z_1} \cdots e^{z_1}$ (or a simple adjustment of this if z_2 is negative), and then, by using $e^{z_1} e^{z_2} = e^{z_1 + z_2}$, we obtain the result $e^{z_1 z_2}$. Indeed, this is called de Moivre's theorem, which essentially states that $(e^z)^n = e^{nz}$ if n is an integer.

Laws of Exponents

Name	The Rule	Example
Product of Powers	$x^m \cdot x^n = x^{m+n}$	$x^3 \cdot x^2 = x^5$
Quotient of Powers	$\frac{x^m}{x^n} = x^{m-n}$ or $\frac{1}{x^{n-m}}$	$\frac{x^7}{x^5} = x^2$
		$\frac{x^7}{x^5} = \frac{1}{x^{-2}}$ or x^2
Power of a Power	$(x^m)^n = x^{mn}$	$(x^4)^6 = x^{24}$
Power of a Product	$(xy)^m = x^m \cdot y^m$	$(xy)^5 = x^5 \cdot y^5$
Power of a Quotient	$\left(\frac{x}{y}\right)^m = \frac{x^m}{y^m}$	$\left(\frac{2}{3}\right)^2 = \frac{2^2}{3^2} = \frac{4}{9}$
Negative Exponent	$x^{-m} = \frac{1}{x^m}$	$x^{-7} = \frac{1}{x^7}$
		$\frac{1}{x^{-5}} = x^5$
Identity Exponent	$x^1 = x$	$8^1 = 8$
Zero Exponent	$x^0 = 1\ (x \neq 0)$	$2^0 = 1$



CX1104: **Linear Algebra for Computing**

Chap. No : **8A.3.0**

Lecture : **DFT Overview**

Topic : **DFT**

Concept : **Seeing DFT as a change of basis**

Instructor: **A/P Chng Eng Siong**

TA: Shreyas Gopal

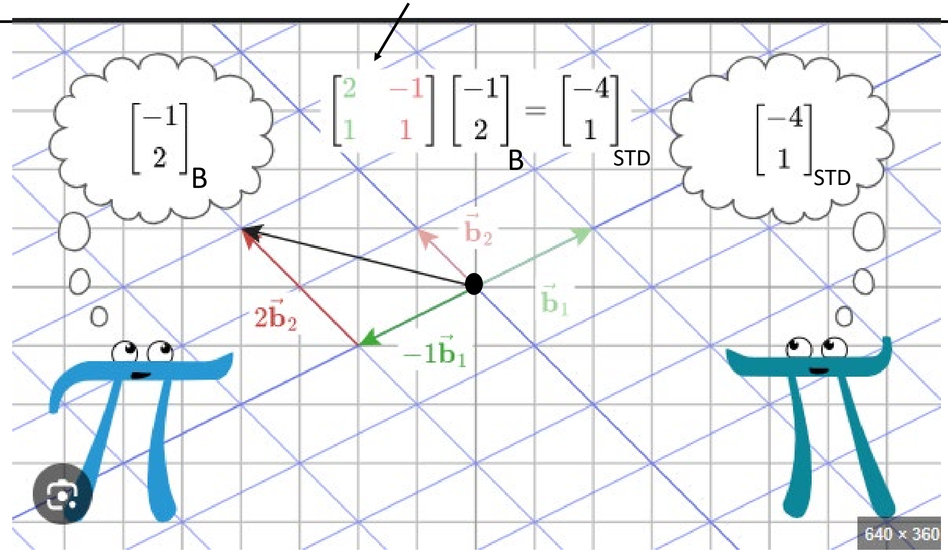
Rev: 23rd July 2024 Last updated: 10 Dec 2025

Content

8A.3.0a: Revision - Change of Basis
8A.3.0b: Examples of Different types of Change of Basis
8A.3.0c: The DFT as a change of basis

8A.3.0a: Revision: What is Change of Basis?

B = change of Basis from B basis to the standard basis



https://www.3blue1brown.com/lessons/change-of-basis

A vector can be represented using different basis.

A change of basis allows you to represent the same vector using different coordinates,
much like how you can describe a location using different coordinate systems
(e.g., latitude/longitude vs. a grid system).

The change of basis formula

Using matrices, this formula can be written

$$\mathbf{x}_{\text{old}} = A\,\mathbf{x}_{\text{new}},$$

where "old" and "new" refer respectively to the initially defined basis and the other basis, $\mathbf{x}_{\text{old}}$ and $\mathbf{x}_{\text{new}}$ are the column vectors of the coordinates of the same vector on the two bases. A is the **change-of-basis matrix** (also called **transition matrix**), which is the matrix whose columns are the coordinates of the new basis vectors on the old basis.

Note: the change of basis matrix must be invertible

$\begin{bmatrix}-4\\1\end{bmatrix}_{STD}$, $\begin{bmatrix}-1\\2\end{bmatrix}_B$

Both vectors indicate the same vector (same direction and length from origin).
The subscripts represent which basis used.

$$[B][x]_B = [x]_{STD}$$

In above, matrix B = change of basis matrix and is square+invertible.

B's j^th columns are the coordinates of the j^th basis vector of B written in the standard basis coordinates.

Typically, $[x]_{STD}$ is simply denoted as $[x]$ without (STD) and its basis is the standard basis

Ref:

1) https://math.stackexchange.com/questions/25779/rigorously-proving-that-a-change-of-basis-matrix-is-always-invertible
2) https://math.stackexchange.com/questions/3005907/how-is-dft-the-change-of-basis-operator

8A.3.0b Different types of change of basis

There are infinitely many basis for a given vector space, Hence any square invertible matrix can be a change of basis matrix.

In this section, we introduce one of the most important change of basis, using the DFT (discrete fourier transform) matrix.

DFT : The DFT basis is a basis in which the columns in the change of coordinate matrix are sequences of complex exponentials.

Other types of change of basis:

1) Discrete Cosine Transform
2) Walsh-Hadamard Transform
3) Haar Transform Matrix
4) Karhunen-Loève Transform (KLT) Matrix
5) Wavelet Transform Matrix

Examples of popular basis

Transform	Type / domain	Basis functions (intuition)	Typical use
Discrete Fourier Transform (DFT)	Orthogonal (unitary), global	Complex exponentials (sinusoids) over the whole support. pages.hmc	Spectral analysis, filtering, modulation, convolution via FFT.
Discrete Cosine Transform (DCT)	Real orthogonal, global	Cosine waves with specific boundary conditions. units +1	Image/audio compression (e.g., JPEG), energy compaction.
Discrete Sine Transform (DST)	Real orthogonal, global	Sine waves with odd-symmetry boundaries. units	PDEs and problems with Dirichlet boundaries.
Walsh–Hadamard Transform (WHT) / Walsh	Real orthogonal	Square/rectangular waveforms taking ±1 values. mathworks +1	Fast binary operations, multiplexing, some compression and coding.
Haar Transform	Real orthogonal, wavelet	Step-like scaling and wavelet functions (piecewise constants). units +1	Simple multiresolution analysis, compression, edge detection.
General Discrete Wavelet Transform (DWT)	Real (often) orthogonal or biorthogonal	Scaled and shifted wavelets with compact or near-compact support. wikipedia +1	Multiresolution signal and image analysis, compression, denoising.
Karhunen–Loève Transform (KLT) / PCA	Data-dependent orthogonal	Eigenvectors of covariance matrix of the data. tecnico.ulisboa +2	Optimal energy compaction, decorrelation, dimensionality reduction.

8A.3.0b Examples of various 4x4 change of Basis matrix

4×4 DFT (unitary up to factor)

A common unnormalized 4-point DFT matrix is

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

Note that DFT matrix's entries are complex numbers and each column can be viewed as a complex exponential rotating at a particular frequency.

One standard form of the 4-point DCT-II matrix (rows are frequency indices $k = 0, \ldots, 3$) is

$$C_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ \sqrt{2} & 0 & -\sqrt{2} & 0 \\ 1 & -1 & -1 & 1 \\ 0 & -\sqrt{2} & 0 & \sqrt{2} \end{bmatrix}.$$

4×4 Walsh–Hadamard (WHT)

The standard 4-point Hadamard/Walsh matrix is

$$H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

4×4 Haar transform (orthonormal)

A common orthonormal 4-point Haar matrix is

$$W_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$

About KLT/PCA and others

For the Karhunen–Loève Transform (KLT/PCA), the 4×4 matrix depends on the 4×4 covariance of the specific data, so there is no single universal numeric matrix to write down.

8A.3.0c) The DFT as a change of Basis

Note:

$x_{STD} = time\ sequence$ – column vector of length N (typically real, but can be complex vector)

$X_{DFT} = frequency\ representation$ – column vector of length N (will be a complex vector)

W is the square (NxN) DFT matrix, whose row k=0..(N-1)

is the complex exponential sequences $e^{\frac{-j2\pi kn}{N}}$, for $n = 0..(N-1)$

$$[W]\, x_{STD} = \quad X_{DFT}$$

DFT forward transform (from time (std) ->frequency(DFT) representation

$$x_{STD} \quad = [\, W^{-1}\,]X_{DFT}$$
$$= (\frac{1}{N} W^{H})X_{DFT}$$

DFT inverse transform (from frequency ->time) representation

$W^{-1} = (\frac{1}{N} W^{H})$ is the inverse DFT matrix. W^{H} is the conjugate transpose of W

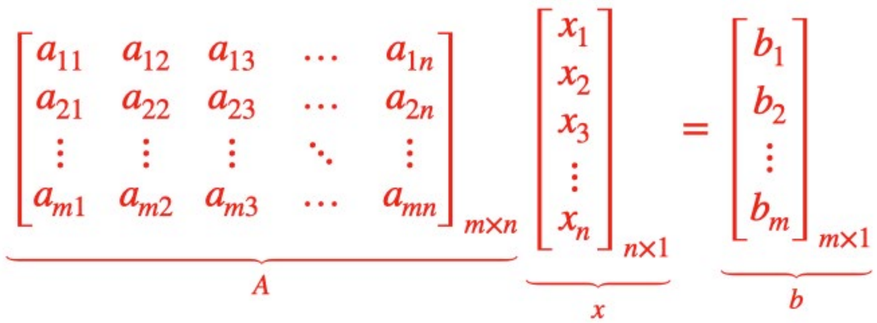
whose columns are scaled complex exponential sequences $\left(\frac{1}{N}\right) e^{\frac{+j2\pi kn}{N}}$

References

References: for Change of Basis

Ref:

1) 3Blue1Brown: https://www.3blue1brown.com/lessons/change-of-basis
2) https://napsterinblue.github.io/notes/stats/lin_alg/change_of_basis/
3) Prof Dave explains: https://www.youtube.com/watch?v=HZa1RwFHgwU
4) Trevor Bazett: https://www.youtube.com/watch?v=KjlTOLhaI9s
5) Lorendo Sadun: https://www.youtube.com/watch?v=1CxBJnrVm3s



CX1104: **Linear Algebra for Computing**

Chap. No : **8A.3**

Lecture : **Complex Numbers**

Topic : **DFT**

Concept : **Understanding complex vectors and DFT Analysis and Synthesis equations**

Instructor: **A/P Chng Eng Siong**

TA: Shreyas Gopal

Rev: 23rdJuly 2024 Last updated: 10 Dec 2025

Content

8A.3.0: DFT overview (read this first!)

8A.3: Complex vectors and DFT

8A.3.1 - Complex vector
- definition
- examples, visualization
- scaling by a complex number

8A.3.2 Inner product and Inverse of complex matrixes

8A.3.3 - Motivation: Decomposing a signal into sum of sinusoids
- DFT is orthogonal decomposition
- The analysis and synthesis equations of DFT
- Example

8A.3.1: Complex vectors: definitions

Given $z \in \mathcal{C}$, let $\boldsymbol{z} \in \mathcal{C}^N$ denote a column vector with *N* complex numbers.

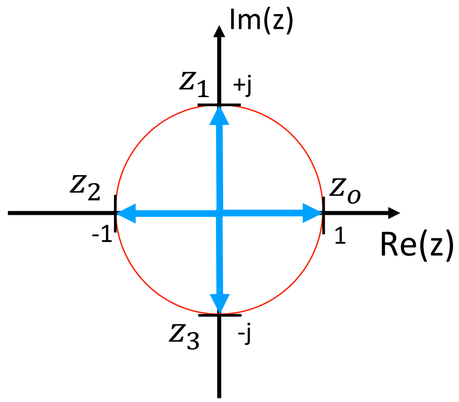
Example: $\boldsymbol{z} \in \mathcal{C}^4$ with elements $z_{n+1} = z_n\left(1e^{\frac{j\pi}{2}}\right)$, and $z_0 = 1$

Alternatively, $z_n = 1e^{\frac{j\pi}{2}n}$, $n = 0..3$

$$z_1 \cdot z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = (r_1 \cdot r_2) e^{i(\theta_1 + \theta_2)}$$

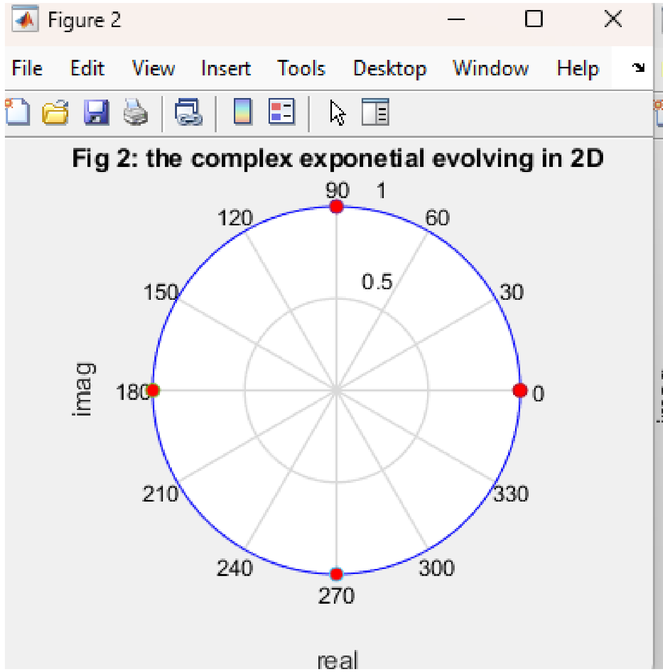
See: 8A.1.2 Multiplication in polar form
Note: multiplication of complex number means length multiplies and angles add.

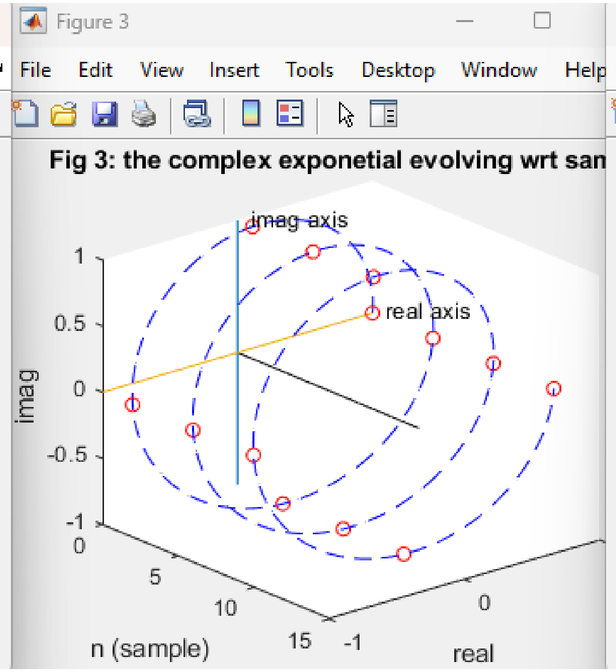
$$\boldsymbol{z} = \begin{bmatrix} z_0 \\ z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1e^{\frac{j\pi}{2}0} \\ 1e^{\frac{j\pi}{2}1} \\ 1e^{\frac{j\pi}{2}2} \\ 1e^{\frac{j\pi}{2}3} \end{bmatrix} = \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix}$$

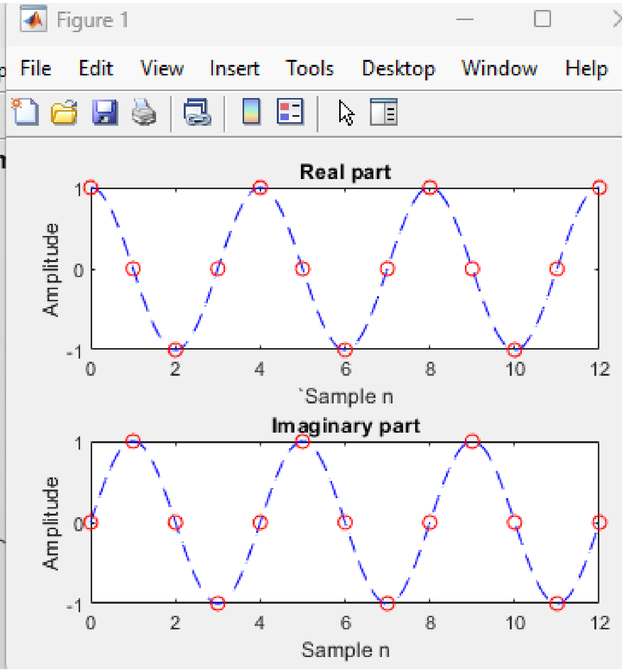


Geometrically: z_n is a complex phasor rotating anti-clockwise at each step by $\frac{\pi}{2}$ radian per step.

8A.3.1: complex vectors - visualization







Given $\omega = \frac{\pi}{2}$,

$$z = 1e^{j\omega},$$

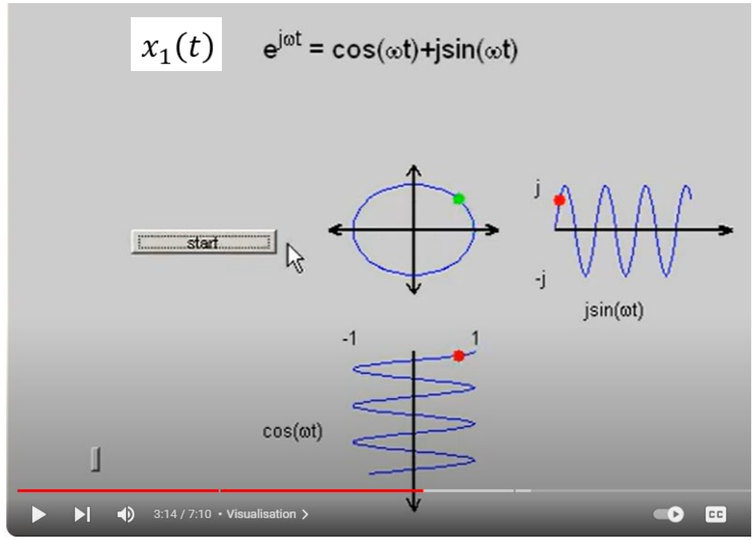
3 views of the complex sequence:

$$z^n = 1e^{j\omega n} = 1e^{j\frac{\pi}{2}n}, \qquad n = 0..12$$

Compressed (zipped) Folder

Matlab code plotting the above

8A.3.1: rotating complex exponentials: more examples



complex exponentials visualisation

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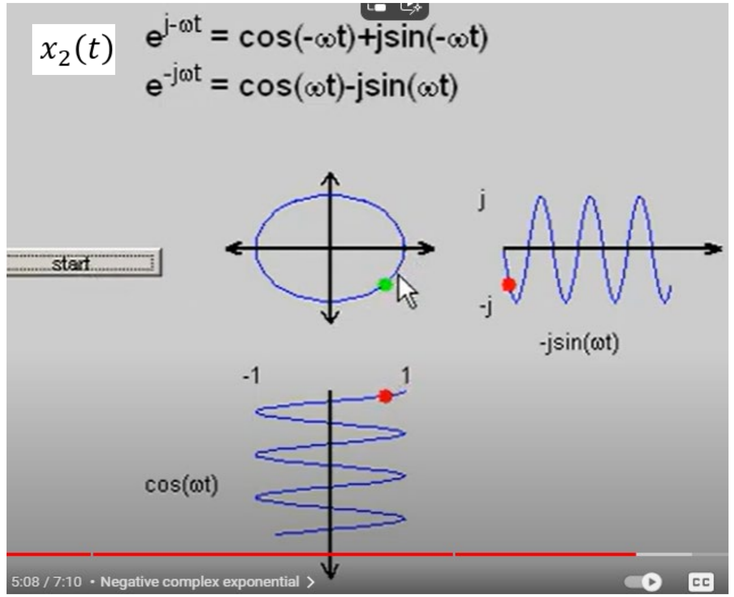


YouTubeLink: https://youtu.be/K_C7htSXORY?si=i_Z5mtR_8aZaHo2V

$$x_1(t) = e^{j(\omega)t}$$

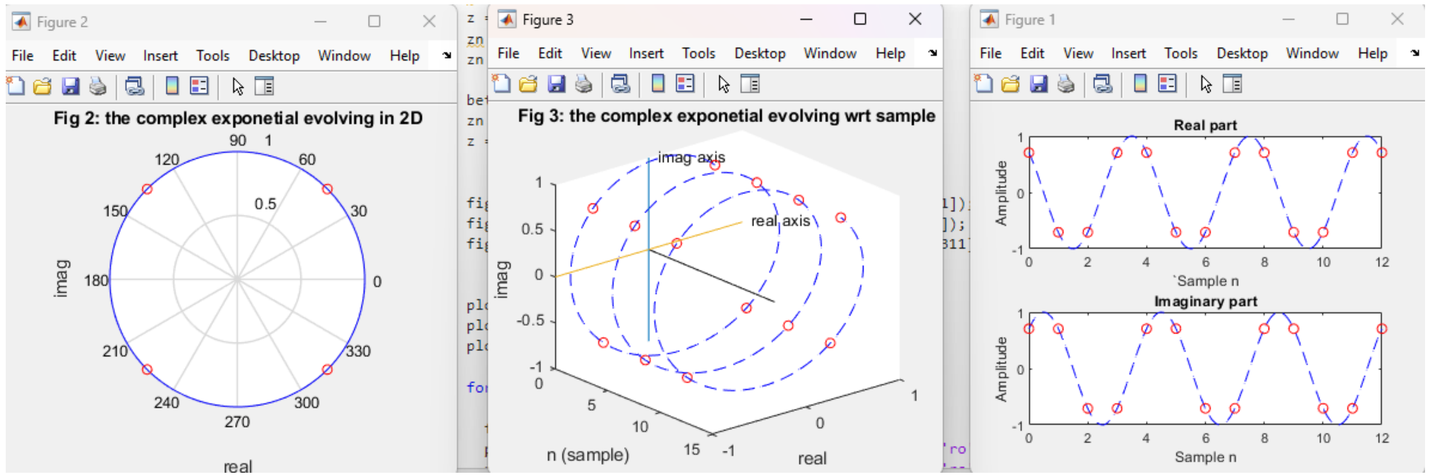
For $\omega = 0..+\pi$, as t increases
then the phasor (green dot) rotates
anticlockwise.

$$x_2(t) = e^{j(-\omega)t}$$

For above, given $(-\omega)$ is between $0..-\pi$,
and as t increases, then the phasor (green dot) rotates
clockwise.

8A.3.1: complex vector scaled by a complex number

$$\boldsymbol{b} = \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix}, \text{ and } \beta = 0.7071 + j0.7071, \text{ what is } \beta\boldsymbol{b}?$$



Ans: $\beta = 0.7071 + j0.7071j = 1e^{\frac{j\pi}{4}}$
the input sequence is scaled by 1,
and phase shifted by $+\frac{j\pi}{4}$.

$$\beta\boldsymbol{b} = (1e^{\frac{j\pi}{4}}) \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix} = (1e^{\frac{j\pi}{4}}) \begin{bmatrix} 1e^{\frac{j\pi}{2}0} \\ 1e^{\frac{j\pi}{2}1} \\ 1e^{\frac{j\pi}{2}2} \\ 1e^{\frac{j\pi}{2}3} \end{bmatrix} = \begin{bmatrix} 1e^{\frac{j1\pi}{4}} \\ 1e^{\frac{j3\pi}{4}} \\ 1e^{\frac{j5\pi}{4}} \\ 1e^{\frac{j7\pi}{4}} \end{bmatrix}$$

Note: easier to view multiplication in the complex exponential form.
Conclusion: since $|\beta| = 1$, the operation $\beta\boldsymbol{b}$ simply phase-shifted the original vector $\boldsymbol{b}$ by j$\frac{\pi}{4}$

Content

8A.3: Complex vectors and DFT

8A.3.1 - Complex vector
- definition
- examples, visualization
- scaling by a complex number

8A.3.2 Inner product and Inverse of complex matrixes

8A.3.3 - Motivation: Decomposing a signal into sum of sinusoids
- DFT is orthogonal decomposition
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- Example

8A.3.2: Inner products definitions

Real Inner Product

Let V be a vector space over a real field $\mathbb{F}$.

A **(real) inner product** is a mapping $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ that satisfies the real inner product axioms:

$(1')$	: Symmetry	$\forall x, y \in V$:	$\langle x, y \rangle = \langle y, x \rangle$
(2)	: Linearity in first argument	$\forall x, y \in V, \forall a \in \mathbb{F}$:	$\langle ax + y, z \rangle = a \langle x, z \rangle + \langle y, z \rangle$
(3)	: Non-Negative Definiteness	$\forall x \in V$:	$\langle x, x \rangle \in \mathbb{R}_{\geq 0}$
(4)	: Positiveness	$\forall x \in V$:	$\langle x, x \rangle = 0 \implies x = \mathbf{0}_V$

Here, field F is the set of real or complex numbers

If $F = Real$, then the familiar dot product is the special case of inner product.

Complex Inner Product

Let V be a vector space over a complex field $\mathbb{F}$.

A **(complex) inner product** is a mapping $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ that satisfies the complex inner product axioms:

(1)	: Conjugate Symmetry	$\forall x, y \in V$:	$\langle x, y \rangle = \overline{\langle y, x \rangle}$
(2)	: Linearity in first argument	$\forall x, y \in V, \forall a \in \mathbb{F}$:	$\langle ax + y, z \rangle = a \langle x, z \rangle + \langle y, z \rangle$
(3)	: Non-Negative Definiteness	$\forall x \in V$:	$\langle x, x \rangle \in \mathbb{R}_{\geq 0}$
(4)	: Positiveness	$\forall x \in V$:	$\langle x, x \rangle = 0 \implies x = \mathbf{0}_V$

1) https://proofwiki.org/wiki/Definition:Inner_Product
2) https://math.stackexchange.com/questions/476738/difference-between-dot-product-and-inner-product

8A.3.2: What does inner product reveal? Example 1: orthogonal projection in real vector space

Orthognal projection: (6.2.2)

$$Proj_{\boldsymbol{b}}\boldsymbol{v} = \left(\frac{\boldsymbol{v}.\boldsymbol{b}}{||\boldsymbol{b}||^2}\right)\boldsymbol{b}$$

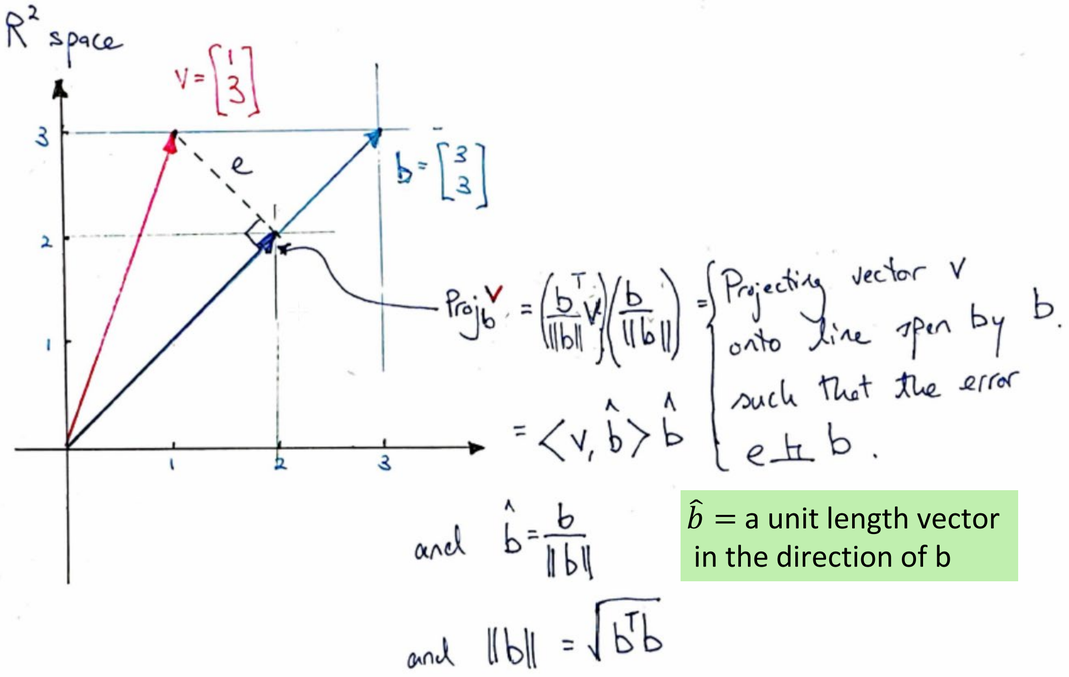
$$Proj_{\boldsymbol{b}}\boldsymbol{v} = <\boldsymbol{v}, \hat{\boldsymbol{b}}>\hat{\boldsymbol{b}}$$

$$<\boldsymbol{v}, \hat{\boldsymbol{b}}> = \hat{\boldsymbol{b}}^{\boldsymbol{T}}\boldsymbol{v} = \sum_{i=0}^{N-1} v_i \hat{b_i}$$

Where $\hat{\boldsymbol{b}} = \frac{\boldsymbol{b}}{||\boldsymbol{b}||}$, $\hat{\boldsymbol{b}}^{\boldsymbol{T}}$ = transpose of $\hat{\boldsymbol{b}}$

and

$$||\boldsymbol{b}|| = \text{norm}(\boldsymbol{b}) = \text{length}(\boldsymbol{b}) = \sqrt{\boldsymbol{b}^{\boldsymbol{T}}\boldsymbol{b}}$$



$\hat{b}$ = a unit length vector in the direction of b

10

Inner product tells you how much of one vector is pointing in the direction of another one. If e is a unit vector then $< f, e >$ is the component of f in the direction of e and the vector component of f in the direction e is $< f, e > e$. The vectors f and e are orthogonal when $< f, e >= 0$, in which case f has zero component in the direction e.

https://en.wikibooks.org/wiki/Linear_Algebra/Orthogonal_Projection_Onto_a_Line
https://math.stackexchange.com/questions/988886/what-does-inner-product-actually-mean

8A.3.2: Example 2: Inner product of complex vectors:

What is $< \boldsymbol{a}, \boldsymbol{b} >$ and $||\boldsymbol{b}||$? For complex vectors

Introduction Canonical inner product of complex vectors

$$< \boldsymbol{a}, \boldsymbol{b} > = \sum_{i=0}^{N-1} a_i \overline{b_i} = \boldsymbol{b^H a}$$

matrix/vector multiplication

Where

$\boldsymbol{a}, \boldsymbol{b} \in \boldsymbol{C^N}$

$\boldsymbol{b^H}$ represents conjugate transpose of $\boldsymbol{b}$

$\overline{b_i}$ represents the $\boldsymbol{b's}$ i^{th} element conjugated,

a_i represents the i^{th} element in vectors $\boldsymbol{a}$

Example:

$$\boldsymbol{a} = \begin{bmatrix} 1+j \\ 2-j \\ 3-1j \\ 4+1j \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ +j \\ -1 \\ -j \end{bmatrix}, \quad \overline{\boldsymbol{b}} = \begin{bmatrix} 1 \\ -j \\ -1 \\ +j \end{bmatrix}, \quad \boldsymbol{b^H} = [1, \ -j, \ -1, \ j \ ]$$

$$\begin{aligned} < \boldsymbol{a}, \boldsymbol{b} > &= (1+j)(1) + (2-j)(-j) + (3-1j)(-1) + (4+1j)(+j) \\ &= (1+j) \ + (-2j+j^2) \ + (-3+j) \ + (4j+j^2) \\ &= -4 \ + 4j \\ &= \boldsymbol{b^H a} \end{aligned}$$

Why 2nd term is conjugated for complex inner product?

Note Norm (b) must be a real number

$$= ||\boldsymbol{b}|| = \sqrt{< \boldsymbol{b}, \boldsymbol{b} >} = \sqrt{\boldsymbol{b^H b}} = \sqrt{\sum_{i=0}^{N-1} b_i b_i}$$

To achieve this for complex vectors, the second entry needs to be conjugated when calculating inner product to ensure that the result is real such that $||\boldsymbol{b}|| = \sqrt{< \boldsymbol{b}, \boldsymbol{b} >} = \sqrt{\boldsymbol{b^H b}}$ will produce a real number.

Example: Find $||\boldsymbol{b}||$

$$< \boldsymbol{b}, \boldsymbol{b} > = \sum_{i=0}^{N-1} b_i \overline{b_i}$$

$$\begin{aligned} &= (1(1)) + (j(-j)) \ + (-1)(-1) \ + (-j)(j) \\ &= 1 - j^2 \ + 1 - j^2 \\ &= 4 \end{aligned}$$

Therefore, $||\boldsymbol{b}|| = \sqrt{< \boldsymbol{b}, \boldsymbol{b} >} = \sqrt{4}$

8A.3.2: Example 2: Inner product of complex vectors:

Find $Proj_{\boldsymbol{b}}\boldsymbol{v} = < \boldsymbol{v}, \hat{\boldsymbol{b}} > \hat{\boldsymbol{b}}$

Given $\boldsymbol{b}$ and $\boldsymbol{v} \in \boldsymbol{C^4}$, where

$$\boldsymbol{b} = \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix}, \; N = 4, \qquad \boldsymbol{b^H} = \; [1, \quad -j, \quad -1, \quad j \;]$$

$$\mathbf{v} = \; (1e^{j\frac{\pi}{4}}) \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix} = \left(1e^{j\frac{\pi}{4}}\right) \boldsymbol{b}$$

Find:

$$Proj_{\boldsymbol{b}}\boldsymbol{v} = < \boldsymbol{v}, \hat{\boldsymbol{b}} > \hat{\boldsymbol{b}}$$

Lets find $\hat{\boldsymbol{b}}$, the unit vector pointing in direction of b first.

$$\hat{\boldsymbol{b}} = \frac{\boldsymbol{b}}{||\boldsymbol{b}||}$$

Because **b**'s element all has length == 1, and it has N elements, then $||\boldsymbol{b}|| = \sqrt{< \boldsymbol{b}, \boldsymbol{b} >} = \sqrt{\boldsymbol{b^H b}} = \sqrt{\boldsymbol{N}}$.

Therefore,

$$\hat{\boldsymbol{b}} = \frac{\boldsymbol{b}}{||\boldsymbol{b}||} = \frac{\boldsymbol{b}}{\sqrt{\boldsymbol{N}}}$$

$$\begin{aligned} < \boldsymbol{v}, \hat{\boldsymbol{b}} > \; &= \hat{\boldsymbol{b}}^H \boldsymbol{v} \\ &= \left(\frac{\boldsymbol{b}}{\sqrt{N}}\right)^H \boldsymbol{v} \\ &= \frac{\boldsymbol{b^H}}{\sqrt{N}} \left(1e^{\frac{j\pi}{4}}\right)\boldsymbol{b}, \qquad \text{// Note: } \boldsymbol{b^H b} = \boldsymbol{N} \\ &= \frac{N}{\sqrt{N}} \left(1e^{\frac{j\pi}{4}}\right) \end{aligned}$$

And $Proj_{\boldsymbol{b}}\boldsymbol{v} = < \boldsymbol{v}, \hat{\boldsymbol{b}} > \hat{\boldsymbol{b}}$

$$= \frac{N}{\sqrt{N}} \left(1e^{\frac{j\pi}{4}}\right) \hat{\boldsymbol{b}}$$

$$= \frac{N}{\sqrt{N}} \left(1e^{\frac{j\pi}{4}}\right) \left(\frac{\boldsymbol{b}}{\sqrt{\boldsymbol{N}}}\right)$$

$$= \left(1e^{\frac{j\pi}{4}}\right) \boldsymbol{b}$$

Conclusions:

Projecting $\boldsymbol{v}$ onto $\boldsymbol{b}$
has no loss of information!

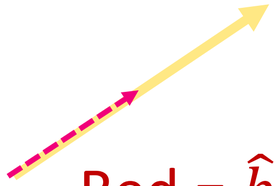
Why: we can think of vectors $\boldsymbol{v}$ and $\boldsymbol{b}$ are vectors in C^N and these 2 vectors have *the same direction but different length!*

Hence, when we use $\boldsymbol{b}$ to represent $\boldsymbol{v}$, we only need to find the correct scaling $< \boldsymbol{v}, \hat{\boldsymbol{b}} >$ to scale $\hat{\boldsymbol{b}}$ to $\boldsymbol{v}$

8A.3.2 Conclusions:

1) When we project a complex vector $\boldsymbol{v}$ to another complex vector $\boldsymbol{b}$
such that both $\boldsymbol{v}$ and $\boldsymbol{b}$ are rotating at the same frequency (but has different phase),
We still can get back the same vector $\boldsymbol{v}$.
Conceptually: vectors $\boldsymbol{v}$ and $\boldsymbol{b}$ have the same direction in C^N vector space.

- The inner product of $\boldsymbol{v}$ with $\widehat{\boldsymbol{b}}$ will reveal the magnitude and phase-shift that's required to construct (synthesize) the vector $\boldsymbol{v}$ from $\widehat{\boldsymbol{b}}$



Red = $\hat{b}$

Yellow = $\boldsymbol{v}$

2) Key-insight of Fourier Transform: the set of vectors $\boldsymbol{b}$ **are complex exponentials and not arbitrary basis.** Fourier transform decompose the original signal $\boldsymbol{v}$ **into sum of scaled complex exponentials with different frequencies.**

Content

8A.3: Complex vectors and DFT

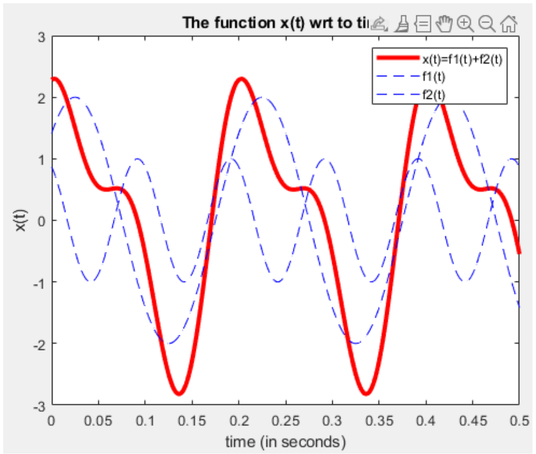
8A.3.1 - Complex vector
- definition
- examples, visualization
- scaling by a complex number

8A.3.2 Inner product and Inverse of complex matrixes

8A.3.3 - Motivation: Decomposing a signal into sum of sinusoids
- DFT is orthogonal decomposition
- The analysis and synthesis equations of DFT
- Example

8A3.3: Motivation: Fourier Transform
Decomposing a signal into sum of sinusoids

$$x(t) = \sum_{\omega} |X(\omega)| \cos(\omega t + \angle X(\omega))$$



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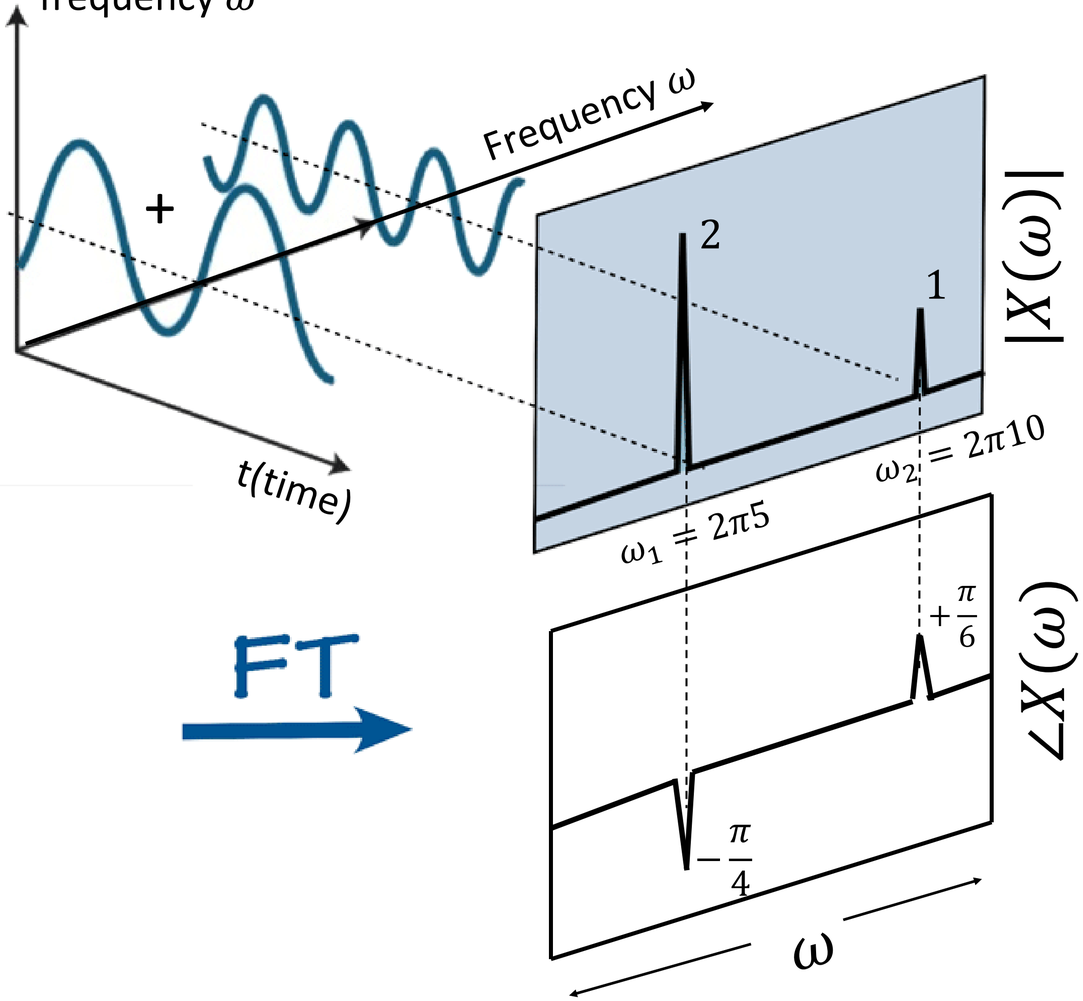
Sinusoids with frequency ω



FT

Warning: the spectrum in this example is based on cosine, but in actual DFT, it is based on complex exponentials

Magnitude spectrum

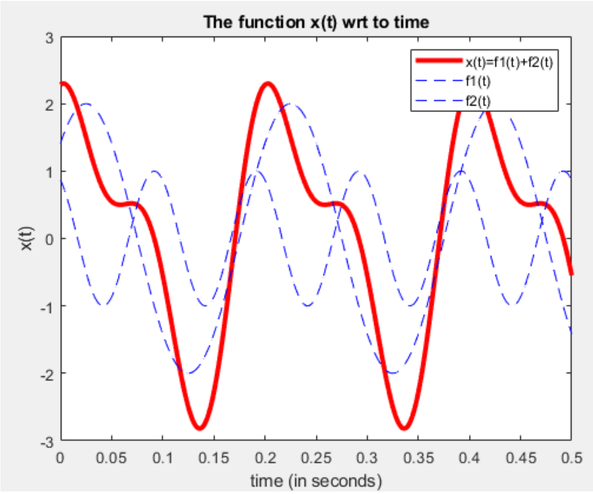
Phase spectrum

Time Domain

Input $x(t) = 2\cos\left(2\pi 5t - \frac{\pi}{4}\right) + 1\cos\left(2\pi 10t + \frac{\pi}{6}\right)$

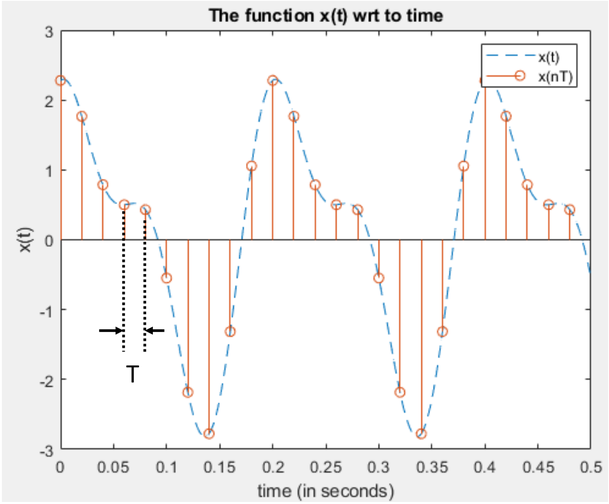
Note: here frequency ω is continuous.
Unit = radian/sec

8A3.3: Example – sampling



$$x(t) = 2\cos\left(2\pi 5t - \frac{\pi}{4}\right) + 1\cos\left(2\pi 10t + \frac{\pi}{6}\right)$$

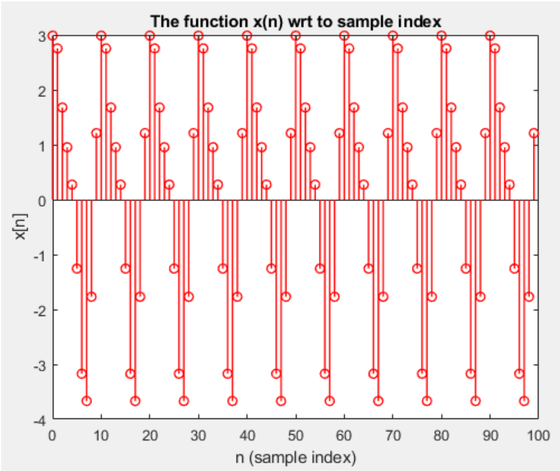
Continuous time



$$x(nT) = 2\cos\left(2\pi 5\left(\frac{n}{50}\right) - \frac{\pi}{4}\right) + 1\cos\left(2\pi 10\left(\frac{n}{50}\right) + \frac{\pi}{6}\right)$$

$$T = \frac{1}{50},\ n \in \mathrm{Z}$$

Sampled version at time step $T = \frac{1}{50}$



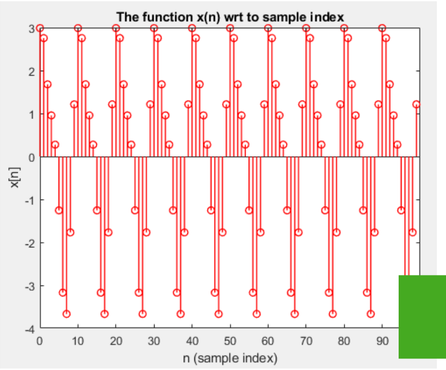
$$x(n) = 2\cos\left(0.2\pi n - \frac{\pi}{4}\right) + 1\cos\left(0.4\pi n + \frac{\pi}{6}\right)$$

Discrete time signal – time is now represented as sample index

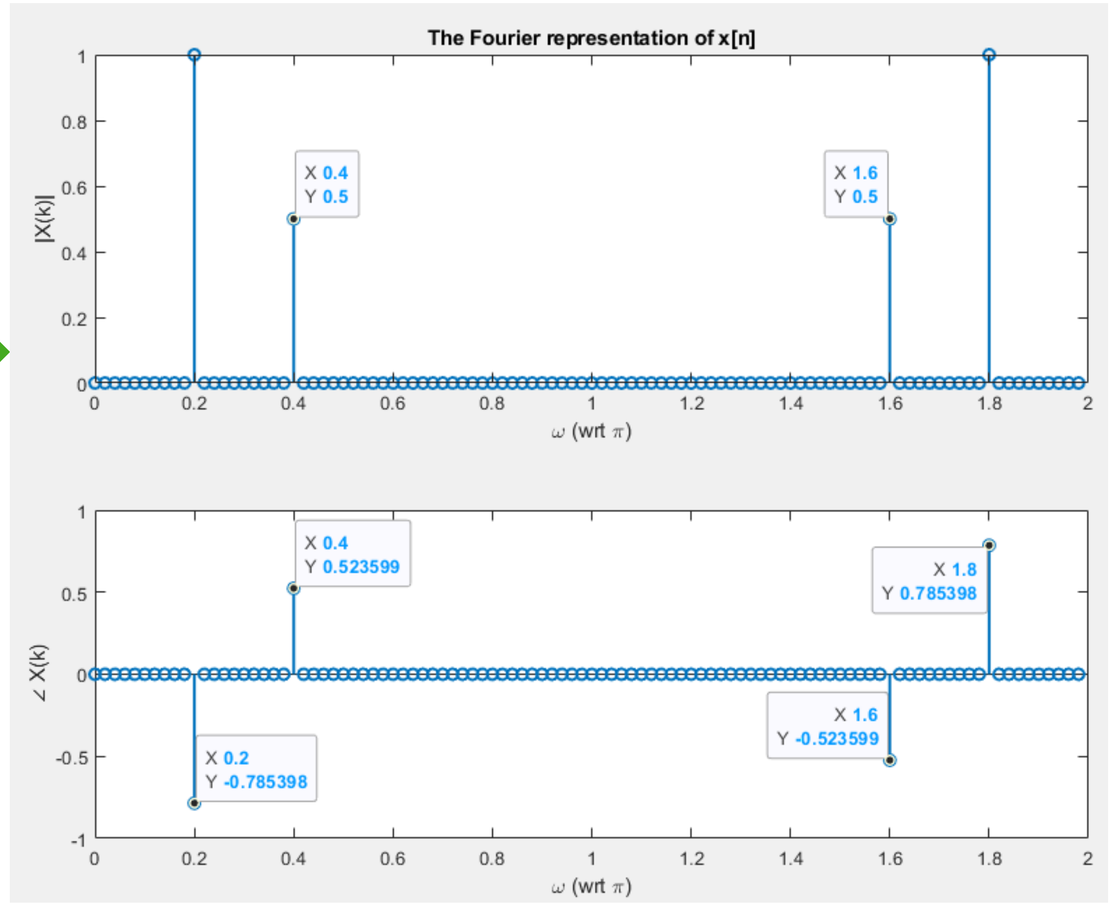
x[n] = Extracted 100 samples

Code= plot_DFT_example3.m

8A3.3: Discrete Time Fourier Transform reveals how x[n] is decomposed to complex exponentials



FT



100 samples of Discrete Time sequence

4 *complex exponentials*

(1) $1e^{j(0.2\pi n-\frac{\pi}{4})}$

(2) $0.5e^{j(0.4\pi n+\frac{\pi}{6})}$

(3) $0.5e^{j(1.6\pi n-\frac{\pi}{6})}$

$= 0.5e^{j(-0.4\pi n-\frac{\pi}{6})}$

$= 0.5e^{-j(0.4\pi n+\frac{\pi}{6})}$

(4) $1e^{j(1.8\pi n+\frac{\pi}{4})}$

$= 1e^{j(-0.2\pi n+\frac{\pi}{4})}$

$= 1e^{-j(0.2\pi n-\frac{\pi}{4})}$

Note: $2\cos\left(0.2\pi n\ -\frac{\pi}{4}\right) = 1e^{j(0.2\pi n-\frac{\pi}{4})} + 1e^{-j(0.2\pi n-\frac{\pi}{4})}$

$And\ :1\cos\left(0.4\pi n+\frac{\pi}{6}\right) = 0.5e^{j(0.4\pi n+\frac{\pi}{6})} + 0.5e^{-j(0.4\pi n+\frac{\pi}{6})}$

Note: here frequency ω is for discrete time,
unit = radian/sample

8A.3.3 DFT is orthogonal decompositon

Review:

Chap. No : **6.2.3**
Lecture : **Orthogonality**
Topic : **Orthogonality**
Concept : **Orthogonal Sets & Orthogonal Basis**

THEOREM 5

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \qquad (j = 1, \ldots, p)$$

PROOF As in the preceding proof, the orthogonality of $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ shows that

$$\mathbf{y}\cdot\mathbf{u}_1 = (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p)\cdot\mathbf{u}_1 = c_1(\mathbf{u}_1\cdot\mathbf{u}_1)$$

Since $\mathbf{u}_1\cdot\mathbf{u}_1$ is not zero, the equation above can be solved for c_1. To find c_j for $j = 2, \ldots, p$, compute $\mathbf{y}\cdot\mathbf{u}_j$ and solve for c_j. ■

Interpretations:

1) The basis is orthogonal, hence the contribution of each $\boldsymbol{u}_{j=1..p}$ is independent of other $\boldsymbol{u}_{j=1..p}$.
2) Therefore, $|c_j|$ tells us how much $\boldsymbol{u}_j$ contributes to $\boldsymbol{y}$ in terms of magnitude.
3) In the DFT representation, we will have DFT coefficients $c_j \in C$ and Fourier basis $\boldsymbol{u}_j \in C^N$, there the value of $\angle c_j$ tells us how much phase shift is required of $\boldsymbol{u}_j$ to phase-align to y.

8A.3.3 Introducing DFT Synthesis equation, $x = \left(\frac{1}{N}\right) W^H X$

DFT Synthesis equation: representing $x[n]$ as sum of scaled+phase-shifted orthogonal complex exponentials vectors

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}, \quad W_N^{-kn} = e^{j\frac{2\pi}{N}kn}, \quad n,k = 0..(N-1)$$

Inverse DFT

Signal vector $\boldsymbol{x} = \left(\frac{1}{N}\right) \boldsymbol{W^H X}$

$X[k]$ represents the required $|.|$ and $\angle$ to scaled columns of W^H to represent $\boldsymbol{x}$

$$\underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\text{Signal vector } \boldsymbol{x}} = \frac{1}{N} \underbrace{\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{j(2\pi/N)} & e^{j(4\pi/N)} & \cdots & e^{j(2(N-1)\pi/N)} \\ 1 & e^{j(4\pi/N)} & e^{j(8\pi/N)} & \cdots & e^{j(4(N-1)\pi/N)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j(2(N-1)\pi/N)} & e^{j(4(N-1)\pi/N)} & \cdots & e^{j(2(N-1)(N-1)\pi/N)} \end{bmatrix}}_{\text{IDFT Matrix } \boldsymbol{W^H} = W_N^{-kn}} \underbrace{\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\text{DFT coeff } \boldsymbol{X}}$$

$n = 0..(N-1)$ (vertical, for x) $\quad k = 0..(N-1)$ (horizontal, for W^H) $\quad k = 0..(N-1)$ (vertical, for X)

This scaling makes $\left(\frac{1}{N}\boldsymbol{W}^H\right)\boldsymbol{W} = \boldsymbol{I}$, hence

$$\boldsymbol{x} = \left(\frac{1}{N}\boldsymbol{W}^H\right)\boldsymbol{X}$$
$$= \left(\frac{1}{N}\boldsymbol{W}^H\right)\boldsymbol{W}\boldsymbol{x}$$
$$= \boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$$

Warning: the place of scaling (1/N) is not uniform. But in DFT, 1/N is placed in the inverse DFT equation

Let $x \in R^N$ be signal vector
Let $X \in C^N$ be DFT coefficients vector

Let $W \in C^{NxN}$ be DFT matrix, and
$W^H \in C^{NxN}$ be IDFT matrix

where W^H is conjugate transpose of W

8A.3.3 The Fourier Basis: columns of $W^H = \mathrm{W_N^{-kn}}$: Example N = 8

The columns of the Synthesis Equation of W^H

The complex exponential at digital angular frequency $\Omega = \frac{2\pi}{N} k \ (\frac{rad}{sample})$

$N = 8,$
$n = 0..(N-1)$

Note:

$$\mathrm{W_N^{-kn}} = e^{j\frac{2\pi}{N}kn}, \qquad n,k = 0..(N-1)$$

$$\mathrm{W_N^{kn}} = e^{-j\frac{2\pi}{N}kn}, \qquad n,k = 0..(N-1)$$

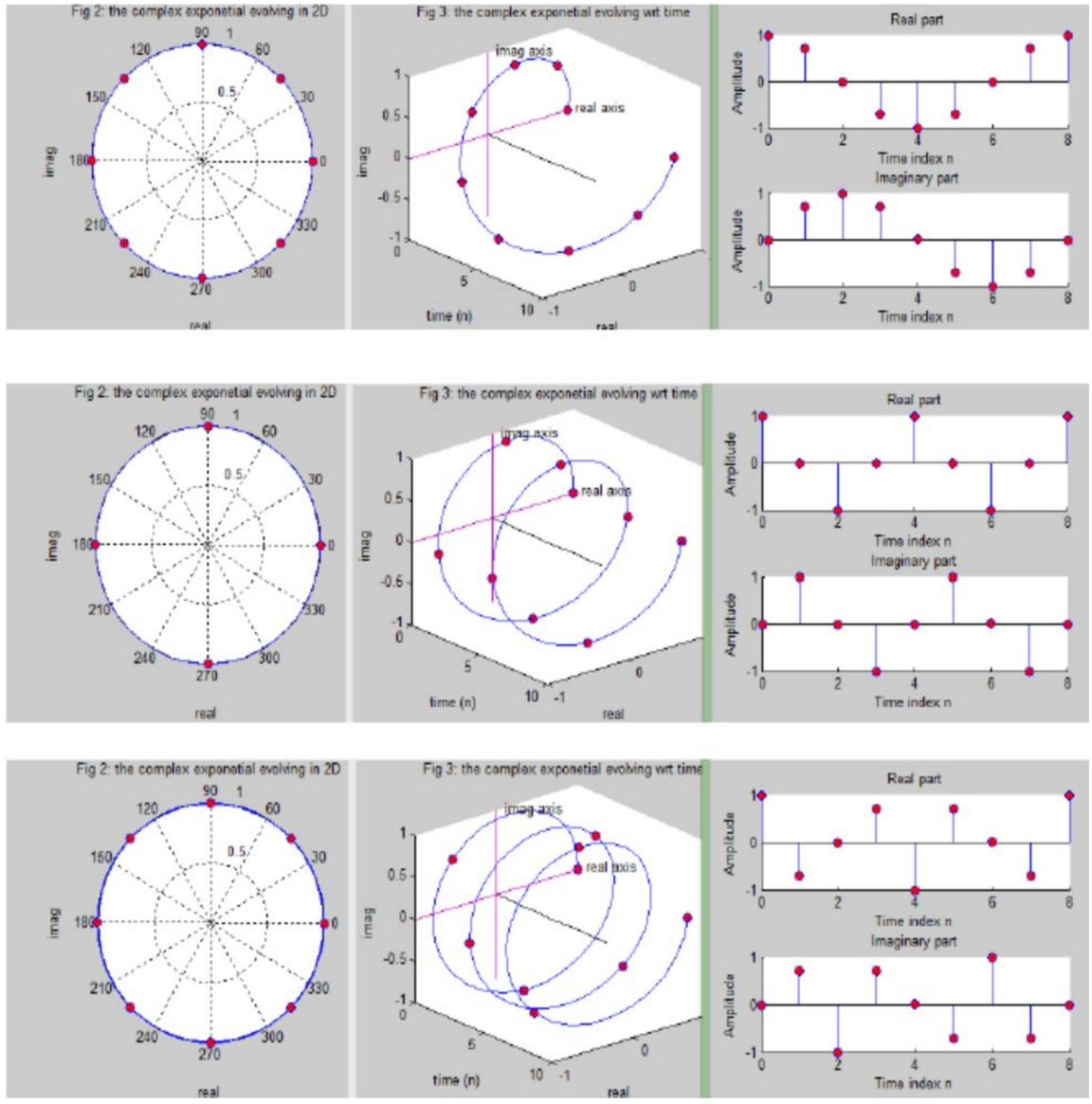
$$W_8^{-1n} = e^{j\frac{2\pi}{8}(1)n}$$

$$W_8^{-2n} = e^{j\frac{2\pi}{8}(2)n}$$

$$W_8^{-3n} = e^{j\frac{2\pi}{8}(3)n}$$

$$W_8^{-1n} = \begin{bmatrix} 1\angle 0.25\pi\ 0 \\ 1\angle 0.25\pi\ 1 \\ 1\angle 0.25\pi\ 2 \\ 1\angle 0.25\pi\ 3 \\ 1\angle 0.25\pi\ 4 \\ 1\angle 0.25\pi\ 5 \\ 1\angle 0.25\pi\ 6 \\ 1\angle 0.25\pi\ 7 \end{bmatrix} \quad n = 0..(N-1) \qquad W_8^{-2n} = \begin{bmatrix} 1\angle 0.5\pi\ 0 \\ 1\angle 0.5\pi\ 1 \\ 1\angle 0.5\pi\ 2 \\ 1\angle 0.5\pi\ 3 \\ 1\angle 0.5\pi\ 4 \\ 1\angle 0.5\pi\ 5 \\ 1\angle 0.5\pi\ 6 \\ 1\angle 0.5\pi\ 7 \end{bmatrix}$$

$$n = 0..(N-1)$$



8A.3.3 The DFT Analysis Equation and DFT Matrix W

DFT Analysis equation: Finding $X[k]$, the DFT coefficients.

DFT coeff $\boldsymbol{X} = \boldsymbol{W}\,\boldsymbol{x}$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, \qquad W_N^{kn} = e^{-j\frac{2\pi}{N}kn}$$

Discrete Fourier Transform

W is the DFT matrix

Note:

$$\boldsymbol{W^H W} = N\boldsymbol{I}$$

$\boldsymbol{I}$ **= identity matrix**

Because:

1) Columns of W are orthogonal
2) Each element of W has |.| = 1
3) Therefore $w_j^H w_k$ = N (when $j == k$)
   and $w_j^H w_k$ = 0 (when $j \neq k$)

Note:
$\boldsymbol{W^H}$ is the conjugate transpose of $\boldsymbol{W}$

$k = 0..(N-1)$ $\qquad$ $n = 0..(N-1)$

$k = 0..(N-1)$

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{-j(2\pi/N)} & e^{-j(4\pi/N)} & \cdots & e^{-j(2(N-1)\pi/N)} \\ 1 & e^{-j(4\pi/N)} & e^{-j(8\pi/N)} & \cdots & e^{-j(4(N-1)\pi/N)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{-j(2(N-1)\pi/N)} & e^{-j(4(N-1)\pi/N)} & \cdots & e^{-j(2(N-1)(N-1)\pi/N)} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$n = 0..(N-1)$

DFT coeff $\boldsymbol{X}$ $\qquad$ DFT Matrix $\boldsymbol{W} = W_N^{kn}$ $\qquad$ Signal vector $\boldsymbol{x}$

8A.3.3) Visualizing the W matrix as rotating phasors

W is the Analysis matrix, also commonly known as the Fourier matrix or the DFT matrix.
Each row of W represents the k^{th} harmonic complex exponential sequence.

Visually, the rows of W are rotating clockwise (when no aliasing occurs)
because its digital frequency at the k-th column is $\frac{-2\pi}{N}k$ (rad/sample). Example below is for N = 8

Ref:
https://en.wikipedia.org/wiki/DFT_matrix

Fig on bottom left representing the W matrix in polar coordinates to visualize the complex exponential rotation.
The circle is the unit circle.

Note the –ve sign in the complex exponential, this makes the sequence rotates clockwise.
This is different to the synthesis matrix W^H whose columns are rotating anti-clockwise.

$n = 0..(N-1)$

$k = 0..(N-1)$

$$W = \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 \\ \omega^0 & \omega^1 & \omega^2 & \omega^3 & \omega^4 & \omega^5 & \omega^6 & \omega^7 \\ \omega^0 & \omega^2 & \omega^4 & \omega^6 & \omega^8 & \omega^{10} & \omega^{12} & \omega^{14} \\ \omega^0 & \omega^3 & \omega^6 & \omega^9 & \omega^{12} & \omega^{15} & \omega^{18} & \omega^{21} \\ \omega^0 & \omega^4 & \omega^8 & \omega^{12} & \omega^{16} & \omega^{20} & \omega^{24} & \omega^{28} \\ \omega^0 & \omega^5 & \omega^{10} & \omega^{15} & \omega^{20} & \omega^{25} & \omega^{30} & \omega^{35} \\ \omega^0 & \omega^6 & \omega^{12} & \omega^{18} & \omega^{24} & \omega^{30} & \omega^{36} & \omega^{42} \\ \omega^0 & \omega^7 & \omega^{14} & \omega^{21} & \omega^{28} & \omega^{35} & \omega^{42} & \omega^{49} \end{bmatrix}$$

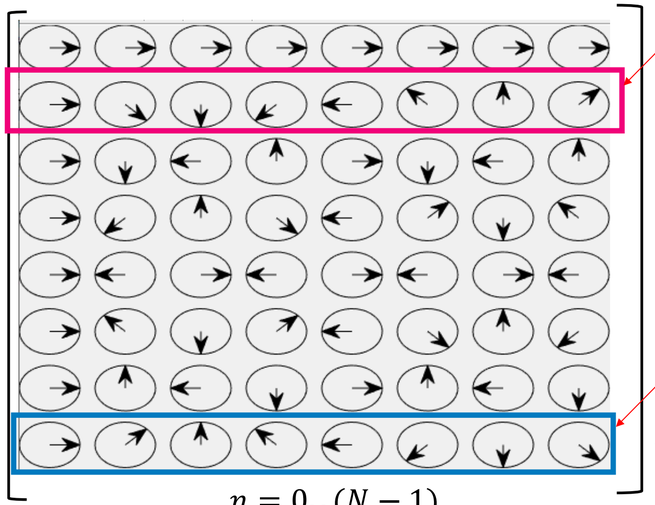
$$w^n = \left(e^{-j\frac{2\pi}{N}}\right)^n$$

-ve radian/sample rotating clockwise

Row 1 of W

$$(W_8^{1n}) = [w^0, \quad w^1, \quad w^2, \quad w^3, \quad \dots, w^7] = [1\angle\left(-\frac{\pi}{4}0\right), 1\angle\left(-\frac{\pi}{4}1\right), 1\angle\left(-\frac{\pi}{4}2\right), 1\angle\left(-\frac{\pi}{4}3\right), \dots, 1\angle\left(-\frac{\pi}{4}7\right)]$$

$W =$



$k = 0..(N-1)$

Row 7 of W (Angle $-\frac{7}{4}\pi \; same \; as \frac{\pi}{4}$), see that row 1 and row 7 are rotating same frequency BUT opposite directions

$$(W_8^{7n}) = \left[1\angle\left(-\frac{\pi}{4}7.0\right), \; 1\angle\left(-\frac{\pi}{4}7.1\right), \; 1\angle\left(-\frac{\pi}{4}7.2\right), \; 1\angle\left(-\frac{\pi}{4}7.3\right), \dots, 1\angle(-\frac{\pi}{4}7.7)\right]$$

$$= \left[1\angle\left(\frac{\pi}{4}1.0\right), \quad 1\angle\left(\frac{\pi}{4}1.1\right), \quad 1\angle\left(\frac{\pi}{4}2\right), \quad 1\angle\left(\frac{\pi}{4}3\right), \quad \dots, \quad 1\angle(\frac{\pi}{4}7)\right]$$

$n = 0..(N-1)$

8A.3.3) Example: calculating DFT and IDFT for 4-pt sequence

Calculate the four-point DFT of the aperiodic signal $x[k]$ considered in Example one.

Solution

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{-j(2\pi/N)} & e^{-j(4\pi/N)} & e^{-j(6\pi/N)} \\ 1 & e^{-j(4\pi/N)} & e^{-j(8\pi/N)} & e^{-j(12\pi/N)} \\ 1 & e^{-j(6\pi/N)} & e^{-j(12\pi/N)} & e^{-j(18\pi/N)} \end{bmatrix}}_{\text{DFT matrix: W}} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{-j(2\pi/4)} & e^{-j(4\pi/4)} & e^{-j(6\pi/4)} \\ 1 & e^{-j(4\pi/4)} & e^{-j(8\pi/4)} & e^{-j(12\pi/4)} \\ 1 & e^{-j(6\pi/4)} & e^{-j(12\pi/4)} & e^{-j(18\pi/4)} \end{bmatrix}}_{\text{DFT matrix: W}} \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3-j2 \\ -3 \\ 3+j2 \end{bmatrix}.$$

Calculate the inverse DFT of $X[r]$ considered in Example two.

Solution

Arranging the values of the DFT coefficients in the DFT vector X, we obtain

$X = [5\ 3-j2\ -3\ \ 3+j2]$.

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{j(2\pi/N)} & e^{j(4\pi/N)} & e^{j(6\pi/N)} \\ 1 & e^{j(4\pi/N)} & e^{j(8\pi/N)} & e^{j(12\pi/N)} \\ 1 & e^{j(6\pi/N)} & e^{j(12\pi/N)} & e^{j(18\pi/N)} \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix}$$

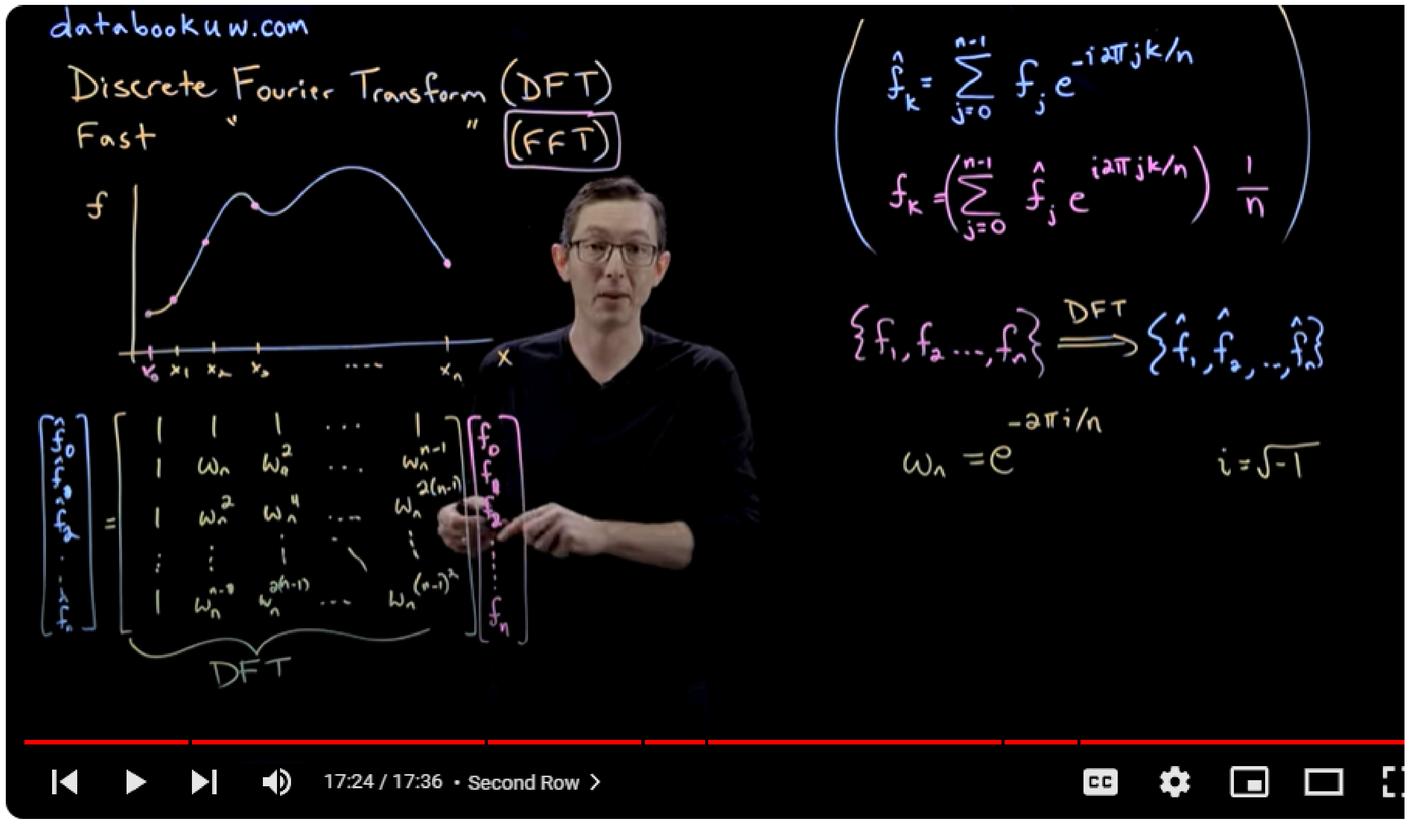
$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{j(2\pi/4)} & e^{j(4\pi/4)} & e^{j(6\pi/4)} \\ 1 & e^{j(4\pi/4)} & e^{j(8\pi/4)} & e^{j(12\pi/4)} \\ 1 & e^{j(6\pi/4)} & e^{j(12\pi/4)} & e^{j(18\pi/4)} \end{bmatrix} \begin{bmatrix} 5 \\ 3-j2 \\ -3 \\ 3+j2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 8 \\ 12 \\ -4 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

The above values for the DT sequence x[k] are the same as the ones obtained in Example On the left.

More work examples: Easy Electronics: N=4, Worked example
https://www.youtube.com/watch?v=dE9g72LlPdM

References

Ref: DFT lecture by Brunton



The Discrete Fourier Transform (DFT)

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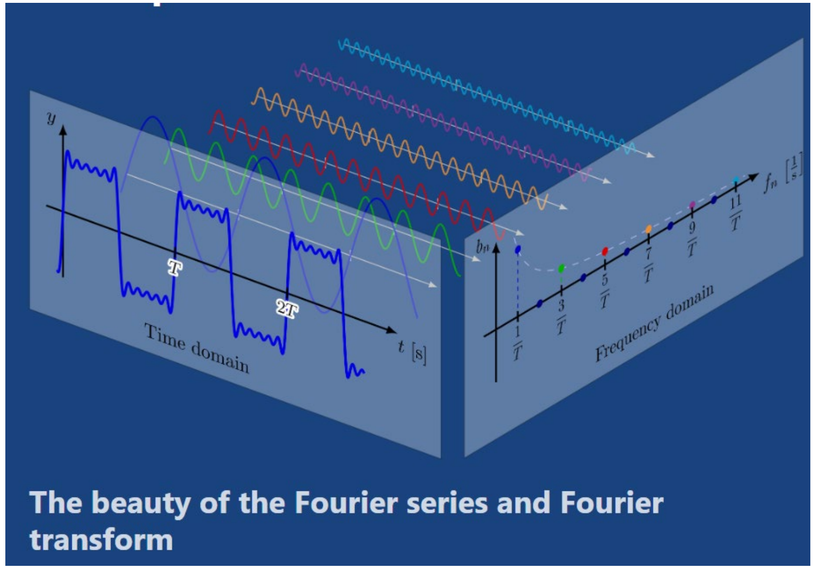
https://youtu.be/nl9TZanwbBk?si=nnJ99MHY64U8NMeV

References:

1) DFT Wikipedia: https://en.wikipedia.org/wiki/DFT_matrix
2) UBC Maths: https://ubcmath.github.io/MATH307/dft/dft.html
3) StatLect: https://www.statlect.com/matrix-algebra/discrete-Fourier-transform

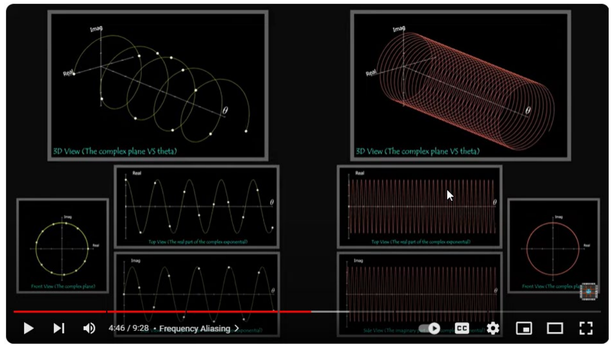
4) Lecture 8 DFT by Lecturer: Dr.Manal Khadhim. University of Iraq
https://uotechnology.edu.iq/dep-electromechanic/typicall/lecture%20interface/lecture/system3-4/4rth_class/SignalsandSystems/lec8.pdf

5)What is DFT matrix : https://www.statlect.com/matrix-algebra/discrete-Fourier-transform
6)What is unitary : https://www.statlect.com/matrix-algebra/unitary-matrix
7) Lecture 8 DFT by Lecturer: Dr.Manal Khadhim. University of Iraq.
https://uotechnology.edu.iq/dep-electromechanic/typicall/lecture%20interface/lecture/system3-4/4rth_class/SignalsandSystems/lec8.pdf
8) Duke: https://dibsmethodsmeetings.github.io/fourier-transforms/



The beauty of the Fourier series and Fourier transform

Ref: visualizing complex exponentials



Discrete And Continuous Time Complex Exponential Signal: a graphical introduction to DSP

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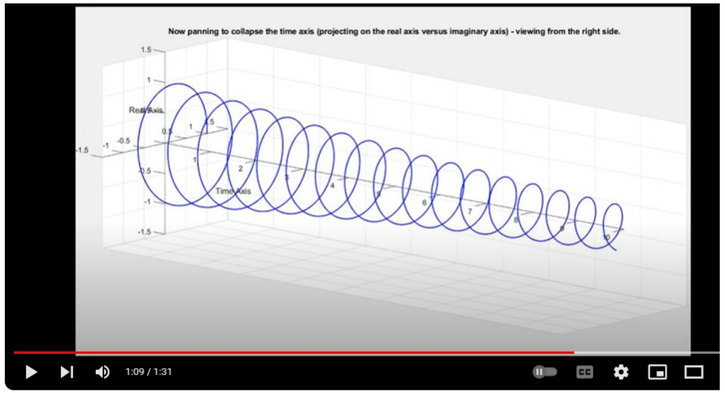

Visualization of a Continuous-Time General Complex Exponential Signal

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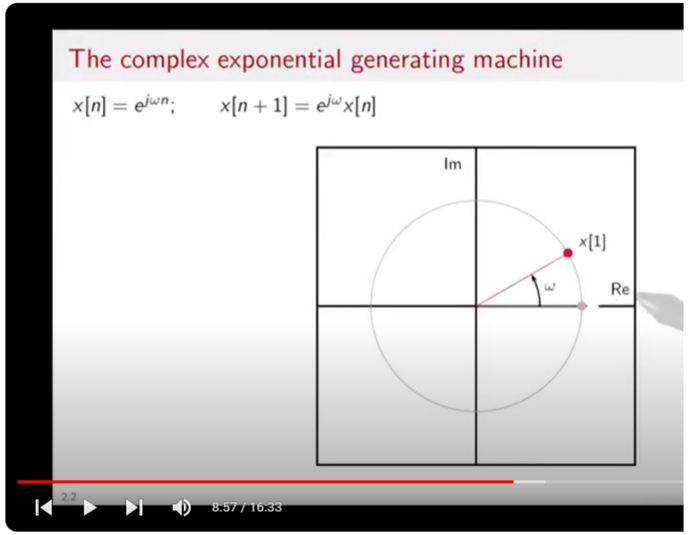

The complex exponential | Digital Signal Processing

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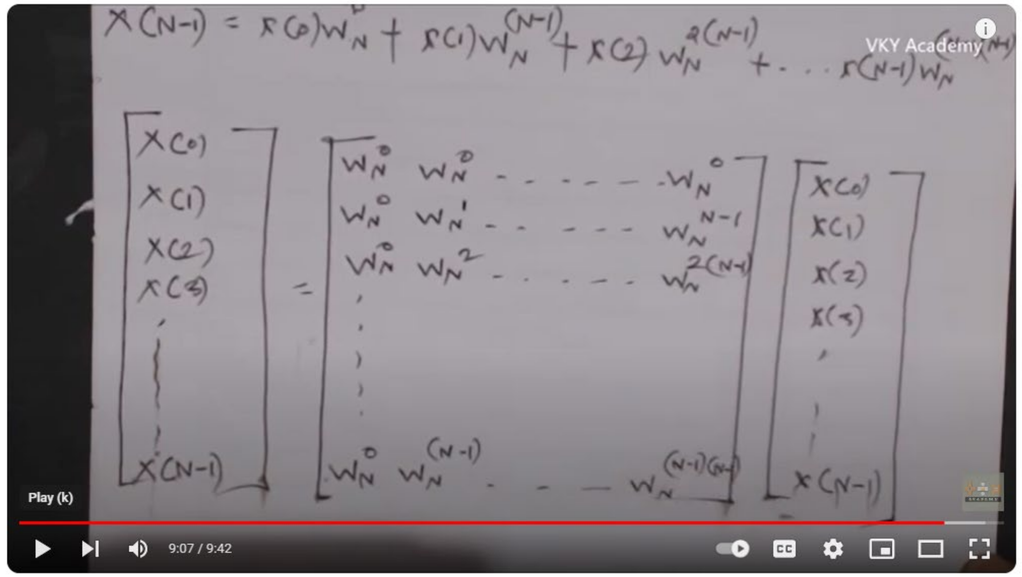
https://www.youtube.com/watch?v=geBSEZJ84Lg
The intuition is great!
But warning: did he draw the complex exponential rotating in wrong direction (?)

Classic explanation by Martin Vetterli

https://www.youtube.com/watch?v=ZzP63nHToa4&list=PLZBpE9maMtF92DepgJDIWCrAhwCFJ_ZPe&index=3

Ref: Video explaining DFT Analysis $X = Wx$



Lecture-4: DFT as Linear Transform (N X N) Linear Transform Matrix (Digital Signal Processing)

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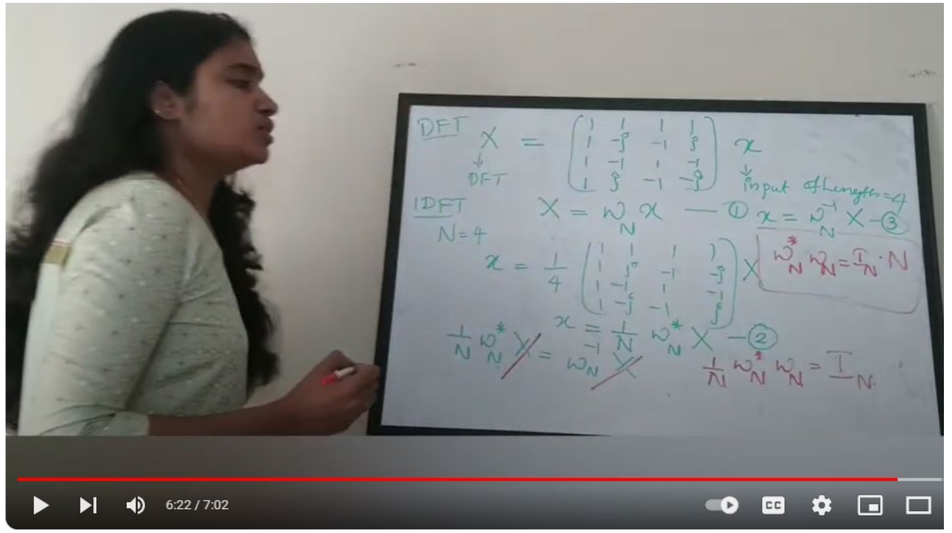
VKY Academy: Deriving matrix form of $X = Wx$
https://www.youtube.com/watch?v=5MRDBYm-Qsw



DFT as a Linear Transformation

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Easy Electronics: N=4, Worked example
https://www.youtube.com/watch?v=dE9g72LlPdM

Ref: Proof that cols of DFT matrix are orthogonal (1)

Proof that the Fourier basis is orthogonal.

Let b_l, b_k represents the col's of ω

$$b_k = \begin{bmatrix} \uparrow \\ e^{-j\frac{2\pi}{N}kn} \\ \downarrow \end{bmatrix}_{n=0..(N-1)} \in C^N \; ; \; \omega = e^{-j\frac{2\pi}{N}}$$

to show orthogonality,

$$\langle b_l, b_k \rangle = \begin{cases} 0 & \text{if } l \neq k \\ N & \text{if } l == k \end{cases}$$

$$= \sum_{n=0}^{N-1} b_{l,n} \overline{b_{k,n}} = b_k^H b_l$$

where $b_{k,n}$ = element n of col vector b_l.

$\overline{b_{k,n}}$ = conjugate of element n of b_l.

Let $\omega = e^{-j\frac{2\pi}{N}}$, $\omega^{ln} = e^{-j\frac{2\pi}{N}ln}$

$$\therefore \langle b_l, b_k \rangle = \sum_{n=0}^{N-1} b_{l,n} \overline{b_{k,n}} = b_k^H b_l$$

$$= \sum_{n=0}^{N-1} \omega^{ln} \omega^{-kn}$$

$$= \sum_{n=0}^{N-1} \omega^{(l-k)n}$$

(A) if $l == k$, $\therefore$

$$\langle b_l, b_l \rangle = b_l^H b_l = \sum_{n=0}^{N-1} \omega^{0n}$$

$$= \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}0}$$

$$= N \quad *$$

Ref: https://math.stackexchange.com/questions/2413218/proof-of-orthonormality-of-basis-of-dft

Ref: Proof that cols of DFT matrix are orthogonal (2)

3

(B) If $(l-k) \neq 0$

then $\sum_{n=0}^{N-1} \omega^{(l-k)n}$ is a finite geometric series

Proof: let $r = \omega^{(l-k)}$

$$\therefore \sum_{n=0}^{N-1} r^n$$

$$S_{N-1} = r^0 + r^1 + r^2 + \ldots + r^{N-1} \quad ; \quad r^0 = 1$$

$$r S_{N-1} = r^1 + r^2 + r^3 + \ldots + r^N$$

$$\therefore S_{N-1} - r S_{N-1} = 1 - r^N$$

$$\therefore S_n = \frac{1-r^N}{1-r} = \frac{1-\omega^{(l-k)N}}{1-\omega^{(l-k)}}$$

$= 0$ // see proof next page.

4

lets work out $\omega^{(l-k)N}$; $l \neq N$

$$\omega^{(l-k)N} = 1 e^{-j\frac{2\pi}{N}(l-k)N} \quad ; \text{ N cancels!}$$

$$= 1 e^{-j2\pi(l-k)} \neq 0.$$

$$= 1 e^{-j2\pi m} \quad ; m = l-k \neq 0$$

$$= 1 \quad \blacksquare$$

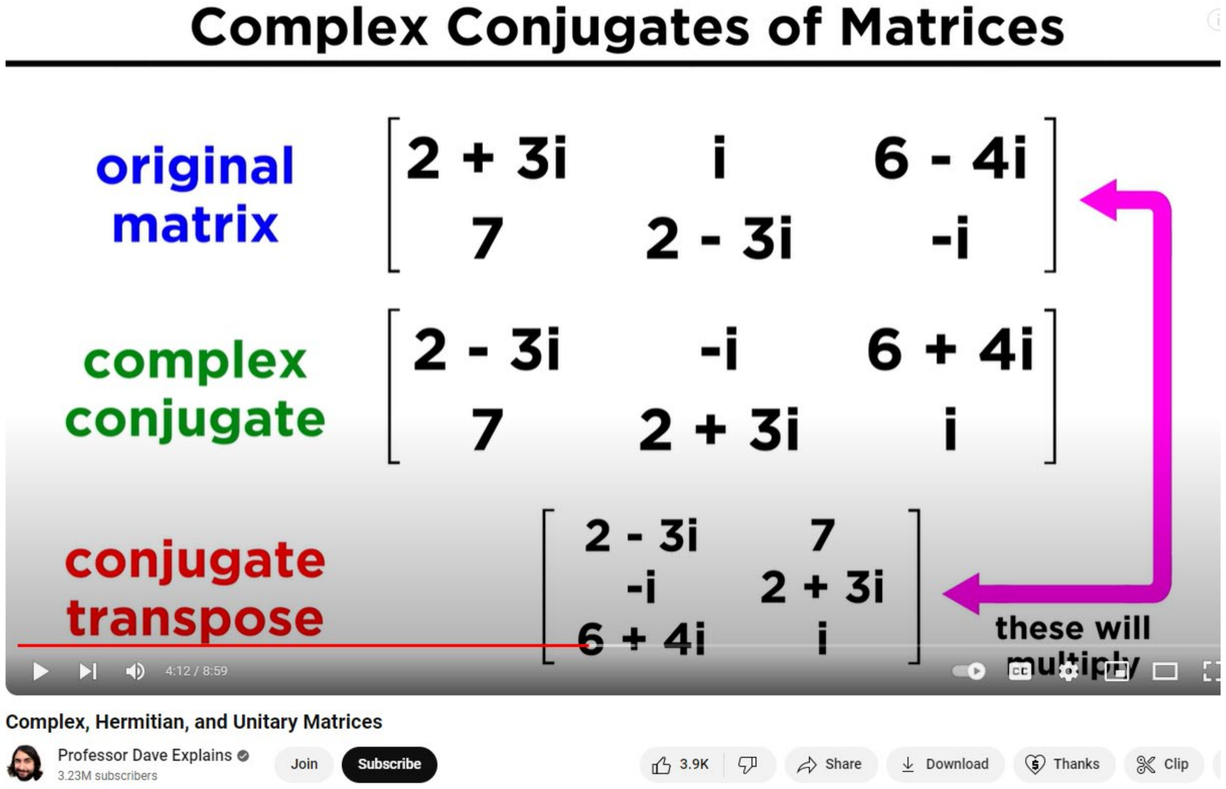
$$\therefore S_n = \frac{1-\omega^{(l-k)N}}{1-\omega^{(l-k)}}$$

$$= \frac{1-1}{1-\underbrace{\omega^{(l-k)}}_{\neq 0}} = \frac{0}{1-\underbrace{1e^{-j\frac{2\pi}{N}(l-k)}}_{\neq 0}}$$

$$= 0. \quad \blacksquare$$

References: conjugate transpose U^H

1) Prof Dave Explains: Complex Matrixes
https://www.youtube.com/watch?v=DUuTx2nbizM



unitary (complex)

- columns form orthonormal vectors

$U^H = U^{-1}$

(conjugate transpose = inverse)

Ref: Basic properties of inner products

Basic properties [edit]

In the following properties, which result almost immediately from the definition of an inner product, x, y and z are arbitrary vectors, and a and b are arbitrary scalars.

- $\langle \mathbf{0}, x \rangle = \langle x, \mathbf{0} \rangle = 0.$
- $\langle x, x \rangle$ is real and nonnegative.
- $\langle x, x \rangle = 0$ if and only if $x = \mathbf{0}.$
- $\langle x, ay + bz \rangle = \overline{a}\langle x, y \rangle + \overline{b}\langle x, z \rangle.$
  This implies that an inner product is a sesquilinear form.
- $\langle x + y, x + y \rangle = \langle x, x \rangle + 2\,\mathrm{Re}(\langle x, y \rangle) + \langle y, y \rangle$, where Re denotes the real part of its argument.

From: Wikipedia
https://en.wikipedia.org/wiki/Inner_product_space#
https://proofwiki.org/wiki/Inner_Product_is_Sesquilinear

Convention variant [edit]

Some authors, especially in physics and matrix algebra, prefer to define inner products and sesquilinear forms with linearity in the second argument rather than the first. Then the first argument becomes conjugate linear, rather than the second. Bra-ket notation in quantum mechanics also uses slightly different notation, i.e. $\langle \cdot | \cdot \rangle$, where $\langle x|y \rangle := (y, x)$.

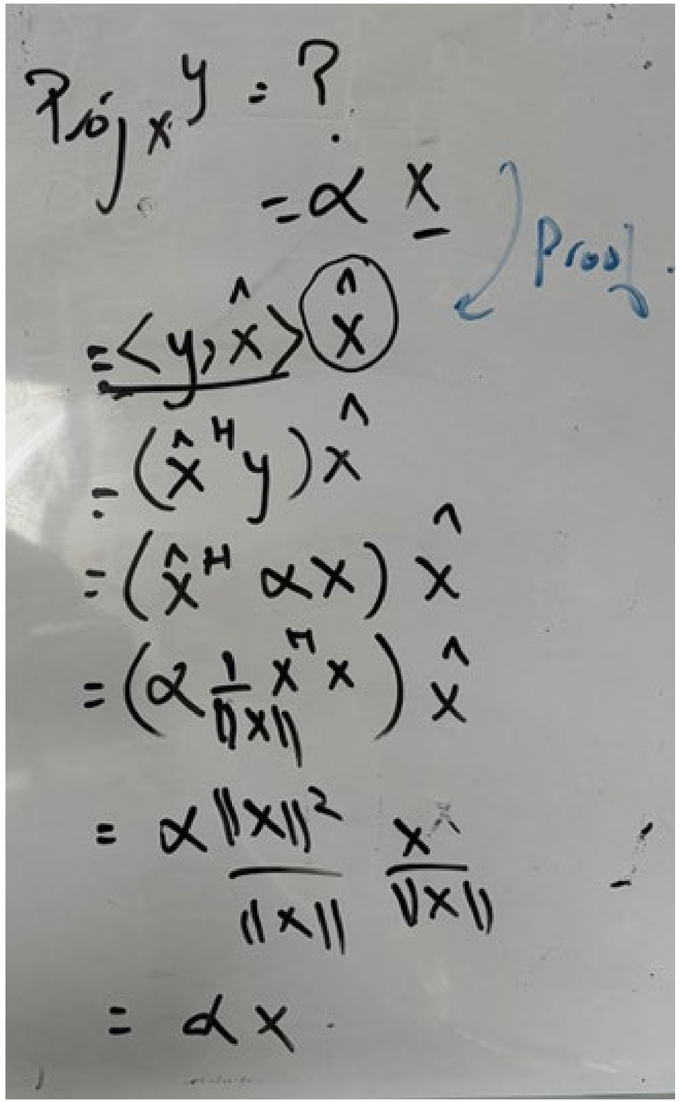
Example: projecting y = αx into x (proof for pg11)

If $\boldsymbol{y} = \alpha\, \boldsymbol{x}$, what is $Proj_x \boldsymbol{y}$?

$where$: $\alpha \in C,\ x \in C^N$

$$\hat{x} = \frac{x}{||x||}$$

$\hat{x}^H$ = conjugate transpose of $\hat{x}$



Developing the Analysis & Synthesis Eq^n of DFT // Chng Eng Siang OCT 2024.

Analysis Eq^n.
DFT

$$\begin{bmatrix} Y(0) \\ Y(1) \\ \vdots \\ Y(N-1) \end{bmatrix} = \begin{bmatrix} & W & \end{bmatrix} \begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix}$$

DFT coeff — $N \times 1$

$N \times N$ DFT matrix.

$N \times 1$. discrete time sequence.

Notation: $\underset{N\times 1}{\underline{Y}} = \underset{N\times N}{W}\ \underset{N\times 1}{\underline{y}}$.

elements of $\underline{Y}$ represents the $y \in R^N$ in the freq domain.

// Note: The W matrix is the DFT matrix in the analysis equation

Synthesis Eq^n:

$\left\{ \frac{1}{N} W^H = W^{-1} \right\}$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix} = \begin{bmatrix} & W^{-1} & \end{bmatrix} \begin{bmatrix} Y(0) \\ Y(1) \\ \vdots \\ Y(N-1) \end{bmatrix}$$

Fourier Basis:

To perform change of coordinate from discrete time to basis $\{ b_k \}_{k=0..N-1}$.

where

$$b_k = \begin{bmatrix} e^{+j\frac{2\pi}{N}k0} \\ e^{+j\frac{2\pi}{N}k1} \\ \vdots \\ e^{+j\frac{2\pi}{N}k(N-1)} \end{bmatrix} \in \mathbb{C}^N$$

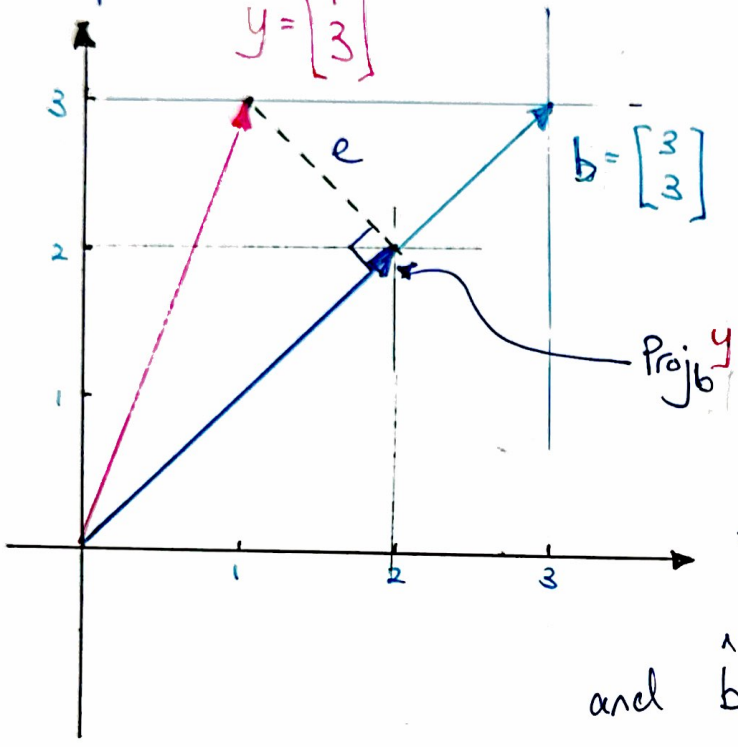
$n = 0..(N-1)$ ↓ $n=0$

i) a vector of N elements, with $n \to 0..(N-1)$ for each element in row n.

ii) change of phase as n increases angular freq by $\left(\frac{2\pi}{N}k\right)$ rad/sample.

Review: Orthogonal Projection to a vector

R^2 space



$$\text{Proj}_b y = \left(\frac{b^T y}{\|b\|}\right)\left(\frac{b}{\|b\|}\right) = \langle y, \hat{b} \rangle \hat{b}$$

Projecting vector v onto line span by b such that the error $e \perp b$.

and $\hat{b} = \frac{b}{\|b\|}$

and $\|b\| = \sqrt{b^T b}$

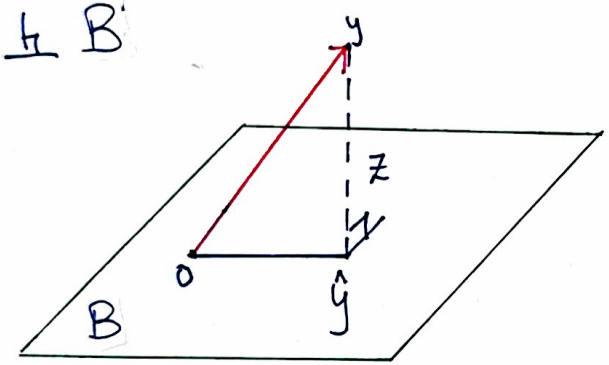
// Ref: 6.2.2 orthogonal Projection.

Review: Orthogonal Projection onto any orthogonal basis $\mathcal{W}$.

// Ref: 6.2.5 orthogonal Decomposition (Theorem 8) (for complex vectors)

$$y = \hat{y} + z$$

and $z \perp B$



$\text{Proj}_B y = \hat{y}$ = orthogonal projection of y onto B

B = vector space with orthogonal basis $\{b_0, b_1, \cdots, b_{N-1}\}$

$$\hat{y} = \text{Proj}_B y = \sum_{k=0}^{N-1} \langle y, \hat{b}_k \rangle \hat{b}_k ,$$

$$\hat{b}_k = \frac{b_k}{\|b_k\|}$$

$$= \sum_{k=0}^{N-1} \left(\frac{b_k^H y}{\|b_k\|} \right) \frac{b_k}{\|b_k\|}$$

; b_k^H = conjugate transpose of col b_k

$$= \sum_{k=0}^{N-1} \frac{b_k^H y}{b_k^H b_k} b_k$$

// complex version of Theorem 8, sec 6.2.5

Note 1: if $y \in B$, then $z = \underline{0}$.

$\hat{y} = y$.

Note 2: if $y \in R^N$ or C^N

and B has an orthogonal basis that spans B,

$\Rightarrow \{b_0, b_1, \cdots, b_{N-1}\}$

then $\text{Proj}_B y = \hat{y} = y$ ■

Discrete Fourier Transform.

1) Fourier Transform propose to use complex exponential basis B.

$$B = \{ b_0, b_1, b_2, \dots, b_{N-1} \}$$

vector space spanned by

$$B = \left\{ \begin{bmatrix} e^{+j\frac{2\pi}{N}00} \\ e^{+j\frac{2\pi}{N}10} \\ e^{+j\frac{2\pi}{N}20} \\ \vdots \\ e^{+j\frac{2\pi}{N}(N-1)0} \end{bmatrix}, \begin{bmatrix} e^{+j\frac{2\pi}{N}01} \\ e^{+j\frac{2\pi}{N}11} \\ e^{+j\frac{2\pi}{N}21} \\ \vdots \\ e^{+j\frac{2\pi}{N}(N-1)1} \end{bmatrix}, \dots \begin{bmatrix} e^{+j\frac{2\pi}{N}0,N-1} \\ e^{+j\frac{2\pi}{N}1,N-1} \\ \vdots \\ e^{+j\frac{2\pi}{N}(N-1)(N-1)} \end{bmatrix} \right\}$$

sample idx $= 0..N-1$ (↓)

b_0 b_1 b_{N-1}

$b_k \in \mathbb{C}^N$, $k = 0..(N-1)$ (→ $k = 0..(N-1)$.)

angular frequency ω_k at col k $= \left(\frac{2\pi}{N} k\right)$ rad/sample.

Discrete

2) The Fourier Transform uses B as the new basis, and B is an orthogonal basis

$$\hat{y} = Proj_B y = \sum_{k=0}^{N-1} \langle y, \hat{b}_k \rangle \hat{b}_k, \quad \hat{b}_k = \frac{b_k}{\|b_k\|}$$

$$\hat{y} = \sum_{k=0}^{N-1} \left(\frac{b_k^H}{\|b_k\|} y \right) \frac{b_k}{\|b_k\|}, \quad \|b_k\|^2 = N$$

$$\hat{y} = \sum_{k=0}^{N-1} \frac{1}{N} \left(b_k^H y \right) b_k$$

since $y \in B$, then $\hat{y} = y$

$$y = \begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix} \in R^N$$

* Interpreting the B matrix

$(N \times N)$

$$B = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ b_0 & b_1 & \cdots & b_{N-1} \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,N-1} \\ b_{1,0} & b_{1,1} & & \vdots \\ b_{2,0} & \vdots & \ddots & \vdots \\ \vdots & & & \\ b_{N-1,0} & b_{N-1,1} & \cdots & b_{N-1,N-1} \end{bmatrix}$$

where $b_{n,k} = e^{+j\frac{2\pi}{N}nk}$

$\therefore$ each col^{n} of b_k is a sequence of complex exponential length = 1, (angular frequency) rotating anti-clockwise at $\frac{2\pi}{N}k$ rad/sample.

Note: compare to $\underbrace{W}_{DFT}$ matrix which is rotating clockwise at $\frac{2\pi}{N}k$ rad/sample

for col^{n} k : } since symmetric matrix.
or row k : }

n idx

k idx

$$b_0 = \begin{bmatrix} e^{+j\frac{2\pi}{N}0,0} \\ e^{+j\frac{2\pi}{N}1,0} \\ \vdots \\ e^{+j\frac{2\pi}{N}(N-1),0} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

b_0 ↑ k idx

a constant value = 1 when $n = 0 \ldots (N-1)$

n idx

k idx

$$b_{k=1} = \begin{bmatrix} e^{+j\frac{2\pi}{N}0,1} \\ e^{+j\frac{2\pi}{N}1,1} \\ e^{+j\frac{2\pi}{N}2,1} \\ \vdots \\ e^{+j\frac{2\pi}{N}(N-1)(1)} \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \\ \\ \\ \\ \\ \end{bmatrix} \quad \begin{matrix} n=0 \\ n=1 \\ n=2 \\ \vdots \\ n=7 \end{matrix}$$

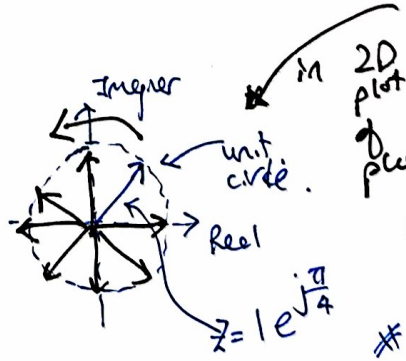
$b_{k=1}$ ↑ k idx

e.g. $N = 8$

rotating anti-clockwise at $\frac{2\pi}{8} = \frac{\pi}{4}$ rad/sample.

for $k = 1$

since $\frac{2\pi}{N}(1) = \frac{2\pi}{8} = \frac{\pi}{4}$ rad/sample



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* Show that the cols of B are orthogonal.

Let b_l, b_k represents the cols of B.

To show orthogonality, since $b_k \in C^N$, we need inner product $\langle \cdot, \cdot \rangle$.

$$\langle b_l, b_k \rangle = \begin{cases} 0 & \text{if } l \neq k \\ N & \text{if } l = k. \end{cases}$$

$$\langle b_l, b_k \rangle = \sum_{n=0}^{N-1} b_{n,l} \overline{b_{n,k}}$$

Note: each element is conjugated!

where $b_{n,k} = e^{+j\frac{2\pi}{N}nk}$

$b_{n,k}$ = element n of col k of B

$$\langle b_l, b_k \rangle = b_k^H b_l \quad // \; b^H = \text{Hermitian} = \text{conjugate transpose!}$$

$$= \overline{b_k}^T b_l$$

Note: when $l = k$, $\overline{b_k}^T b_k = N$ ☒

Let $\omega = e^{+j\frac{2\pi}{N}}$, $\omega^{-nk} = e^{-j\frac{2\pi}{N}nk}$

$$\langle b_l, b_k \rangle = \sum_{n=0}^{N-1} \omega^{+nl} \omega^{-nk}$$

$$= \sum_{n=0}^{N-1} \omega^{(l-k)n}$$

if $(l-k) \neq 0$ then

$$\langle b_l, b_k \rangle = \frac{1-\omega_0^N}{1-\omega_0} = 0$$

bcos:

$\omega_0 = \omega^{(l-k)} \neq 0$

$= \left(1 e^{j\frac{2\pi}{N}}\right)^{(l-k)} \neq 0.$

$\omega_0^N = 1 e^{j\frac{2\pi}{N}(l-k)N}$

$= 1 e^{j2\pi(l-k)} \neq 0$

$= 1$ ☒

$S_n = ar^0 + ar^1 + \ldots + ar^{n-1}$

$= \sum_{k=0}^{N-1} ar^k$

$= \frac{a - ar^n}{1-r}$

Proof:

$S_n = ar^0 + ar^1 + \ldots + ar^{n-1}$

$rS_n = ar^1 + ar^2 + \ldots + ar^n$

$S_n - rS_n = ar^0 - ar^n$

$S_n(1-r) = a - ar^n$

$\therefore S_n = \frac{a - ar^n}{1-r}$

$= a\left(\frac{1-r^n}{1-r}\right).$

∴ Synthesis Eq?

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix} = \frac{1}{N} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ b_0 & b_1 & \cdots & b_{N-1} \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \begin{bmatrix} \leftarrow \bar{b}_0 \rightarrow \\ \leftarrow \bar{b}_1 \rightarrow \\ \vdots \\ \leftarrow \bar{b}_{N-1} \rightarrow \end{bmatrix} \begin{bmatrix} Y(0) \\ Y(1) \\ \vdots \\ Y(N-1) \end{bmatrix}$$

B (N×N) matrix.

W matrix (N×N) (N×1)

$$b_k^H y = Y(k)$$

Note

b_k^H = conjugate transpose

$\bar{b}_k$ = conjugate

and the DFT coeff is

$$\begin{bmatrix} Y(0) \\ Y(1) \\ \vdots \\ Y(N-1) \end{bmatrix} = \begin{bmatrix} \leftarrow \bar{b}_0 \rightarrow \\ \leftarrow \bar{b}_1 \rightarrow \\ \vdots \\ \leftarrow \bar{b}_{N-1} \rightarrow \end{bmatrix} \begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix}$$

analysis eq?

$$Y = W \cdot y$$

Y: DFT coeff

DFT matrix W.

y: Discrete time seq.

Synthesis Eq?

$$y = \left(\frac{1}{N} W^{-H}\right) Y$$

Note: . is symmetric.

W^H = conjugate transpose

$$Y = W y$$

each raw is complex exponential rotating clockwise at $\omega = \frac{2\pi}{N} k$ rad/sample

$$\begin{matrix} \text{raw } 0 \rightarrow \\ \text{raw } 1 \rightarrow \\ \bar{b}_k \rightarrow \\ \text{raw } (N-1) \rightarrow \end{matrix} \begin{bmatrix} e^{-j\frac{2\pi}{N}0,0} & e^{-j\frac{2\pi}{N}1,0} & e^{-j\frac{2\pi}{N}2,0} & \cdots & e^{-j\frac{2\pi}{N}(N-1),0} \\ e^{-j\frac{2\pi}{N}0,1} & e^{-j\frac{2\pi}{N}1,1} & & \cdots & e^{-j\frac{2\pi}{N}N-1,1} \\ \vdots & \vdots & & & \vdots \\ e^{-j\frac{2\pi}{N}0,N-1} & -j\frac{2\pi}{N}1,N-1 & & & e^{-j\frac{2\pi}{N}(N-1)(N-1)} \end{bmatrix}$$

SC1004 Linear Algebra for Computing

Tutorial 8A (Complex Numbers)

Complex numbers

Q1) Evaluate $z = (2-i)^2 + \frac{7-4i}{2+i} - \frac{1}{1+i}$, and give the result in rectangular and complex exponential form.

Q2) Given $y = -2x^2 + 3x - c$, where c is a real number, answer the following questions:

a) Find c for the case when there is only 1 repeated root.
b) For $c = -2$, fine the roots of the equation
c) When $c \to +\infty$, there will only be (repeated/complex/real)? Roots
d) When $c \to -\infty$, there will only be (repeated/complex/real)? Roots

Q3) Evaluate z^4 given $z = \overline{(1+2i)}$

Q4) Find the values of a, b for $(a-bi)^2(3+5i) = \overline{(-6-24i)}$.

DFT

Q5)

a) Given $x = [2,0,0,3]^T$, find DFT coefficients X with $N = 4$.
b) Given $X = N\begin{bmatrix} 0 \\ 3e^{-j\frac{\pi}{4}} \\ 0 \\ 0 \end{bmatrix}$, find x via IDFT with $N = 4$, and suggest the equation describing $x[n]$ by inspection.
c) Given $X = N\begin{bmatrix} 2 \\ 3e^{-j\frac{\pi}{4}} \\ 0 \\ 0 \end{bmatrix}$, find solution of x found in Q5b, suggest the equation describing $x[n]$ by inspection.

// To be Discussed in the lecture

Q6) The following questions to help students have intuition of DFT.

For $N = 16$,

a) Describe the DFT Analysis matrix W .
b) Provide the equation representing the k^{th} row of W and hence list the first 4 terms.
c) How is the k^{th} row of W related to the DFT coefficient $X(k)$. Provide an interpretation.
d) Given $x(n) = 3e^{j(\frac{2\pi}{16}n + \frac{\pi}{3})}$, what is $X(k)$?
e) Given $x(n) = 3\sin(\frac{2\pi}{16}n + \frac{\pi}{3})$, what is $X(k)$? How is this answer differ to Q6d.

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